



PREPARED FOR
AELIA
AELIA HOLDINGS LIMITED

CONFIDENTIAL · SYSTEM REQUIREMENTS & DESIGN DOCUMENT

AELIA Creator Commerce Platform → System Requirements & Design Document

The single consolidated specification for the AELIA Creator Commerce Platform. It covers requirements, delivery methodology, technical architecture, the pluggable integration framework, API, data model, UI/UX, security and data protection, payment integration, test and UAT, and the deployment and support runbook. The platform is built on the Codevertex ecosystem, and every integration can be reconfigured from within AELIA.

Marketing Objective → Creator Match → Campaign → Content → Commerce → Measurement → Relationship

MVP Pilot · Month 3

Version 1 · Month 6

Go · Next.js PWA · PostgreSQL · Redis · NATS

Pluggable Integrations

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DOCUMENT VERSION

1.0 (Baseline for Development)

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CLASSIFICATION

Confidential

Table of Contents

Document Control	4
Resolved Decisions Register	5
Glossary	6
PART A · PRODUCT & REQUIREMENTS	7
1. Introduction & Platform Overview	7
2. Stakeholders & Actors	8
3. Business Objectives & Product Principles	9
4. Functional Requirements	10
5. User Journeys	19
6. Non-Functional Requirements	20
7. Release Scope & Future Enhancements	21
PART B · METHODOLOGY & DELIVERY	24
8. Delivery Methodology	24
9. Governance, Roles & Responsibilities	25
10. Delivery Roadmap: MVP and Version 1	26
PART C · TECHNICAL ARCHITECTURE SPECIFICATION	30
11. Architecture Principles	30
12. System Context & Container Architecture	31
13. Internal Module Design	33
14. Codevertex Ecosystem Integration Map	35
15. Event-Driven Design	37
PART D · PLUGGABLE INTEGRATION FRAMEWORK	39
16. Integration Framework Overview	39
17. Provider Ports & Adapters	40
18. Configuration Model	44
19. Webhooks & Callback URLs	45
20. Default Codevertex Profile & Provider Swap Scenarios	47
PART E · API SPECIFICATION	50
21. API Conventions	50
22. Endpoint Catalogue	51
23. Event Catalogue	54
24. Shared Service & External API Documentation Register	56
PART F · DATABASE & DATA MODEL SPECIFICATION	58
25. Data Architecture	58

26. Entity-Relationship Diagrams	59
27. Entity Dictionary, Ownership & Retention	61
PART G · WORKFLOWS & STATE MACHINES	64
28. Campaign Lifecycle	64
29. Content, Review & Dispute Workflows	67
PART H · UI/UX DESIGN SPECIFICATION	68
30. Experience Architecture	68
PART I · SECURITY & DATA PROTECTION SPECIFICATION	71
31. Security Architecture	71
32. Data Protection & Privacy	73
PART J · PAYMENT INTEGRATION SPECIFICATION	75
33. Payments, Ledger & Tax	75
PART K · TEST & UAT PLAN	79
34. Test Strategy	79
35. User Acceptance Testing & Acceptance Criteria	81
PART L · DEPLOYMENT & SUPPORT RUNBOOK	83
36. Deployment Architecture	83
37. Operations & Support	85
38. Technical Risks, Assumptions & Dependencies	86
39. Appendix A · Requirements Traceability Matrix	87
40. Appendix B · References & Sign-Off	88

Document Control

Item	Detail
Document	AELIA Creator Commerce Platform: System Requirements & Design Document (SRDD)
Version	1.0, the baseline for development
Date	28 September 2026
Status	For AELIA product-owner sign-off before development starts on 5 October 2026
Classification	Confidential
Prepared by	Codevertex Africa Limited (Technical Lead: Titus Owuor)
Prepared for	Aelia Holdings Limited (Product Owner: Erick Wanga)
Platform	AELIA Creator Commerce: a brand and creator marketplace for influence and content
Technology stack	Go backend services; Next.js Progressive Web App; PostgreSQL; Redis; NATS JetStream; Codevertex shared libraries
Ecosystem services used by default	Codevertex SSO (auth-api), Notifications, Subscriptions, Treasury (payments, payouts, ledger, eTIMS), MarketFlow AI runtime (Vera)
New components	<code>aelia-api</code> , <code>aelia-worker</code> , <code>aelia-ui</code> , and a dedicated AELIA PostgreSQL database
Delivery	MVP pilot live at month 3 (week 13); Version 1 live at month 6 (week 24); handover at week 26

Purpose of this document

This SRDD is the single consolidated specification for the AELIA Creator Commerce Platform. It merges the approved product requirements, the technical feasibility findings and the delivery plan into one baseline. It also includes all eight technical documents listed in §5.4 of the requirements specification as numbered Parts. Commercial arrangements between the parties are governed by a separate agreement and are out of scope here.

Source documents and traceability

Source document	Content taken into this SRDD	Where in this SRDD
AELIA System Requirements & MVP Specification v1.0 (Sep 2026)	Platform model, users, objectives, 18 functional modules, journeys, MVP scope, technical/data/trust requirements, ownership, API documentation standard, pilot targets, Definition of Done, next technical documents	Parts A, E, I, K, L
Codevertex Technical Partnership & Feasibility Report (21 Sep 2026)	Ecosystem reuse assessment, reference-plus-cache pattern, integration architecture, transaction swimlane, 10-step campaign sequence, payout schedules, Campaign Composer lanes, withholding tax, data protection, content disclosure, phased roadmap, technical risks	Parts A, C, D, G, J
Codevertex Project Proposal (26 Sep 2026): technical and delivery content only	Scope by release, team and responsibilities, agile methodology, sprint cycle, Definition of Done, change control, phases and gates, work breakdown, delivery KPIs, support model	Parts A, B, K, L

The eight technical documents, as Parts of this SRDD

Spec §5.4 document	Purpose	This SRDD
Technical Architecture Specification	How the platform will be built	Part C and Part D
API Specification	How systems will communicate	Part E
Database / Data Model Specification	How platform data will be structured	Part F
UI/UX Design Specification	Screens, workflows and user experience	Part H (workflows in Parts A and G)
Security & Data Protection Specification	Security and privacy controls	Part I
Payment Integration Specification	Payment and transaction architecture	Part J
Test & UAT Plan	Testing and acceptance	Part K
Deployment & Support Runbook	Operations, maintenance and support	Part L

Resolved Decisions Register

These pre-development decisions (requirements specification §5.3) are resolved for the purposes of this design. Items governed by the separate commercial agreement are referenced but not detailed.

#	Decision	Resolution applied in this SRDD
D1	Legal structure and ownership model	Governed by the separate agreement. The system design is entity-agnostic: the operating entity, its merchant accounts and its tax registration are configuration values.
D2	Operating entity (Aelia Holdings or a dedicated entity)	Same as D1. The operating entity is modelled as the AELIA platform tenant, and a future separate entity is only a configuration change.
D3	Payment provider and transaction architecture	Payments run through licensed rails only (Paystack, M-Pesa) via the Codevertex Treasury service , using merchant accounts held in AELIA's name. AELIA never operates escrow or a stored-value balance. See Part J.
D4	Tax treatment and withholding logic	A configurable tax engine. The default rule is 5% digital-content withholding on gross creator fees, and rates and categories are effective-dated settings. Withholding certificates are issued monthly.
D5	Codevertex shared services reused	SSO (auth-api), Notifications, Subscriptions (entitlements), Treasury only (not the wider ERP suite), the MarketFlow AI runtime (Vera), and the shared Go and UI libraries. Each is bound through the pluggable integration framework in Part D and can be replaced.
D6	Data controller and processor	AELIA is the data controller and Codevertex is the data processor under a written data-processing agreement. See Part I.
D7	Commission and pricing model	Governed by the separate agreement. The system models commission rates, plan tiers and fees as configurable, effective-dated settings with no hard-coded values.
D8	Pilot brands and creators	Pilot targets are set in §7.3. Recruitment is led by AELIA.
D9– D10	Brand Agreement and Creator Agreement	Legal text is prepared by AELIA's counsel. The platform versions each agreement, records acceptance and links it to every campaign (Part F, entity <code>AGREEMENT_TEMPLATE</code>).
D11	Infrastructure, SLA, support and maintenance	Codevertex-managed Kubernetes hosting, with the SLAs in §6 and the support model in Part L.

#	Decision	Resolution applied in this SRDD
D12	Source code, data, account and access arrangements	These follow the ownership and access matrix in §31.4: AELIA-controlled domain, repository access, database, analytics, merchant accounts and backups.
D13	Technology stack	Resolved by Codevertex as the platform standard: a Go backend (chi, Ent, Atlas), a Next.js PWA, PostgreSQL, Redis, NATS JetStream, and the Codevertex shared libraries and event standards.
D14	Mobile channel	A responsive, installable PWA for MVP and v1. Native Android and iOS apps are a Later enhancement on the same APIs.

Glossary

Term	Meaning
Brand	An organisation that commissions creator campaigns. It can have several team members.
Creator	An individual who offers influence (publishing to their own audience) and/or content (producing assets for the brand).
Creator Passport / Brand Passport	The verified professional profile of each side. It covers identity, capability, commercial terms and track record.
Influence campaign	The creator publishes content to their own audience on behalf of a brand.
Content (UGC) campaign	The creator produces content for the brand's use, with or without publishing it themselves.
AELIA Match	Creator recommendations for a brief. The rule is “the platform recommends, the brand decides”.
Campaign Composer	The option screen that shows the Human Creator, Hybrid and AI-Generated lanes for a brief.
Campaign Workspace	The single place where a campaign's brief, creators, agreements, messages, deliverables, approvals, payments and performance are managed.
Funding	A brand's payment for a campaign, collected through a licensed payment rail.
Payout schedule	The agreed release timing for the creator's share: full on approval, a starting fee then the balance, or full upfront.
Withholding (WHT)	Tax deducted from a creator's gross fee and remitted to KRA by AELIA as payer.
Port / Adapter	A port is an interface AELIA code depends on (for example <code>PaymentProvider</code>). An adapter is a concrete implementation of it (for example Codevertex Treasury, direct Paystack, or a generic REST provider).
Integration profile	The set of provider bindings for every capability. The Codevertex default profile is seeded at install.
Tenant	A Codevertex auth-api tenant. AELIA operates as a platform tenant (slug <code>aelia</code>). Brands and creators are AELIA-domain entities, not Codevertex tenants.
Outbox	A transactional table from which domain events are published to NATS, so that no event is lost.
MVP v1 Later	Release tags. MVP means the month-3 pilot. v1 means version 1 at month 6. Later means a future enhancement that is not scheduled in the 6-month plan.

Part A · Product & Requirements

What the platform must do for brands, creators and AELIA operations, specified for the fully developed platform, with each requirement tagged to the release that delivers it.

1. Introduction & Platform Overview

1.1 Purpose

AELIA Creator Commerce is a technology-enabled marketplace connecting **brands** with **creators** for influence and content. It provides the tools to match, collaborate, contract, pay, measure and build repeat commercial relationships.

Core commercial journey: Marketing Objective → Creator Match → Campaign → Content → Commerce → Measurement → Relationship.

Core MVP principle: connect a real brand with a suitable creator, manage the campaign, facilitate payment, measure the outcome, and create the basis for repeat business.

1.2 Platform model

Component	Requirement
Influence Marketplace	Creators publish content to their own audiences on behalf of brands.
Content Marketplace	Creators produce content for brands, whether or not they publish it to their own audiences.
Brand Side	Brands discover creators, create campaigns, collaborate, manage content, fund campaigns and measure results.
Creator Side	Creators build professional profiles, discover opportunities, deliver campaigns, get paid and build commercial relationships.
AELIA Operations	Verification, matching support, campaign administration, payment monitoring, disputes, reporting and platform management.
Fulfilment lanes	Each brief can be fulfilled by an external creator, by AELIA Creative & Digital (the in-house production arm), or through the AI-generated lane. AELIA is accountable in all three.

1.3 Scope boundary

This SRDD covers the **Creator Commerce Platform** and its dedicated web application and domain. The general Aelia Brand House website, which introduces AELIA Marketing, AELIA Creative & Digital (with its Academy) and this platform, is a separate build and is outside this specification. The platform links to it only through public navigation.

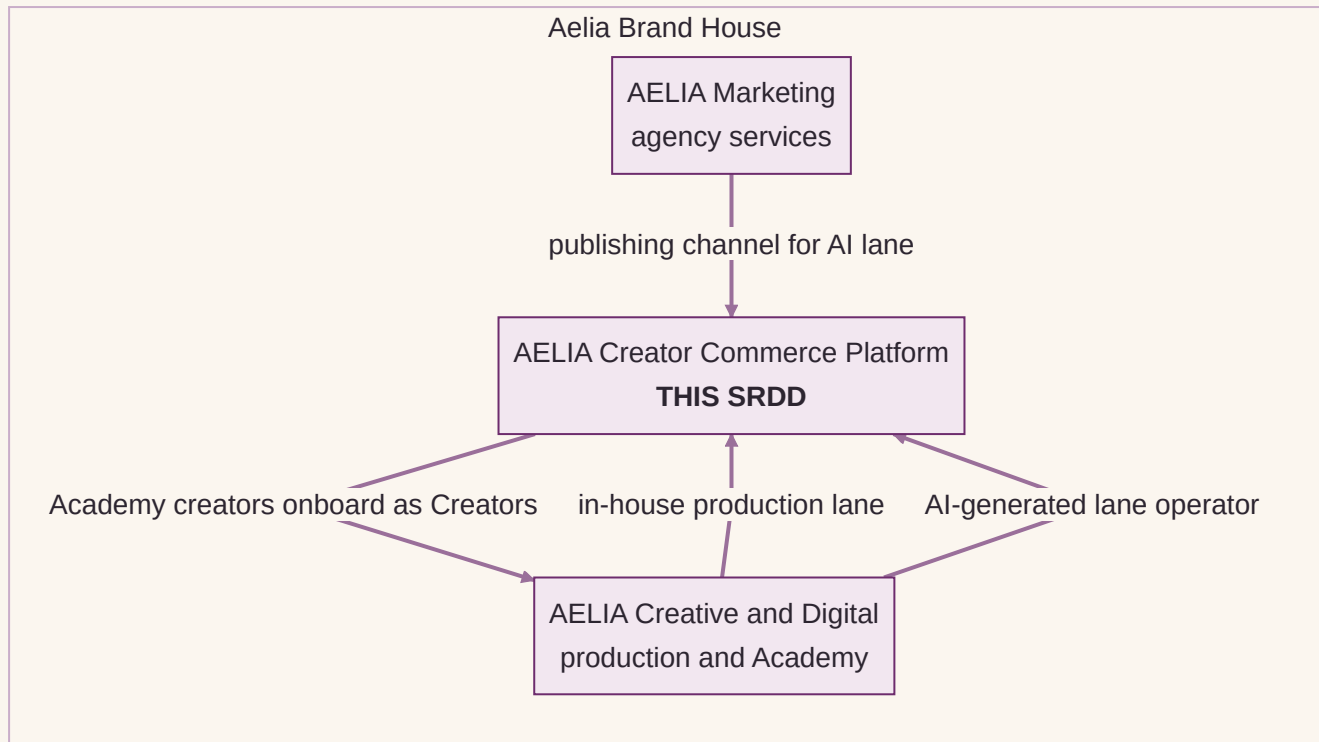


Figure 1. The platform within the Aelia Brand House. Only the Creator Commerce Platform is in scope.

2. Stakeholders & Actors

2.1 Human actors

Actor	Primary role	Key capabilities	Codevertex identity
Brand Owner	Creates and manages creator campaigns and the brand's team	Register the organisation, verify, brief, shortlist, agree, fund, approve, review, rebook	SSO user with the AELIA role <code>brand_owner</code>
Brand Member	A team member acting for a brand	Permissions delegated by the Brand Owner (for example approver or viewer)	SSO user with the AELIA role <code>brand_member</code>
Creator	Offers influence, content and creative services	Passport, socials, rates, availability, apply or accept, deliver, get paid, review	SSO user with the AELIA role <code>creator</code>
AELIA Admin	Operates the marketplace and assures quality	Verification, curated matching, campaign administration, moderation, disputes	SSO user with the AELIA role <code>aelia_admin</code>
Finance / Administration	Monitors money and reconciliation	Verify bank-transfer proofs, approve payouts, reconcile, issue withholding certificates	SSO user with the AELIA role <code>aelia_finance</code> ; OTP required on approvals
Super Administrator	System-level configuration	Integration profiles, tax and fee settings, roles, feature flags, audit	SSO user with the AELIA role <code>super_admin</code> ; 2FA enforced
Agency Manager Later	Manages several brands from one account	Portfolio switching, consolidated reporting, bulk campaign tools	SSO user with the AELIA role <code>agency_manager</code>

A single person may hold several roles, for example a creator who also owns a small brand. The platform supports one login with a role switcher.

2.2 System actors

System actor	Role in AELIA
Identity Provider (default: Codevertex SSO)	Authentication, sessions, MFA, user profile
Notification Provider (default: Codevertex Notifications)	Email, SMS, WhatsApp and push delivery
Billing / Entitlements Provider (default: Codevertex Subscriptions)	Brand plans, feature gates, usage limits
Payment & Payout Provider (default: Codevertex Treasury over Paystack and M-Pesa)	Funding collection, payouts, GL posting, eTIMS
AI Runtime (default: MarketFlow Vera)	Brief builder assistance, AELIA Match scoring and explanations
Social Platforms (Meta, TikTok, YouTube, LinkedIn)	Creator account verification and audience and performance metrics
AI Video Vendors (HeyGen, Creatify)	Generation for the AI-content lane

2.3 Use-case overview



Figure 2. Use-case overview by actor.

3. Business Objectives & Product Principles

#	Business objective (spec §1.4)	Product capability that serves it	Measured by
01	Connect brands with relevant creators	Passports, discovery, AELIA Match	Shortlists per brief, match acceptance rate

#	Business objective (spec §1.4)	Product capability that serves it	Measured by
02	Turn marketing objectives into structured creator campaigns	Brief builder, Campaign Composer, workspace	Briefs published and completed
03	Create professional commercial opportunities for creators	Opportunity feed, invitations, agreements	Invitations and acceptances per creator
04	Facilitate campaign collaboration and payment	Workspace, licensed-rail funding, payouts	Funded campaigns, on-time payouts
05	Measure campaign and commercial outcomes	Performance tracking, reporting	Reports generated, metrics captured
06	Build two-sided reputation and trust	Verification, two-way reviews, disputes	Review coverage, dispute rate
07	Enable repeat bookings and relationships	Saved creators, rebooking	Repeat-booking rate
08	Build creator-commerce data and intelligence	Event stream, analytics store, match feedback loop	Match quality over time
09	Generate sustainable platform revenue	Configurable commission, fees and plans captured in the ledger	Ledger-reconciled platform revenue

Product principles

1. **The platform recommends and the brand decides.** Matching explains its reasoning and never auto-books.
2. **Human in the loop for MVP.** AELIA staff confirm matches and introductions before money or a creator's time is committed.
3. **Money only moves on licensed rails.** AELIA records intents, splits and payouts and never holds client money.
4. **Compliance by default.** Withholding tax, data-protection consent and sponsored-content disclosure are built in from the MVP.
5. **KES-native, multi-currency-ready.** Currency, country and tax rules are per-account settings from day one.
6. **Everything configurable.** Rates, fees, plans, templates and integration providers are settings, not code.
7. **Differentiate on integration.** The differentiators are KES contracts and payouts, automatic withholding, cross-platform matching from one brief, a reconcilable audit trail, and the Influence plus Content dual marketplace.

4. Functional Requirements

Requirements are grouped by platform module (spec Part 2), plus the modules the feasibility review added: tax, AI content lane, integrations hub and compliance. They are specified for the **fully developed platform**. Each requirement carries a release tag, and only **MVP** and **v1** items are scheduled in the 6-month delivery plan (Part B). **Later** items make up the Future Enhancements Backlog (§7.4). Priorities follow MoSCoW: every **MVP** item is a *Must*, **v1** items are *Should*, and **Later** items are *Could*.

4.1 User Registration & Verification (REG)

ID	Requirement	Release	Acceptance criteria
REG-01	Brands, creators and AELIA staff sign up and sign in through the configured Identity Provider (default Codevertex SSO, OIDC + PKCE).	MVP	A new user completes SSO sign-up and lands on role onboarding; the AELIA user is provisioned just-in-time on first valid token.

ID	Requirement	Release	Acceptance criteria
REG-02	Role selection at onboarding (Brand or Creator); one person may hold both, with a role switcher.	MVP	A user can create a Brand and a Creator profile under one login and switch context without re-authenticating.
REG-03	Brand organisation registration: legal name, registration number, KRA PIN, industry, country, address, billing contact.	MVP	Required fields validated; KRA PIN format checked; organisation in <code>pending_verification</code> .
REG-04	Creator registration: legal name, national ID or passport, date of birth (18+ gate), phone, email, KRA PIN, payout method.	MVP	Under-18 blocked; KRA PIN mandatory before any payout; payout method validated against provider.
REG-05	Document upload for verification (certificate of incorporation, ID) to encrypted object storage.	MVP	Files ≤ 10 MB, PDF/JPG/PNG, virus-scanned, visible only to owner and verifiers.
REG-06	Admin verification queue with approve / reject / request-more-info and reason codes; verified badge on Passport.	MVP	Decision is audited and notifies the applicant; unverified brands cannot fund and unverified creators cannot be booked.
REG-07	Brand team management: invite members by email, assign brand roles (owner, approver, finance, viewer).	MVP	Invitee accepts via SSO; permissions enforced on every brand-scoped endpoint.
REG-08	Two-step login (MFA) enforced for AELIA Admin, Finance and Super Admin; optional for others.	v1	Staff roles cannot reach admin routes without an MFA-asserted session (<code>amr</code> claim).
REG-09	Automated KYC/KYB checks via a pluggable verification provider (ID and company registry lookups).	Later	Provider result is attached to the verification case; manual override retained.
REG-10	Social sign-in (Google, Apple) through the Identity Provider.	Later	Enabled per IdP configuration with no AELIA code change.

4.2 Creator Passport (CPP)

ID	Requirement	Release	Acceptance criteria
CPP-01	Profile: display name, bio, photo, location, languages, niches (taxonomy), skills, content formats.	MVP	Passport completeness score shown; publishable at ≥ 70%.
CPP-02	Platforms and audience entered manually: handle, URL, follower count, average views, engagement rate, audience geography, age and gender split.	MVP	Values marked “self-declared” until verified.
CPP-03	Rates and packages: per-deliverable rate cards (post, reel, story, video, UGC asset, usage-rights add-ons) in account currency.	MVP	Packages selectable in briefs and agreements; the currency follows account settings.
CPP-04	Availability calendar and capacity (maximum concurrent campaigns).	MVP	Matching excludes unavailable creators for the brief window.
CPP-05	Portfolio: uploaded samples and external links, tagged by niche and format.	MVP	Media stored in object storage, with thumbnails generated.
CPP-06	Usage-rights preferences: default licence terms offered (duration, territory, paid-media allowance, exclusivity).	MVP	Defaults prefill the agreement and can be edited per deal.
CPP-07	Campaign history and on-platform track record: completed campaigns, on-time rate, approval rate, average rating.	v1	Computed from platform events and not editable by the creator.

ID	Requirement	Release	Acceptance criteria
CPP-08	Connected social accounts via OAuth (Instagram, Facebook, TikTok, YouTube, LinkedIn) with verified metrics refresh.	v1	Verified badge per platform; tokens encrypted; metrics refreshed daily; revocation honoured.
CPP-09	Academy credential badge for AELIA Academy graduates.	v1	Badge set by admin or by an integration event.
CPP-10	Deeper audience analytics (audience quality, fake-follower signals, brand-affinity).	Later	Scores sourced through a pluggable analytics provider.

4.3 Brand Passport (BPP)

ID	Requirement	Release	Acceptance criteria
BPP-01	Organisation profile: logo, description, industry, locations, website, social links, verification status.	MVP	Visible to invited and applied creators.
BPP-02	Commercial reputation: payment reliability (on-time funding and approval), campaign count, creator ratings.	v1	Computed from ledger and review events.
BPP-03	Brand guidelines library (tone, do/don't, assets) reusable across briefs.	v1	Attachable to any brief.
BPP-04	Agency portfolio: one agency account managing several brand passports.	Later	Agency users switch brand context; consolidated reporting.

4.4 Creator Discovery (DIS)

ID	Requirement	Release	Acceptance criteria
DIS-01	Search creators by keyword, niche, platform, location, language, follower band, engagement band, rate band and availability.	MVP	Results in < 1 s p95 for 10k creators; filters combinable.
DIS-02	Creator cards and a full Passport view with portfolio and rates (visible only to verified brands).	MVP	Contact details hidden until an agreement exists.
DIS-03	Shortlists: save creators to named lists per campaign or globally.	MVP	Shortlists shared with brand team members.
DIS-04	Opportunity discovery for creators: open briefs matching their niche and platform.	MVP	Creators see only open, funded-intent briefs they qualify for.
DIS-05	Admin-curated shortlist attached to a brief by AELIA staff.	MVP	The brand sees an "AELIA curated" label and the admin's note per creator.
DIS-06	Saved searches and alerts for new matching creators.	v1	Notification when a new verified creator matches.

4.5 Creator Matching: AELIA Match (MAT)

ID	Requirement	Release	Acceptance criteria
MAT-01	Human-curated matching: an AELIA Admin reviews each brief and publishes a shortlist with rationale.	MVP	Every submitted brief has a curated shortlist within the configured SLA (default 2 business days).
MAT-02	Rules-based scoring across niche, platform, geography, audience fit, performance, budget fit, reliability, suitability and availability, with configurable weights.	v1	Every brief gets a ranked shortlist with a per-factor score breakdown; weights editable by Super Admin.

ID	Requirement	Release	Acceptance criteria
MAT-03	Explanations: “why this creator” text generated by the AI runtime from the score factors.	v1	Explanation shown per recommendation; no automated booking.
MAT-04	Human confirmation step: an admin confirms or adjusts the brand’s selection before agreement.	MVP	Selection cannot move to agreement without an admin confirmation record.
MAT-05	Feedback loop: brand accept/reject reasons are captured to tune weights.	v1	Reasons stored and reported in admin match-quality analytics.
MAT-06	Predictive matching model trained on platform outcomes.	Later	Delivered through the same AIMatchProvider port; A/B tested against the rules engine.

4.6 Campaign Creation & Campaign Composer (CMP)

ID	Requirement	Release	Acceptance criteria
CMP-01	Brief builder: objective, target audience, platforms, deliverables (type, quantity, specs), timeline, budget, content requirements, usage rights, disclosure requirements.	MVP	Brief validates required fields; saved as draft, then submitted.
CMP-02	Campaign type: Influence, Content (UGC) or both.	MVP	Deliverable templates adapt to the type.
CMP-03	Budget and currency, with an automatic fee preview (commission, taxes) from current settings.	MVP	Preview matches ledger computation for the same inputs.
CMP-04	Creator invitation (direct) and open call (applications) modes.	MVP	Invited creators get notifications; open-call applications are listed for the brand.
CMP-05	AI Brief Builder: assisted drafting of objective, deliverables and messaging from a short prompt.	v1	Suggestions are editable; nothing is saved without brand confirmation.
CMP-06	Campaign Composer: side-by-side options per brief across the lanes: Human Creator, AELIA Creative & Digital in-house production, Hybrid, and AI-Generated.	v1	Each lane shows indicative cost and timeline computed from settings and rate cards.
CMP-07	Campaign templates and duplicate-campaign.	v1	Duplicate copies the brief without agreements or payments.
CMP-08	Multi-market campaigns (several countries or currencies in one campaign).	Later	Per-market sub-budgets and tax rules.

4.7 Campaign Workspace (WSP)

ID	Requirement	Release	Acceptance criteria
WSP-01	A single workspace per campaign showing brief, participants, agreements, messages, deliverables, deadlines, content, approvals, payments and performance.	MVP	All tabs available and permission-scoped.
WSP-02	Deliverable tracker with due dates, status and overdue flags.	MVP	Overdue items trigger reminders at configurable offsets.
WSP-03	Activity timeline (audit of state changes) visible to participants.	MVP	Every state transition appears with actor and time.
WSP-04	Per-creator sub-workspaces in multi-creator campaigns.	MVP	Creators see only their own thread, deliverables and payout.

ID	Requirement	Release	Acceptance criteria
WSP-05	Shared files area with version history.	v1	Previous versions retained and downloadable.

4.8 Content Marketplace, Submission & Approval (CNT)

ID	Requirement	Release	Acceptance criteria
CNT-01	Creators submit drafts (video, image, copy, links) per deliverable.	MVP	Upload up to 1 GB via resumable, direct-to-storage upload; preview generated.
CNT-02	Brand review: approve, or request revision with comments, within the agreed revision allowance.	MVP	Revision counter enforced; exceeding the allowance needs creator consent or a change fee.
CNT-03	Approval locks the approved version (hash recorded) and triggers payout eligibility.	MVP	Approved asset is immutable; the SHA-256 is stored on the approval record.
CNT-04	Publication proof for influence deliverables: live post URL and screenshot; platform verification where the API permits.	MVP	Proof is required before a deliverable is marked “published”.
CNT-05	Sponsored-content disclosure check: required disclosure tag present (for example #ad, a paid-partnership label).	MVP	Checklist is mandatory at submission; admin brand-safety report path available.
CNT-06	Time-coded comments on video drafts.	v1	Comments pinned to timestamps.
CNT-07	Content library for brands with licence metadata (rights, expiry, territory).	v1	Expiry reminders sent before the licence ends.
CNT-08	Automated brand-safety and content moderation scanning.	Later	Via a pluggable moderation provider; flags go to the admin queue.

4.9 Contracts & Rights (AGR)

ID	Requirement	Release	Acceptance criteria
AGR-01	Versioned platform terms (Brand Terms, Creator Terms, Privacy Policy, Acceptable Use) with recorded acceptance (user, version, time, IP).	MVP	No brief submission or application without acceptance of the current terms.
AGR-02	Per-campaign engagement records: AELIA–Brand campaign order and AELIA–Creator engagement, linked by one campaign reference.	MVP	Both records are generated from agreed terms; each party accepts its own record.
AGR-03	Terms captured: deliverables, fees, deadlines, revisions, cancellation, ownership, usage, territory, duration, exclusivity, payout schedule, disclosure.	MVP	Fields are structured, not free text only; rendered to PDF with a hash.
AGR-04	Content-rights chain: the creator licences rights to AELIA, and AELIA passes them to the brand, recorded per deliverable.	MVP	The rights record is visible on each approved asset.
AGR-05	Amendments with re-acceptance and version history.	v1	Prior versions immutable; the diff is visible.
AGR-06	Qualified electronic signature via a pluggable e-signature provider.	Later	The provider is selectable in the Integrations Hub.

4.10 Communication & Notifications (COM)

ID	Requirement	Release	Acceptance criteria
COM-01	Campaign-scoped messaging between brand, creator and AELIA staff, with attachments.	MVP	Messages delivered in real time (SSE) with read receipts; no off-campaign DMs.
COM-02	In-platform notification centre (bell) for all state changes.	MVP	Unread counter; mark-all-read; deep links.
COM-03	Email and SMS notifications for key events via the configured Notification Provider.	MVP	Templates <code>aelia/*</code> for every event in the catalogue (§23).
COM-04	WhatsApp notifications where AELIA has an active WhatsApp configuration.	v1	Meta-approved templates with button links.
COM-05	Web push via the PWA.	v1	Opt-in prompt; delivery via the configured push channel.
COM-06	Per-user notification preferences by channel and category.	v1	Transactional and security messages cannot be disabled.
COM-07	Contact-detail masking in messages until an agreement exists.	MVP	Phone numbers and emails redacted with a notice.

4.11 Payments (PAY)

ID	Requirement	Release	Acceptance criteria
PAY-01	Campaign funding by card or M-Pesa through the configured Payment Provider (default Codevertex Treasury).	MVP	Funding intent created; brand pays; campaign moves to <code>funded</code> only on a verified provider event.
PAY-02	Bank transfer with proof-of-payment upload and Finance verification.	MVP	Finance approves with OTP; the ledger posts on approval; rejection notifies the brand.
PAY-03	Per-transaction commission ledger (double-entry) for funding, commission, withholding, payouts, refunds and fees.	MVP	Every movement balances; the ledger reconciles to provider statements to the shilling.
PAY-04	Creator payouts via M-Pesa (B2C or Pochi) or bank, per payout schedule: full on approval, starting fee then balance, or full upfront (eligibility-gated).	MVP	Payout amount = gross share – withholding – fees; idempotent; status visible to both sides.
PAY-05	Refunds and cancellations per agreement terms (full, partial, pro-rata).	MVP	Refund routed through the original rail; ledger reversal posted.
PAY-06	Transaction statuses and receipts on brand and creator dashboards.	MVP	Receipts downloadable as PDF.
PAY-07	Configurable fee settings: commission rates by plan or campaign type, AI-lane fees, processing-fee pass-through.	v1	Effective-dated; changes audited; never retroactive to funded campaigns.
PAY-08	Bulk payouts and a payout approval batch for Finance.	v1	Batch approval with OTP; partial-failure handling.
PAY-09	Multi-currency collection and payout (UGX, TZS, RWF, NGN and others).	Later	Currency per account and campaign; FX recorded at transaction.
PAY-10	Affiliate and promo-code commissions paid to creators.	Later	See AFF module.

4.12 Tax & Compliance (TAX)

ID	Requirement	Release	Acceptance criteria
TAX-01	Configurable tax engine: work categories, rates, thresholds and effective dates; default rule is digital-content withholding at 5% of the gross creator fee.	MVP	Tax computed at payout from the rule effective on the payout date; the rule version is stored on the payout.
TAX-02	KRA PIN capture and validation for creators and brands before the first payout or funding.	MVP	Payout blocked without a validated PIN.
TAX-03	Withholding register and a remittance export for Finance.	MVP	Monthly export totals equal the ledger WHT account balance.
TAX-04	Monthly withholding certificates per creator (PDF), downloadable and emailed.	v1	A certificate is issued for every creator with WHT in the month.
TAX-05	eTIMS invoice for AELIA's commission and fees via Treasury eTIMS.	v1	Invoice transmitted; failures retried and flagged.
TAX-06	VAT treatment of platform fees as a configurable rule.	v1	VAT applied per rule; shown on receipts.
TAX-07	Country tax packs for regional expansion.	Later	Tax rules loaded per country configuration.

4.13 Performance Tracking (PRF)

ID	Requirement	Release	Acceptance criteria
PRF-01	Manual performance entry by the creator (reach, views, engagement, clicks) with screenshot evidence.	MVP	Required for influence deliverables at +7 days after publication.
PRF-02	API-based metrics for connected accounts (views, likes, comments, shares, saves, reach) where platform permissions allow.	v1	Metrics refreshed daily for 30 days after publication.
PRF-03	Tracked links (UTM plus short link) per creator and deliverable, to count clicks and conversions.	v1	Clicks attributed to the creator and campaign.
PRF-04	Leads, conversions and sales via brand pixel or server-to-server postback.	Later	Postback URL per campaign; signed.

4.14 Reputation (REP)

ID	Requirement	Release	Acceptance criteria
REP-01	Two-sided reviews after completion: the brand rates the creator and the creator rates the brand (1–5 stars across criteria, plus text).	v1	Double-blind: revealed when both submit or after 14 days.
REP-02	Reputation scores feed Passports and matching.	v1	Scores recomputed on each new review.
REP-03	Review moderation and a right of reply.	v1	Flagged reviews go to the admin queue.

4.15 Rebooking (RBK)

ID	Requirement	Release	Acceptance criteria
RBK-01	Saved creators and “work again” lists.	MVP	Available from any completed campaign.

ID	Requirement	Release	Acceptance criteria
RBK-02	Rebook flow: new campaign pre-filled with the same creator, terms and deliverables.	v1	Completed in ≤ 3 steps; admin confirmation optional for trusted pairings.
RBK-03	Retainers and recurring campaigns (monthly deliverables).	Later	Recurring schedule with per-cycle funding.

4.16 Reporting & Analytics (RPT)

ID	Requirement	Release	Acceptance criteria
RPT-01	Basic dashboards: brand (campaigns, spend, status), creator (earnings, pending payouts), admin (queues, volumes).	MVP	Loaded in < 2 s p95.
RPT-02	Campaign performance report per campaign (deliverables, metrics, cost per engagement).	v1	Exportable as PDF and CSV.
RPT-03	Financial reports: funding, commission, WHT, payouts and refunds by period; reconciliation report.	MVP	Totals equal ledger balances.
RPT-04	Platform analytics: GMV, active brands and creators, repeat-booking rate, AI-lane orders, match acceptance.	v1	Computed from the event store; daily refresh.
RPT-05	Self-service BI and data export to AELIA's own warehouse.	Later	Scheduled export via the data connector.

4.17 Administration (ADM)

ID	Requirement	Release	Acceptance criteria
ADM-01	Verification queue and user management (suspend, reinstate, role changes).	MVP	Every action audited with reason.
ADM-02	Campaign administration: view any campaign, curate shortlists, confirm matches, intervene in state.	MVP	Interventions require a reason and are visible in the timeline.
ADM-03	Payment monitoring: funding, payouts, failed transactions, manual-payment verification.	MVP	Failures alert Finance within 5 minutes.
ADM-04	Dispute case management: open, evidence, messages, decision, financial outcome (release, partial refund, full refund).	v1	The decision posts the correct ledger entries; parties notified.
ADM-05	Moderation: flagged content, reviews and messages; brand-safety reports.	v1	SLA timers shown on the queue.
ADM-06	Platform configuration: taxonomies, fee and tax settings, templates, feature flags, match weights.	MVP	Effective-dated changes; audit trail.
ADM-07	Integrations Hub (see INT).	MVP	See Part D.
ADM-08	Full audit log explorer with export.	v1	Filter by actor, entity and time; tamper-evident.

4.18 AI-Generated Content Lane (AIC)

ID	Requirement	Release	Acceptance criteria
AIC-01	Managed-service request: a brand selects the AI lane; the order is fulfilled by AELIA Creative & Digital off-platform, then delivered and approved in the workspace.	MVP	The order follows the same brief, funding and approval flow; operator is an internal role.
AIC-02	Integrated AI video vendor (HeyGen or Creatify first) via the <code>AIVideoVendor</code> port: script or product URL to draft clip.	v1	Clip generated, stored and routed to human review before delivery.
AIC-03	Automatic cost per clip (vendor cost, operator time, infrastructure share) and brand price from settings.	v1	Cost and price recorded on every order.
AIC-04	Publishing through AELIA Marketing channels as an add-on.	v1	Publication proof recorded like CNT-04.
AIC-05	Hybrid lane: creator brief plus AI-assisted cut-downs, variants and dubbing.	v1	Extra cuts itemised per deliverable.
AIC-06	Cinematic tier (Sora, Veo, Kling, Runway) gated behind human review.	Later	Added as new adapters with no core change.

4.19 Integrations Hub (INT)

ID	Requirement	Release	Acceptance criteria
INT-01	Codevertex default integration profile seeded and active at install for identity, notifications, entitlements, payments and payouts, AI runtime.	MVP	A fresh environment works end-to-end with no manual provider set-up beyond secrets.
INT-02	Super Admin can view every capability's bound provider, health, last error and configuration.	MVP	Status page shows health checked every 60 s.
INT-03	Register an alternative provider for any capability via a built-in adapter or the declarative REST connector, and bind it with fallback.	v1	"Test connection" and conformance checks must pass before activation.
INT-04	Inbound webhook endpoints per provider with configurable signature verification.	MVP	Unsigned or invalid requests rejected with 401 and logged.
INT-05	Outbound webhooks and callback subscriptions for AELIA events with HMAC signing, retries and a delivery log.	v1	Deliveries retried with exponential backoff for 24 h; manual replay.
INT-06	Credentials stored encrypted; only fingerprints displayed; rotation without downtime.	MVP	Secrets never returned by any API.
INT-07	Per-organisation provider overrides (for example an agency bringing its own payment account).	Later	Binding scope <code>organisation</code> .

4.20 Future platform modules

ID	Requirement	Release	Notes
MOB-01	Native Android app	Later	Built on the same public API and IdP; push via FCM.
MOB-02	Native iOS app	Later	As MOB-01; App Store accounts owned by AELIA.
AFF-01	Affiliate commerce: creator storefront links, promo codes, conversion-based commission	Later	Reuses tracked links (PRF-03) and the ledger.

ID	Requirement	Release	Notes
GEO-01	Regional expansion (Uganda, Tanzania, Rwanda, then wider Africa): country as a tenant with its own currency, tax pack, rails and entity	Later	Supported by the multi-tenant design (\$12.4).
AUT-01	Advanced automation: auto-shortlist, auto-reminders, SLA escalations and rule builder	Later	NATS-driven workflow rules.
AIX-01	Advanced AI-generated content, predictive performance and budget optimiser	Later	Via AI ports.

5. User Journeys

5.1 Brand journey (16 stages)



Figure 3. Brand journey. Stages 5–8 include the AELIA human confirmation step (MAT-04).

5.2 Creator journey (16 stages)



Figure 4. Creator journey. In MVP, stage 4 is manual entry; OAuth verification follows in v1.

5.3 How one campaign moves through the platform

Step	What happens	System behaviour
1	The brand posts a brief: objective, audience, platform, deliverables, budget. No money moves.	Campaign draft → submitted ; event <code>aelia.campaign.submitted</code> .
2	AELIA Match generates options: matching creators, in-house production where the budget fits, and the AI lane where it fits.	MVP: admin-curated shortlist. v1: rules-based ranking plus Composer lanes.
3	The brand picks one option or shortlists several. This is a preference, not a booking.	Selection recorded as proposed .
4	An AELIA reviewer confirms the pick against the brief or suggests an adjustment.	Admin confirmation record required (MAT-04).
5	AELIA contacts the creator and arranges an introduction with the brand. In-house and AI lanes skip to production once funded.	Invitation plus notification; introduction logged in the workspace.
6	Terms are agreed: deliverables, deadline, fee, usage rights and payout schedule. The brand accepts the AELIA campaign order and the creator accepts the AELIA engagement.	Two linked engagement records; agreed .
7	The brand funds the campaign through the licensed rail, or by bank transfer with proof for Finance to verify.	Payment intent created in Treasury; funded only on a verified event.
8	The commission is set aside immediately. The creator's share follows the agreed payout schedule.	Ledger entries posted; a starting fee is released if scheduled.
9	The creator produces and submits; the brand reviews, requests permitted revisions, and approves.	Submission and revision loop; approved version hash-locked.

Step	What happens	System behaviour
10	The remaining fee is paid out with withholding applied. Both sides review; the brand can rebook.	Payout with WHT; completed ; review window opens.

6. Non-Functional Requirements

99.5% Monthly platform availability from pilot launch	< 0.5 s p95 API and page response time	4 h / 24 h Recovery time / maximum data-loss window	≥ 70% Test coverage on payment, commission and tax code
0 Open high or critical security findings at launch	To the shilling Monthly ledger reconciliation accuracy	2 h Critical support response, any day	85% Committed sprint work delivered per sprint

ID	Category	Requirement	Release
NFR-01	Availability	99.5% monthly availability for <code>aelia-api</code> and <code>aelia-ui</code> ; <code>replicaCount: 2</code> and a <code>PodDisruptionBudget</code> of <code>minAvailable: 1</code> .	MVP
NFR-02	Performance	95% of API requests under 500 ms; dashboards under 2 s; search under 1 s at 10k creators and 1k concurrent users.	MVP
NFR-03	Scalability	Stateless services behind HPA; NATS consumers use queue groups; object storage for media; no local disk state.	MVP
NFR-04	Recoverability	RTO 4 h and RPO 24 h, with a restore proven by drill before v1. Daily database backups and quarterly restore tests.	v1
NFR-05	Data integrity	Every money movement is double-entry, idempotent and reconcilable; outbox for every domain event; no lost events.	MVP
NFR-06	Security	OWASP ASVS L2 controls; TLS 1.2+; encryption at rest for secrets and PII documents; zero open high or critical findings at launch.	MVP
NFR-07	Privacy	Kenya Data Protection Act 2019 compliance: consent log, DSR workflow, retention schedule (Part I).	MVP
NFR-08	Usability	Mobile-first responsive PWA, installable to the home screen; every core journey completes on a 360 px-wide screen.	MVP
NFR-09	Accessibility	WCAG 2.1 AA for all brand and creator screens.	v1
NFR-10	Localisation	English at launch; i18n-ready strings; currency, date and number formats per account; Kiswahili Later .	MVP
NFR-11	Configurability	Rates, fees, taxes, plans, templates, match weights and integration providers are runtime settings with audit and effective dates.	MVP
NFR-12	Interoperability	Every external dependency is behind a port with at least one alternative adapter path (Part D).	MVP
NFR-13	Observability	Structured logs with <code>request_id</code> , <code>tenant_id</code> and <code>user_id</code> ; health and readiness endpoints; alerting to the on-call channel within 5 minutes.	MVP

ID	Category	Requirement	Release
NFR-14	Maintainability	Codevertex Go and Next.js standards; code review by a second engineer; generated API docs; runbooks.	MVP
NFR-15	Portability and ownership	Container images and Helm values reproducible on AELIA-controlled infrastructure; source in a repository AELIA can access.	MVP
NFR-16	Auditability	Tamper-evident audit log of admin, finance and configuration actions retained for 7 years.	MVP

7. Release Scope & Future Enhancements

7.1 Release definitions

Release	Target	Definition
MVP Pilot	Week 13 (month 3)	A real brand funds a real campaign, a real creator delivers and is paid, and the ledger reconciles to the shilling, in production. Matching is admin-curated; socials are entered manually; the AI lane is a managed service.
v1 Version 1	Week 24 (month 6)	Rules-based AELIA Match, Passport v2 with connected social metrics, reviews and rebooking, reporting, withholding certificates, fee settings, the integrated AI content lane, the full admin console, and hardening, security review, load test and UAT.
Later Enhancements	Not scheduled	Native apps, predictive AI, affiliate commerce, regional expansion, advanced automation and further integrations. These are specified here so the architecture accommodates them.

7.2 Requirements by release

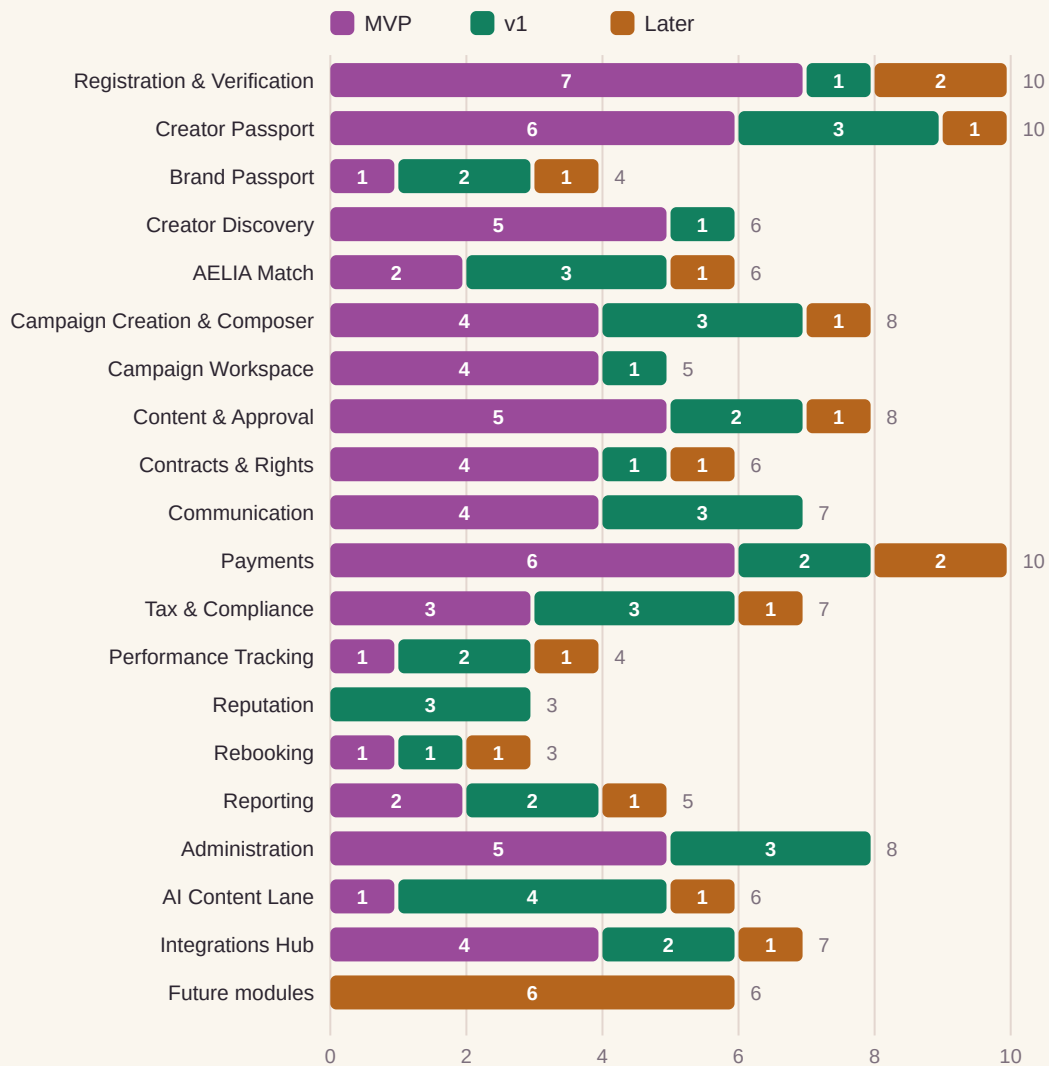


Figure 5. Functional requirements per module by release (129 requirements: 65 MVP, 42 v1, 22 Later).

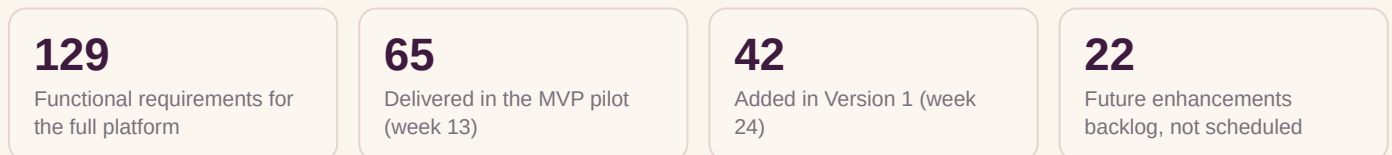


Figure 6. Share of the full platform delivered within the six-month plan: 107 of 129 requirements (83%).

7.3 Pilot validation targets

Indicator	Target
Brands onboarded and verified	8–10
Creators onboarded with a verified Passport	15–20
Real campaigns run end to end	At least 5
Paid end-to-end transactions through the platform	At least 3
Content delivered on time	90% or more
Brands intending to rebook	At least 2

Indicator	Target
User feedback	Collected, triaged each sprint and reviewed
Economic validation data	Campaign value, platform revenue, creator payout, payment costs and support costs, reported from the ledger

7.4 Future Enhancements Backlog (not scheduled in the 6-month plan)

Enhancement	Requirement IDs	Architectural provision already in this design
Native Android and iOS apps	MOB-01, MOB-02	Public versioned API; OIDC public clients; push via the Notification port
Predictive AI matching and AI optimisation	MAT-06, AIX-01	<code>AIMatchProvider</code> port; event history retained for training
Full affiliate ecosystem	AFF-01, PRF-04, PAY-10	Tracked links, signed postbacks, ledger account types
Regional expansion	GEO-01, PAY-09, TAX-07, CMP-08	Tenant-per-country, currency on every amount, tax packs
Advanced automation	AUT-01	NATS event stream and a rule-engine hook
Automated KYC/KYB, e-signature, moderation, audience analytics	REG-09, AGR-06, CNT-08, CPP-10	New capabilities registered in the Integrations Hub
Agency accounts and per-organisation provider overrides	BPP-04, INT-07	Organisation-scoped bindings and roles
Retainers and recurring campaigns	RBK-03	Campaign schedule entity
BI export and more AI vendors and formats	RPT-05, AIC-06	Outbound webhooks and data connector; <code>AIVideoVendor</code> port

Part B · Methodology & Delivery

How the MVP and Version 1 scope is delivered in 26 weeks: the method, the governance, the plan, and the gates AELIA signs off.

8. Delivery Methodology

8.1 Approach

Delivery is **agile (Scrum-based) in 13 two-week sprints**. A working demo closes every sprint, and five formal gates (G1–G5) require AELIA sign-off before the next stage. Requirements and system design (this SRDD) are completed **before** development starts on **5 October 2026**. Build phases then overlap, so building, testing and pilot validation run side by side instead of in sequence. This is how the concept's prove-the-workflow-manually-first principle fits into a 6-month plan.

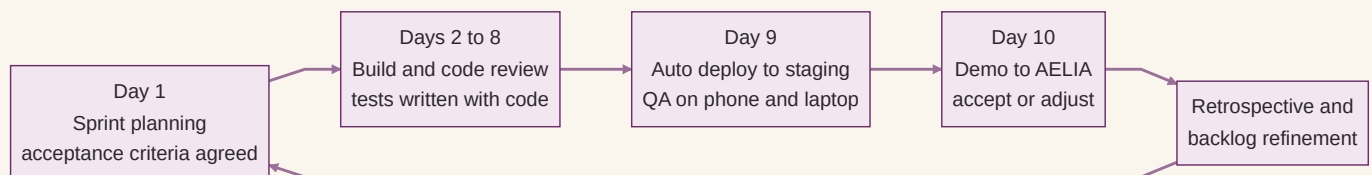


Figure 7. The ten-working-day sprint cycle, repeated 13 times across 26 weeks.

8.2 Engineering practices

Practice	Standard
Source control	Git with protected <code>main</code> ; feature branches; pull requests reviewed by a second engineer
CI	Lint, unit and integration tests, Ent and Atlas migration check, container build and vulnerability scan on every PR
CD	GitOps: merge to <code>main</code> builds an image, CI bumps <code>image.tag</code> , ArgoCD syncs staging automatically; production promotion is a reviewed tag
Testing	Test-first for money, tax and state-machine code; contract tests for every integration adapter (Part K)
Documentation	OpenAPI generated from handlers; ADRs for significant decisions; runbooks kept with the code
Environments	Development (local and preview), Staging (production-like, sandbox providers), Production
Feature flags	Incomplete v1 features are shipped dark behind flags so MVP production stays stable

8.3 Definition of Done (every feature)

Requirements spec §5.2 and the delivery DoD combined: **BUILD → TEST → SECURITY CHECK → DOCUMENTATION → UAT → AELIA APPROVAL**.

Requirement	Completion condition
Functionality	Meets the acceptance criteria agreed at sprint planning and the FR in §4
UI / UX	Screens and workflow completed; checked on a phone and a laptop
Code quality	Reviewed by a second engineer; automated tests pass in CI

Requirement	Completion condition
Money-code coverage	Payment, commission and tax code has at least 70% test coverage
Integration	Required APIs tested against sandbox providers; adapter conformance suite passes
Security	Access and permissions tested; no known high or critical issue open
Error handling	Key failure scenarios tested (provider timeout, duplicate webhook, invalid signature, insufficient permission)
Analytics	Required domain events emitted and visible in the event log
Documentation	API docs, user help and runbook updated
Deployment	Deployed to staging automatically
UAT and approval	Demonstrated to AELIA and accepted by the designated product owner

8.4 Change control

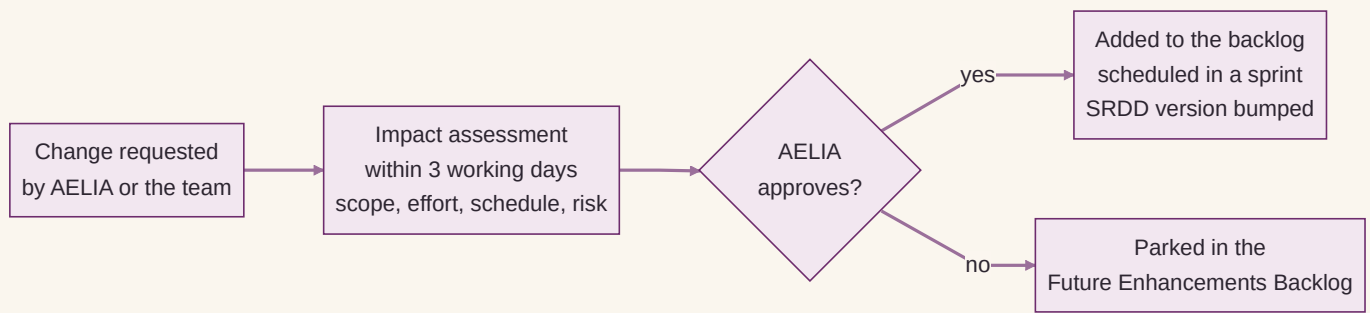


Figure 8. Change control. Approved changes update this SRDD (minor version) and the traceability matrix.

9. Governance, Roles & Responsibilities

9.1 Governance cadence

Forum	Cadence	AELIA	Codevertex	Purpose
Steering	Monthly	Board representative (Erick Wanga)	Director (Titus Owuor)	Scope, gates, risks, go/no-go
Product	Weekly	Product owner	Technical lead	Backlog order, decisions, acceptance
Delivery	Fortnightly	Pilot operations (Creative & Digital, Marketing)	Delivery team	Sprint demo and feedback
Status	Every Friday	Product owner	Technical lead	Written update: done, next, blocked

A shared project board is open to AELIA at all times. The named contact for urgent questions is the technical lead.

9.2 Delivery team

Member	Role	Responsibilities
Titus Owuor	Technical lead, backend and infrastructure	Refines requirements and owns this SRDD; architecture, payments, commission and tax logic, security, infrastructure, releases

Member	Role	Responsibilities
Steve Oginga	Backend engineer	Onboarding, Passports, campaigns, messaging, payouts, tax, reporting, and the integration adapters for Codevertex services
Meshack Isava	Frontend engineer	Design system in AELIA's brand, brand and creator journeys, dashboards, admin console, installable PWA

9.3 RACI

Responsibility	AELIA	Titus	Steve	Meshack
Product priorities and backlog order	A	C	C	C
Critical technical decisions	A (jointly)	R, A (jointly)	C	C
Backend services and integrations	I	A	R	I
Web app and user experience	C	A	C	R
Infrastructure, security, releases	I	A, R	C	C
Payment, commission and tax logic	C	A, R	R	I
Pilot recruitment and onboarding	A, R	C	I	I
User acceptance testing	A, R	R	C	C

R does the work · A is accountable and signs off · C is consulted · I is kept informed.

10. Delivery Roadmap: MVP and Version 1

10.1 Phases and gates

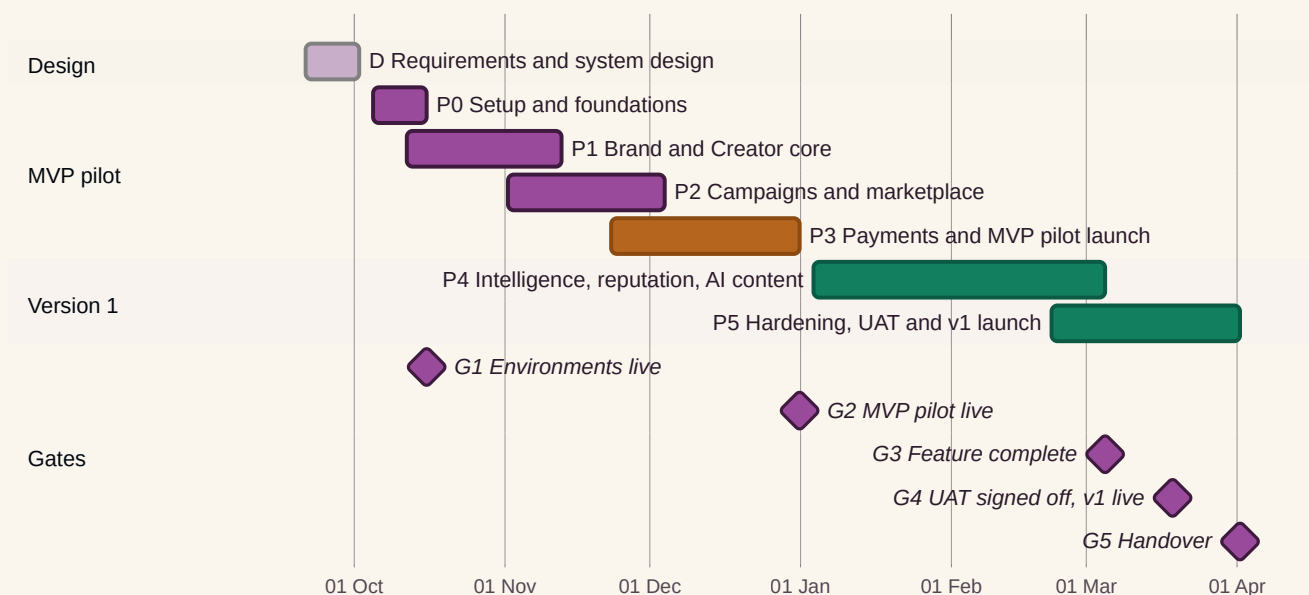


Figure 9. Delivery plan from 5 October 2026 (week 1). Payments (P3) start in week 8, so the pilot goes live with real transactions at week 13.

Phase	Weeks	Deliverables	Exit criteria and gate
D · Requirements and system design	Before 5 Oct	This SRDD; screen flows; a ready build backlog	SRDD signed off by AELIA, so developers start with nothing left to guess
P0 · Setup and foundations	1–2	Dev, staging and production environments on AELIA-controlled accounts; CI/CD; monitoring; AELIA design system; SSO working; Codevertex default integration profile seeded	G1: every change deploys to staging automatically; a test brand and a test creator can sign in
P1 · Brand and Creator core	2–6	Brand onboarding and verification; creator onboarding; Creator Passport v1; Brand Passport v1; admin verification queue; user management	A brand and a creator register, are verified by an admin and see their Passports
P2 · Campaigns and marketplace	5–9	Brief builder with content rights; discovery and curated shortlist; workspace and messaging; submission, revision, approval; notifications at every step	A full campaign runs in staging from brief to approved content, with notifications at each step
P3 · Payments and MVP pilot launch	8–13	M-Pesa and card funding; commission ledger; payouts with starting fee and balance; bank transfer with proof; configurable tax engine; dashboards; production go-live	G2: a real campaign is funded, delivered, approved and paid out in production, and the ledger reconciles to the shilling
P4 · Intelligence, reputation, AI content	14–22	Rules-based matching; Passport v2 with social connections; brand reporting; WHT certificates; reviews and rebooking; fee settings; AI content lane with automatic cost per clip; alternative-provider registration and outbound webhooks; pilot improvements	G3: every brief gets a ranked shortlist; every AI clip is costed automatically; pilot feedback triaged every sprint
P5 · Hardening, UAT, v1 launch	21–26	Admin console for disputes, moderation and audit; security and data-protection review; load test; backup and restore drill; UAT; documentation and training; launch and two weeks of close support	G4: UAT signed off and v1 live (week 24). G5: handover complete; zero open high or critical findings; restore proven within 4 hours (week 26)

10.2 Work breakdown (planned team days)

Phase	Work package	Titus	Steve	Meshack	Days
D	Requirements refinement; system design; screen flows and backlog	10	–	–	10
P0	Environments, deployment, AELIA-owned accounts, secrets, monitoring	7	1	–	8
P0	Design system, app shell, landing and sign-up pages, installable app base	–	–	6	6
P0	Single sign-on, roles and permissions	2	3	2	7
P1	Brand onboarding, organisation profile, verification workflow	1	5	4	10
P1	Creator onboarding and Creator Passport v1	1	5	5	11
P1	Brand Passport v1	–	2	2	4
P1	Admin foundations: verification queue, users, configuration	1	3	4	8
P2	Brief builder with content rights	1	3	4	8
P2	Discovery, search and curated shortlist	–	2	3	5
P2	Workspace and messaging v1	–	3	4	7
P2	Content submission, revisions, approval	–	3	4	7

Phase	Work package	Titus	Steve	Meshack	Days
P2	Notifications at each campaign step	–	1	–	1
P3	M-Pesa and card payments, campaign funding	6	3	3	12
P3	Commission ledger and reconciliation	6	3	–	9
P3	Creator payouts, starting-fee and balance schedules	1	3	2	6
P3	Bank transfer with proof upload and finance verification	–	3	3	6
P3	Configurable tax engine	4	1	1	6
P3	Transaction statuses and basic dashboards	–	1	2	3
P3	Pilot deployment, hardening, go-live support	4	2	2	8
P4	Rules-based matching v1 and refinement	7	6	2	15
P4	Passport v2: track record and platform history	–	3	4	7
P4	Reporting, performance data, campaign insights	–	4	7	11
P4	Monthly withholding certificates	1	1	1	3
P4	Messaging, discovery and brief enhancements from pilot	–	3	3	6
P4	Two-sided ratings and reviews	–	3	4	7
P4	Rebooking flow	–	3	2	5
P4	Revenue tracking and configurable fee settings	–	3	2	5
P4	AI content lane: AI video tools, automatic cost per clip	7	3	3	13
P4	Social account connections for Passport metrics	4	2	2	8
P4	Pilot support and iteration	3	3	3	9
P5	Admin console: disputes, moderation, audit log	–	5	6	11
P5	Security and data-protection review, fixes, access review	6	3	–	9
P5	Load testing, backup and restore drills	4	2	–	6
P5	UAT with pilot brands and creators, fixes	3	4	5	12
P5	Documentation, runbooks, admin guide, handover training	4	3	2	9
P5	Launch and two weeks of close post-launch support	3	3	3	9
Total	All work packages	86	101	100	287

Integrations Hub scope within the plan. The Codevertex default profile, adapters, inbound webhooks and credential vault are built inside P0 (SSO), P2 (notifications) and P3 (payments), as part of those packages. Alternative-provider registration, the declarative REST connector and outbound webhooks (INT-03, INT-05) are delivered in P4 under “pilot support and iteration” and “revenue tracking and fee settings”, with the conformance suite in P5.

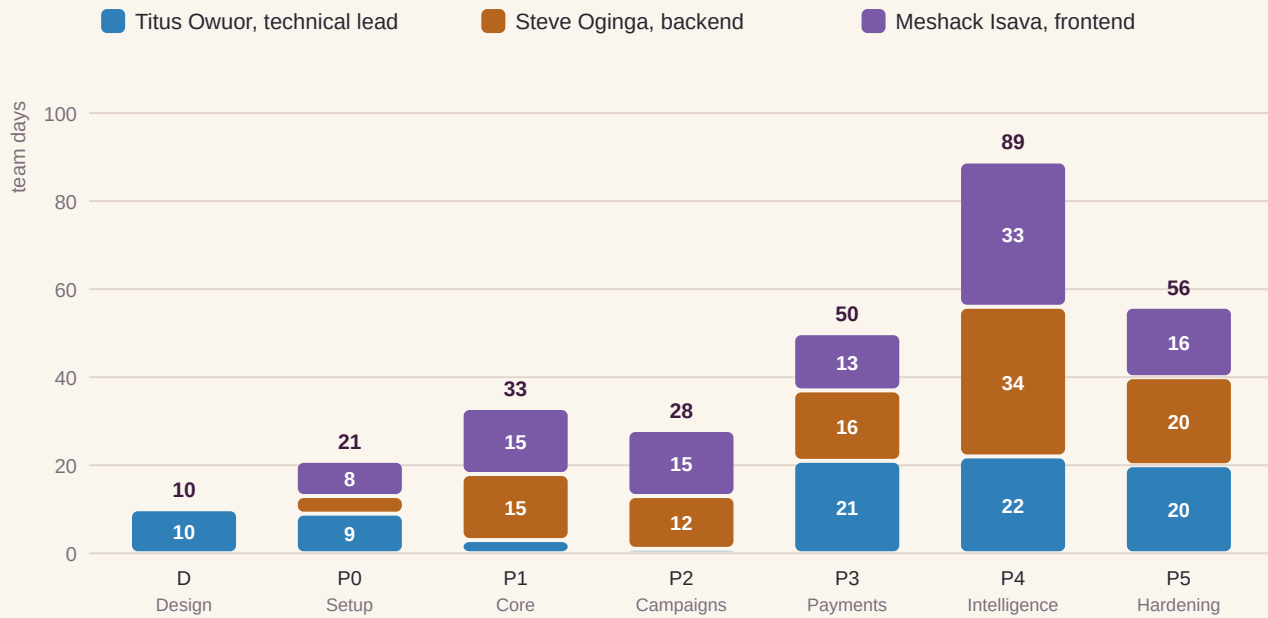


Figure 10. Planned team days per phase. No one is planned above 90% in any window: whole-project loading is Titus 70.5%, Steve 82.8% and Meshack 82%, leaving room to absorb the unexpected without moving a gate.

Part C · Technical Architecture Specification

How the platform is built: new AELIA services with their own data, running on and integrated with the Codevertex ecosystem through configurable ports.

11. Architecture Principles

#	Principle	What it means for AELIA
AP-1	AELIA owns its application and data	A dedicated <code>aelia</code> PostgreSQL database, not shared with any other product; AELIA-controlled domain, repository access and backups.
AP-2	New domain, shared plumbing	The brand-to-creator marketplace is new code. Identity, notifications, entitlements, payments, payouts and AI runtime are consumed from Codevertex services instead of being rebuilt.
AP-3	Reference plus cached display, never duplicate	Cross-service data is stored as a nullable UUID reference plus a small display snapshot (for example <code>sso_user_id</code> with <code>display_name</code> and <code>email</code> ; <code>payment_intent_id</code> with <code>amount</code> and <code>status</code>). Live details are fetched when needed.
AP-4	Ports and adapters for every external dependency	Domain code depends on interfaces (<code>IdentityProvider</code> , <code>NotificationProvider</code> , <code>BillingProvider</code> , <code>PaymentProvider</code> , <code>PayoutProvider</code> , and others). Codevertex services are the default adapters, and any adapter can be replaced by configuration (Part D).
AP-5	Event-driven, reliably	Every state change writes a domain event to a transactional outbox, published to NATS JetStream under <code>aelia.*</code> using the <code>shared-events</code> envelope. Consumers are idempotent.
AP-6	Money moves only on licensed rails	AELIA records intents, splits, ledger entries and payouts. Funds are collected and disbursed by licensed providers through Treasury.
AP-7	Link, don't rebuild	Where a Codevertex UI already does the job (for example Treasury's GL and eTIMS views for Finance), AELIA links to it instead of recreating it.
AP-8	Stateless, horizontally scalable services	Session state lives in tokens and Redis; media in object storage; HPA-ready deployments with at least 2 replicas.
AP-9	Secure by default	Every mutating route passes rate limit → authentication → subscription gate → JIT provisioning → context → permission. Secrets are encrypted and never returned.
AP-10	Configuration over code	Tax, fees, plans, templates, match weights and provider bindings are effective-dated, audited runtime settings.

12. System Context & Container Architecture

12.1 System context

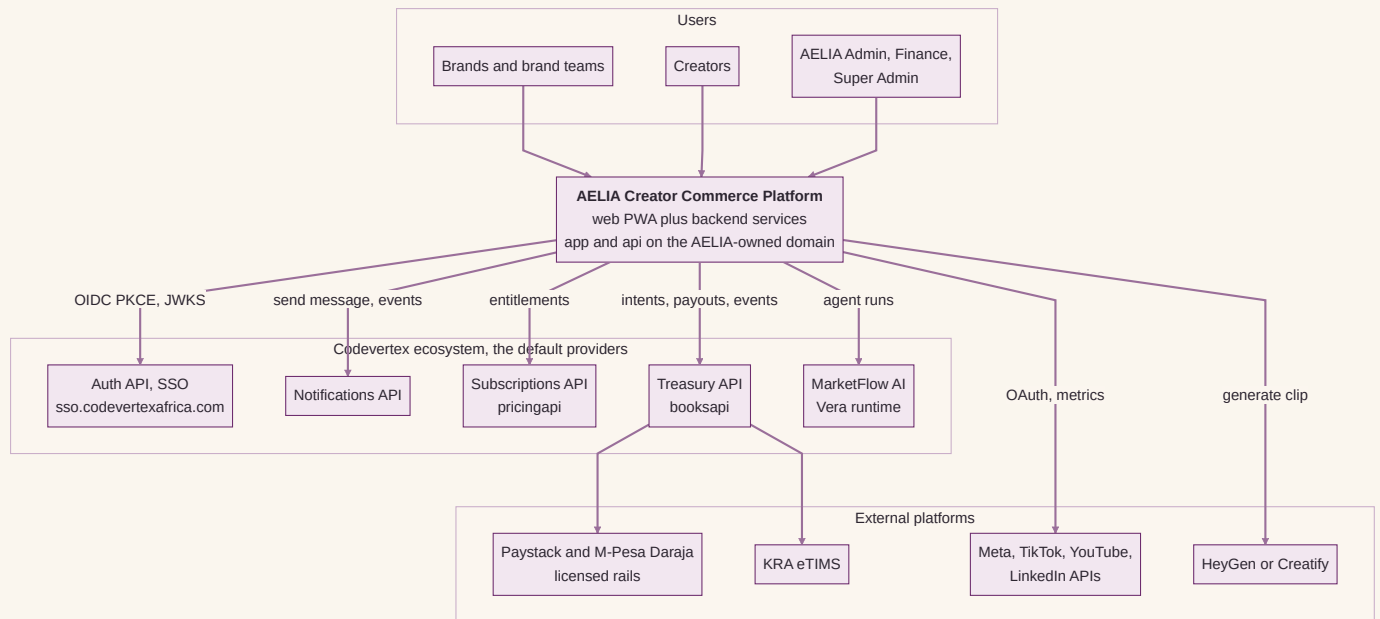


Figure 11. System context. Every arrow from AELIA crosses a configurable port (Part D).

12.2 Containers

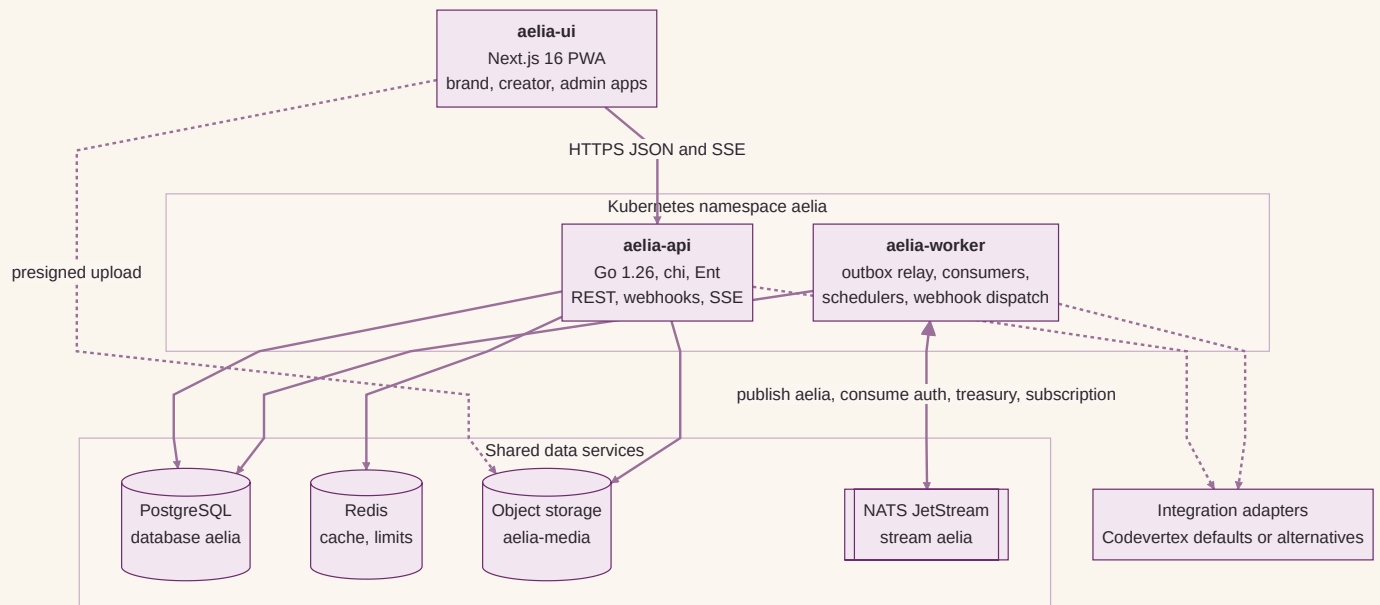


Figure 12. Container view. The API and worker share one codebase and the same integration adapters, deployed as two workloads.

Container	Technology	Responsibilities	Scaling
aelia-ui	Next.js 16 (App Router), React 19, Tailwind v4, <code>next-pwa</code> , TanStack Query, <code>@bengo-hub/shared-ui-lib</code>	Brand, creator and admin experiences; SSO login; payment modal or redirect; PWA install and offline shell; web push	2+ replicas, stateless

Container	Technology	Responsibilities	Scaling
aelia-api	Go 1.26, chi v5, Ent, Atlas, <code>httpware</code> , <code>shared-auth-client</code> , <code>shared-service-client</code> , <code>shared-ratelimit</code> , <code>pagination</code> , <code>cache</code>	REST API; auth middleware chain; domain services; inbound webhooks; SSE for messaging and notifications; presigned uploads	HPA on CPU and latency; 2+ replicas
aelia-worker	Go (same module), <code>shared-events</code>	Outbox relay to NATS; consumers of <code>auth.*</code> , <code>treasury.*</code> , <code>subscription.*</code> and provider webhooks; schedulers (reminders, metrics refresh, certificates, reconciliation); outbound webhook delivery	Queue-group consumers; 2+ replicas
PostgreSQL	PostgreSQL 16 through PgBouncer (<code>pgbouncer.infra.svc.cluster.local:6432</code>); migrations use a direct <code>POSTGRES_MIGRATE_URL</code>	System of record for all AELIA data	Cluster instance; daily backups
Redis	Shared Redis, key prefix <code>aelia:</code>	Cache-aside (tenant details, entitlements, integration config), rate limits, idempotency keys, SSE fan-out	Shared
Object storage	S3-compatible (default MinIO; configurable)	Verification documents (private), portfolio and campaign media, agreement PDFs, certificates	Bucket per environment
NATS JetStream	Shared cluster, <code>EVENTS_NATS_URL</code>	Stream <code>aelia</code> (subjects <code>aelia.></code>); durable consumers on other streams	Shared

12.3 Technology stack

Layer	Choice	Notes
Backend language	Go 1.26	Codevertex standard; direct use of the shared toolkit
HTTP	chi v5, <code>httpware</code> (request ID, structured zap logs, recovery, CORS)	Routes under <code>/api/v1/{tenant}/...</code>
Data access	Ent v0.14 and Atlas versioned migrations	<code>migrate</code> runs in the container entrypoint before <code>serve</code>
Database	PostgreSQL 16	<code>numeric(18,2)</code> for money plus an ISO-4217 currency column on every amount
Cache	Redis 7 via the shared <code>cache</code> package (<code>Aside[T]</code> , TTL tiers 5 min / 1 min / 30 s)	
Messaging	NATS JetStream via <code>shared-events</code> (envelope, outbox, idempotency store)	
Frontend	Next.js 16, React 19, TypeScript, Tailwind v4, <code>next-pwa</code> , <code>pnpm</code>	AELIA design system over shared UI primitives
Auth client	<code>shared-auth-client</code> (JWT validation via JWKS, API-key validation, subscription gating helpers)	
S2S client	<code>shared-service-client</code> (circuit breaker, retries with backoff, 10 s default timeout, OTel spans)	

Layer	Choice	Notes
API docs	OpenAPI 3, generated; Swagger UI at <code>/v1/docs</code>	External tags filtered for public docs
Containers	<code>golang:1.26-alpine</code> build → <code>alpine:3.20</code> runtime, non-root 100:101	<code>/healthz</code> , <code>/readyz</code> , <code>/metrics</code>
Deployment	k3s, ArgoCD app-of-apps, generic Helm chart <code>charts/app</code> , cert-manager, Cloudflare	Part L

12.4 Tenancy model

AELIA runs as a **platform tenant** in Codevertex auth-api (slug `aelia`). Brands, creators and campaigns are **AELIA-domain entities** inside that tenant; they are not Codevertex tenants. Brand teams are managed in AELIA's own `organisation_members` .

- **Every table is tenant-scoped.** Every AELIA table carries `tenant_id` , and every query filters by it (the Codevertex rule).
- **Regional expansion** Later **is configuration only.** Each country operation (for example `aelia-ke` , `aelia-ug`) becomes its own tenant with its own currency, tax pack, merchant accounts and integration profile. There is no schema change.
- **Codevertex services see one tenant.** Downstream services see AELIA as a single tenant. Treasury holds AELIA's merchant configuration, and Notifications holds AELIA's sender identities.
- **Tenant is always explicit on S2S calls.** Outbound calls carry `X-Tenant-ID` (the UUID, never the slug), so API-key calls never silently resolve to the platform-owner tenant.

13. Internal Module Design

`aelia-api` and `aelia-worker` form a **modular monolith**: one Go module with strict bounded-context packages that communicate through service interfaces and domain events. There are no cross-module table joins, so a module can later be split into its own service if load requires it.

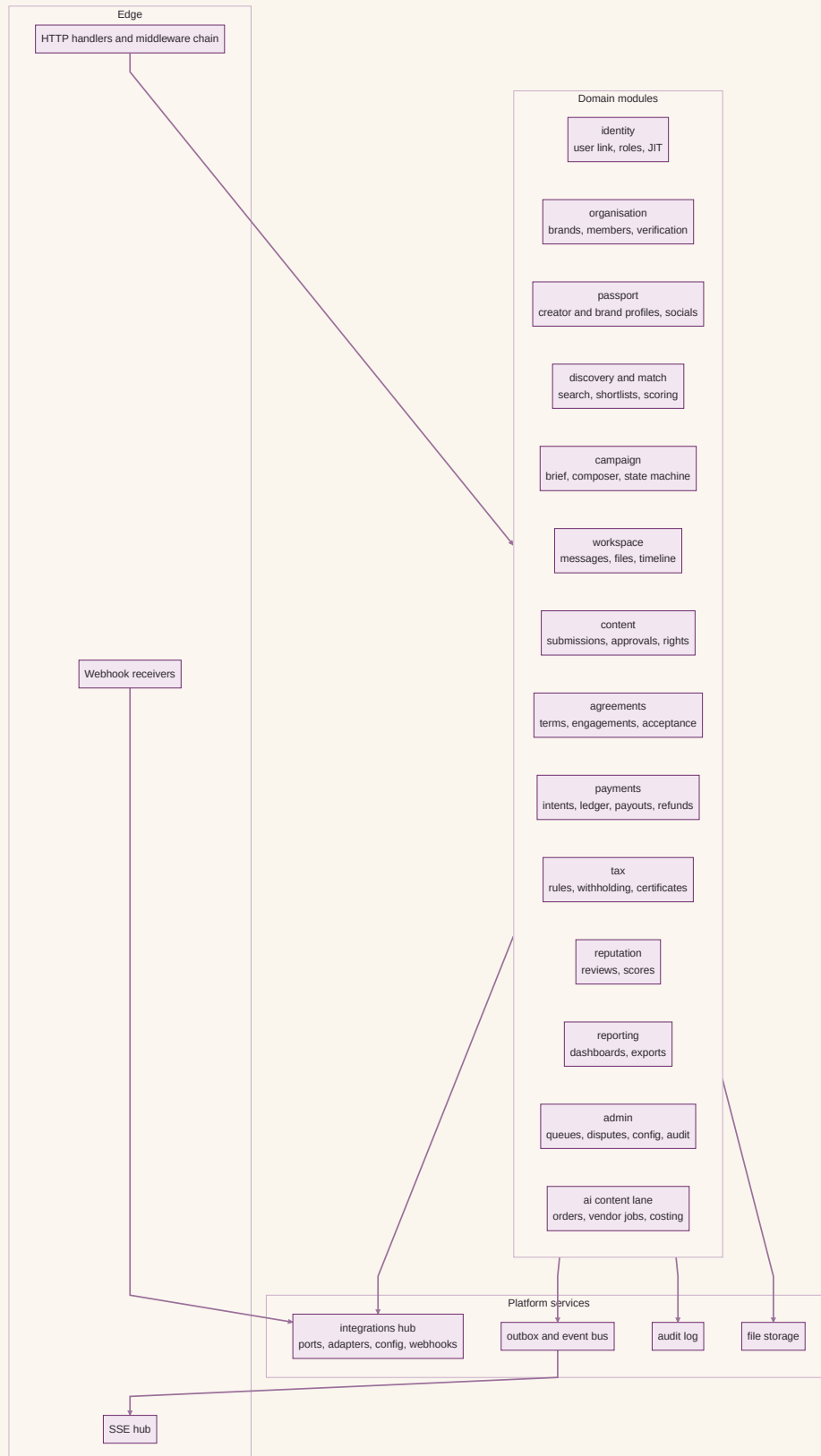


Figure 13. Bounded contexts inside the AELIA backend.

Module	Owns (tables)	Publishes	Depends on ports
identity	users , user_roles , identity_links	aelia.user.*	IdentityProvider
organisation	organisations , organisation_members , verification_cases , documents	aelia.organisatio n.* , aelia.verificatio n.*	ObjectStorage, KYCProvider Later

Module	Owns (tables)	Publishes	Depends on ports
passport	creator_profiles , creator_rates , creator_availability , portfolio_items , social_accounts , social_metrics , brand_profiles	aelia.passport.*	SocialPlatformConnector
discovery and match	shortlists , shortlist_items , match_runs , match_scores , match_weights	aelia.match.*	AIMatchProvider
campaign	campaigns , campaign_deliverables , campaign_participants , applications	aelia.campaign.*	BillingProvider (entitlements)
workspace	threads , messages , attachments , timeline_events	aelia.message.*	NotificationProvider
content	submissions , submission_versions , review_comments , approvals , publication_proofs , content_rights	aelia.content.*	ObjectStorage
agreements	agreement_templates , terms_acceptances , engagements	aelia.agreement.* *	ESignatureProvider Later
payments	funding_intents , ledger_accounts , ledger_entries , payouts , refunds , payment_proofs	aelia.funding.* , aelia.payout.* , aelia.refund.*	PaymentProvider, PayoutProvider
tax	tax_rules , tax_withholdings , tax_certificates	aelia.tax.*	InvoicingProvider (eTIMS)
reputation	reviews , reputation_scores	aelia.review.*	—
reporting	read models, metric_snapshots	—	—
admin	disputes , dispute_events , settings , feature_flags , moderation_items	aelia.dispute.* , aelia.settings.*	—
ai content lane	ai_orders , ai_jobs , ai_cost_lines	aelia.ai_order.*	AIVideoVendor
integrations hub	integration_providers , integration_bindings , integration_credentials , webhook_endpoints , webhook_subscriptions , webhook_deliveries , integration_health	aelia.integration .*	all ports

14. Codevertex Ecosystem Integration Map

This is the **default** wiring, seeded as the Codevertex integration profile. Each row can be replaced through Part D.

Codevertex service	Reuse mode	What AELIA uses	Interface (default adapter)
Auth API (SSO): sign-on, client portal, users	REUSE AS-IS	OIDC Authorization Code + PKCE login for all users; JWT (RS256, JWKS) validation; profile; MFA for staff; API-key validation for S2S	<code>GET /api/v1/authorize</code> , <code>POST /api/v1/token</code> , <code>POST /api/v1/auth/refresh</code> , <code>GET /api/v1/auth/me</code> , <code>/.well-known/openid-configuration</code> ; OIDC client <code>aelia-ui</code> seeded with <code>/{tenant}/auth/callback</code> and <code>/auth/callback</code> ; consumes <code>auth.user.created</code> and <code>auth.user.login</code> on plain NATS
Notifications API: email, SMS, WhatsApp, push	REUSE AS-IS	Transactional messages for every campaign and payment state change; templates namespaced <code>aelia/*</code> ; AELIA sender identities as tenant provider settings	<code>POST /{tenantId}/notifications/messages</code> with <code>channel</code> , <code>template</code> , <code>to</code> , <code>data</code> , <code>Idempotency-Key</code> ; event-driven path where Notifications subscribes to <code>aelia.></code> ; push via <code>/api/v1/push/web-config</code> and <code>/api/v1/push/tokens</code>
Subscriptions API: plans, entitlements, gating	REUSE, EXTEND	The AELIA tenant's own entitlement to the Treasury service only ; AELIA's brand plan catalogue as consumer-scoped plans and features (for example <code>aelia_max_active_campaigns</code> , <code>aelia_ai_lane_member_pricing</code>)	<code>GET /api/v1/tenants/{tenantId}/subscription</code> ; <code>GetEntitlements</code> / <code>ConsumerHasFeature</code> S2S; JWT <code>sub_*</code> claims; events <code>subscription.*</code> . Extension: consumer-scoped plans keyed <code>aelia:brand:{id}</code>
Treasury API: payments, payouts, GL, eTIMS	REUSE, EXTEND	Payment intents for funding; Paystack and M-Pesa collection on AELIA-owned merchant configuration; manual bank-transfer confirmation; payouts to creators; GL auto-posting; eTIMS for commission invoices	<code>POST /api/v1/{tenant}/payments/intents</code> (<code>source_service: "aelia"</code>) , <code>/intents/{id}/initiate</code> , <code>/confirm-manual</code> , <code>/check-status</code> ; pay page <code>books.codevertexafrica.com/pay</code> or <code>TreasuryPaymentModal</code> ; events <code>treasury.payment.succeeded</code> , <code>treasury.payment.failed</code> , <code>treasury.payout.completed</code> , <code>treasury.refund.completed</code> . Extensions: <code>creator</code> payee type, three-way split, S2S payout API with idempotency key (§33.3)
MarketFlow AI (Vera runtime)	REUSE AS-IS	Agent loop for the AI Brief Builder and for AELIA Match explanations and scoring, with AELIA tools exposed to the agent	Agent-run API with AELIA tool definitions; runs scoped to the AELIA tenant
Shared Go toolkit	REUSE AS-IS	<code>shared-auth-client</code> , <code>shared-service-client</code> , <code>shared-events</code> , <code>httpware</code> , <code>cache</code> , <code>shared-ratelimit</code> , <code>pagination</code>	Go modules
Shared UI library	REUSE AS-IS	<code>SSOLoginModal</code> , <code>TreasuryPaymentModal</code> (iframe + <code>postMessage</code>), layout primitives	<code>@bengo-hub/shared-ui-lib</code>

Codevertex service	Reuse mode	What AELIA uses	Interface (default adapter)
Platform infrastructure	REUSE AS-IS	k3s cluster, ArgoCD, Helm chart, cert-manager TLS, Cloudflare edge, PgBouncer, Redis, NATS, MinIO, backups, <code>fleet-health-watcher</code> alerting	<code>devops-k8s apps/aelia-api , apps/aelia-worker , apps/aelia-ui</code>
MarketFlow CRM social connect	REUSE PATTERN	The OAuth “connect your account” handshake used for Facebook Pages and TikTok. AELIA builds its own connectors with profile and insights scopes for Instagram, Facebook, TikTok, YouTube and LinkedIn, and stores tokens encrypted.	New <code>SocialPlatformConnector</code> adapters
HR service, wider ERP suite	NOT USED	AELIA consumes Treasury only	–

Treasury-only consumption. AELIA's Codevertex tenant is entitled to the Treasury service alone, not the wider ERP suite (HR, payroll, inventory, POS). Finance users reach Treasury screens (GL, eTIMS, gateway configuration) through SSO deep links from the AELIA admin console. AELIA's own finance views (ledger, payouts, withholding) live in `aelia-ui`.

14.1 Reference-plus-cache examples

AELIA record	Reference column	Cached snapshot	Live source
users	<code>sso_user_id</code> (UUID), <code>idp_issuer</code> , <code>idp_subject</code>	<code>display_name</code> , <code>email</code> , <code>avatar_url</code>	IdentityProvider <code>/me</code> or <code>userinfo</code>
funding_intents	<code>provider_intent_id</code> (Treasury <code>payment_intent_id</code>)	<code>amount</code> , <code>currency</code> , <code>status</code> , <code>provider_reference</code>	PaymentProvider <code>check-status</code> , payment events
payouts	<code>provider_payout_id</code>	<code>amount</code> , <code>status</code> , <code>rail</code> , <code>completed_at</code>	PayoutProvider events
organisations	<code>crm_contact_id</code> (nullable; only if AELIA chooses to enrich a MarketFlow record)	<code>name</code>	MarketFlow <code>GET /api/v1/contacts/{id}</code>
ai_jobs	<code>vendor_job_id</code>	<code>status</code> , <code>vendor_cost</code>	AIVideoVendor status

15. Event-Driven Design

15.1 Conventions (Codevertex event standards)

- Stream and subjects.** The stream is `aelia`, with subjects `aelia.{aggregate}.{event}` built by `shared-events` from `aggregate_type` and `event_type`. The event type is passed **without** the service prefix (`NewEvent("campaign.funded", "aelia", ...)`), which avoids dead subjects of the form `aelia.aelia.*`.
- Envelope** (`shared-events`): `id`, `tenant_id`, `aggregate_type`, `aggregate_id` (UUID), `event_type`, `payload`, `timestamp`, `version: "1.0"`.

- **Outbox.** Each domain write inserts an `outbox_events` row in the same transaction, holding the **full envelope**. The worker relays it to NATS and later prunes it.
- **Idempotency.** Consumers claim `(event_id, consumer)` in `processed_events` (`ON CONFLICT DO NOTHING`) **before** any irreversible side effect (payout, refund, notification). Payment events are keyed on the envelope `id` , never on the provider reference.
- **Consumers.** Subscriptions are durable queue groups (`QueueSubscribe`), so replicas share the load. `auth.*` subjects arrive on plain NATS rather than JetStream and use `nc.Subscribe` .
- **Entitlement check in consumers.** A consumer checks that the tenant is entitled before acting, and acks and drops the event otherwise.
- **Coverage.** `aelia-api` is added to `tools/event-subject-coverage.sh` , so orphan publishes and dead subscriptions are reported.

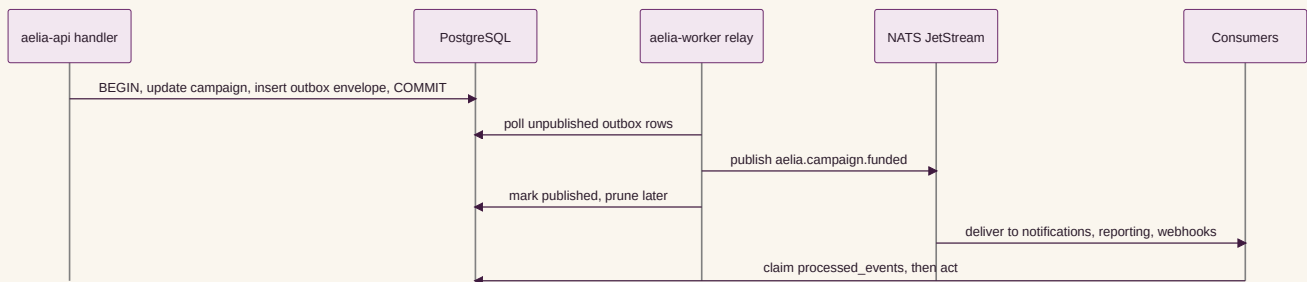


Figure 14. Transactional outbox and idempotent consumption.

Part D · Pluggable Integration Framework

How every external service AELIA depends on is configured from inside the AELIA platform. Codevertex services load by default, and any capability can be switched to another provider through a programming interface or an API interface, with webhooks and callback URLs, without changing AELIA's domain code.

16. Integration Framework Overview

16.1 Requirement

AELIA must be able to integrate with another SSO, notification, subscription or billing, or finance and treasury provider, or any other external service. It does so by applying a **programming interface** (a code-level adapter implementing an AELIA port) or an **API interface** (a declarative REST connector configured in the admin console). It then configures the new service's credentials, webhooks and callback URLs. The existing Codevertex integrations remain the **default configuration** and load automatically for any Codevertex service plugged into AELIA's workflows.

16.2 Design: ports, adapters, profiles

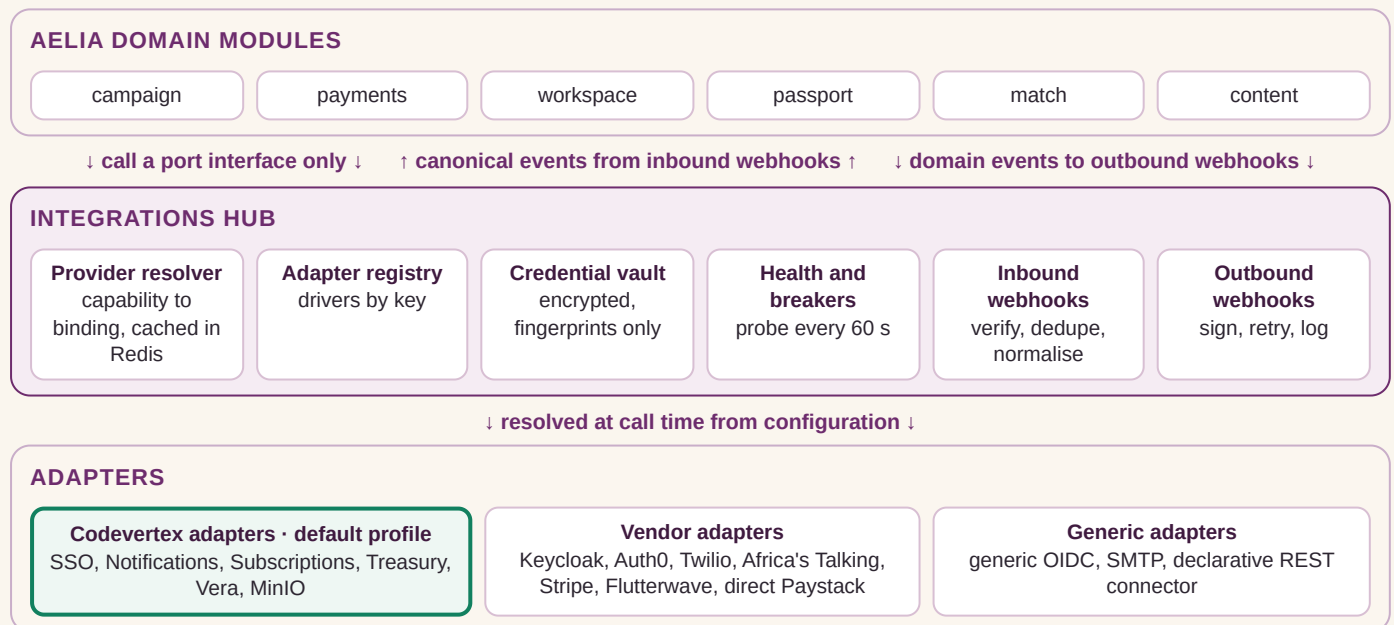


Figure 15. Domain code only ever calls a port. The resolver picks the bound adapter at call time from configuration.

Concept	Definition
Capability	A kind of external service AELIA needs, such as <code>identity</code> , <code>notifications.email</code> , <code>notifications.sms</code> , <code>notifications.whatsapp</code> , <code>notifications.push</code> , <code>billing</code> , <code>payments.collect</code> , <code>payments.payout</code> , <code>invoicing.tax</code> , <code>ai.runtime</code> , <code>ai.video</code> , <code>social.<platform></code> , <code>storage.objects</code> , <code>kyc</code> Later , <code>esign</code> Later , <code>moderation</code> Later
Port	A Go interface in <code>internal/integrations/ports</code> that domain code depends on, one per capability
Driver	An adapter implementation registered under a key, such as <code>codevertex.sso</code> , <code>generic.oidc</code> , <code>vendor.twilio</code> , <code>generic.rest</code>

Concept	Definition
Provider	A configured instance of a driver: its endpoints, options, credentials and webhooks. Several providers can use the same driver, for example two SMTP accounts
Binding	Capability plus scope mapped to a primary provider and an optional fallback provider, with priority and effective date
Profile	A named set of bindings. <code>codevertex-default</code> is seeded at install; others can be cloned from it
Scope	<code>platform</code> (the AELIA tenant). <code>organisation</code> Later lets an agency or brand bring its own provider

16.3 Two ways to add a provider

Path	Who uses it	When	How
Programming interface (adapter SDK)	Engineers	The provider needs custom logic: an SDK, a complex auth flow, or multi-step orchestration	Implement the port interface in Go, register the driver, pass the conformance suite (§34.4), then ship it in a release. After that it is configurable from the admin console like any other driver
API interface (declarative REST connector)	Super Admin, with engineering support if needed	The provider exposes a conventional REST/JSON API	Configure in the Integrations Hub: base URL, auth scheme, operation mappings (request and response templates), status mapping, webhook verification and event mapping. Test the connection and the conformance checks, then activate. No deployment needed

17. Provider Ports & Adapters

17.1 Port catalogue

Capability	Port	Key operations	Default adapter (Codevertex profile)	Alternative adapters
Identity	<code>IdentityProvider</code>	<code>AuthorizeURL</code> , <code>ExchangeCode</code> , <code>Refresh</code> , <code>VerifyToken</code> (JWKS), <code>UserInfo</code> , <code>Logout</code> , <code>ValidateAPIKey</code>	<code>codevertex.sso</code> (auth-api OIDC + PKCE)	<code>generic.oidc</code> (Keycloak, Auth0, Azure AD / Entra ID, Google, Okta, Cognito); <code>vendor.firebase</code> Later
Notifications	<code>NotificationProvider</code>	<code>Send(message)</code> , <code>SendBatch</code> , <code>Status(id)</code> , <code>RegisterPushToken</code>	<code>codevertex.notifications</code> (template keys <code>aelia/*</code>)	<code>generic.smtp</code> , <code>vendor.africastalking</code> , <code>vendor.twilio</code> , <code>vendor.meta_whatsapp</code> , <code>vendor.fcm</code> , <code>generic.rest</code> , <code>generic.webhook</code>
Billing / entitlements	<code>BillingProvider</code>	<code>GetEntitlements(subject)</code> , <code>HasFeature</code> , <code>GetLimit</code> , <code>CreateSubscription</code> , <code>ChangePlan</code> , <code>Cancel</code>	<code>codevertex.subscriptions</code>	<code>builtin.plans</code> (plans in AELIA's own tables), <code>vendor.stripe_billing</code> , <code>generic.rest</code>

Capability	Port	Key operations	Default adapter (Codevertex profile)	Alternative adapters
Payment collection	PaymentProvider	CreateIntent , Initiate(method) , ConfirmManual , CheckStatus , Refund	codevertex.treasury (Paystack and M-Pesa via Treasury)	vendor.paystack (direct), vendor.flutterwave , vendor.mpesa_daraja (direct), vendor.pesapal , generic.rest
Payouts	PayoutProvider	CreateRecipient , Payout(idempotencyKey) , BulkPayout , Status , Balance	codevertex.treasury (Paystack Transfers and M-Pesa B2C via Treasury)	vendor.paystack_transfer s , vendor.mpesa_b2c , vendor.flutterwave , generic.rest
Tax invoicing	Invoicing Provider	IssueInvoice , Status , CreditNote	codevertex.treasury_etims	vendor.kra_etims (direct OSCU/VSCU), generic.rest
AI runtime	AIMatchProvider , AIAssistant	ScoreCandidates , Explain , DraftBrief	codevertex.vera	builtin.rules (no AI), vendor.anthropic , vendor.openai , generic.rest
AI video	AIVideoVendor	CreateJob(script or url) , Status , FetchAsset , CostQuote	– (v1: vendor.heygen or vendor.creatify)	vendor.arcads , vendor.runway Later , generic.rest
Social platforms	SocialPlatformConnector	AuthorizeURL , ExchangeCode , Profile , AudienceInsights , MediaInsights(url) , Revoke	builtin.meta (Instagram and Facebook Graph), builtin.tiktok , builtin.youtube , builtin.linkedin	generic.rest for new platforms
Object storage	ObjectStorage	PresignPut , PresignGet , Delete , Head	codevertex.minio	vendor.s3 , vendor.gcs , vendor.azure_blob , vendor.r2
KYC/KYB, e-signature, moderation Later	KYCProvider , ESignatureProvider , ModerationProvider	Per port	–	Vendor adapters or generic.rest

17.2 Programming interface (Go)

```
// internal/integrations/ports/payments.go
type PaymentProvider interface {
    Capabilities() PaymentCaps           // methods, currencies, split support,
    refunds
    CreateIntent(ctx context.Context, in CreateIntentInput) (*Intent, error)
    Initiate(ctx context.Context, intentID string, m Method, payer Payer) (*Initiation, error)
    ConfirmManual(ctx context.Context, intentID string, proof ManualProof) (*Intent, error)
    CheckStatus(ctx context.Context, intentID string) (*Intent, error)
    Refund(ctx context.Context, in RefundInput) (*Refund, error)
    // Translate a verified provider callback into canonical AELIA events.
    ParseWebhook(ctx context.Context, req VerifiedWebhook) ([]CanonicalEvent, error)
    Health(ctx context.Context) HealthStatus
}

// Every driver registers itself once; configuration decides which one is used.
func init() {
    registry.Register("codevertex.treasury", registry.Driver{
        Capabilities: []string{"payments.collect", "payments.payout", "invoicing.tax"},
        ConfigSchema: treasuryConfigSchema,           // JSON Schema rendered as the admin form
        New:          newTreasuryAdapter,             // func(cfg ProviderConfig, secrets
        SecretReader) (any, error)
    })
}

// Domain code never names a vendor:
pay := hub.Payments(ctx, tenantID)           // resolves binding, applies breaker and
fallback
intent, err := pay.CreateIntent(ctx, ports.CreateIntentInput{
    ReferenceType: "aelia_campaign_funding", ReferenceID: campaign.ID,
    Amount: money.KES(250000), IdempotencyKey: "fund:" + campaign.ID.String(),
})
```

- **Canonical types.** Ports speak AELIA's canonical types (money as integer minor units plus currency, canonical statuses `pending`, `processing`, `succeeded`, `failed`, `reversed`). Adapters own all translation.
- **Error taxonomy.** Adapters return typed errors: `ErrRetryable`, `ErrRejected`, `ErrAuth`, `ErrNotSupported`. These drive retry, fallback and admin alerts.
- **Capability flags.** Each adapter declares what it supports (for example `SupportsSplit`, `SupportsBulkPayout`, `SupportsWhatsApp`), and the UI hides options the bound provider cannot do.
- **Frontend counterparts.** `aelia-ui` has matching client adapters:
 - **Login:** `SSOLoginModal` for `codevertex.sso`, or a standard OIDC redirect for `generic.oidc`.
 - **Payments:** `TreasuryPaymentModal` for `codevertex.treasury`, or a hosted-checkout redirect, or an embedded widget, as the provider's descriptor declares.
 - **Selection:** the UI reads which adapter to use from `GET /api/v1/{tenant}/integrations/client-config`, which exposes public settings only.

17.3 API interface: declarative REST connector (`generic.rest`)

A provider with a conventional REST API is integrated by configuration. The connector definition is stored as versioned JSON (edited as a form or as YAML in the admin console) and validated against the port's operation contract.

```

driver: generic.rest
capability: payments.collect
name: Flutterwave (via REST connector)
base_url: https://api.flutterwave.com/v3
auth: { scheme: bearer, secret_ref: FLW_SECRET_KEY }
timeouts: { connect_ms: 3000, request_ms: 10000 }
retry: { max_attempts: 3, backoff: exponential, retry_on: [429, 502, 503, 504] }
operations:
  CreateIntent:
    method: POST
    path: /payments
    headers: { Idempotency-Key: "{{ .IdempotencyKey }}" }
    body: |
      { "tx_ref": "{{ .ReferenceID }}", "amount": "{{ .Amount.Major }}",
        "currency": "{{ .Amount.Currency }}", "redirect_url": "{{ .CallbackURL }}",
        "customer": { "email": "{{ .Payer.Email }}", "phonenumber": "{{ .Payer.Phone }}" } }
    response:
      provider_intent_id: $.data.id
      checkout_url: $.data.link
  CheckStatus:
    method: GET
    path: /transactions/verify_by_reference?tx_ref={{ .ReferenceID }}
    response:
      status: { path: $.data.status, map: { successful: succeeded, failed: failed, pending:
pending } }
      provider_reference: $.data.flw_ref
webhook:
  verification: { scheme: shared_secret_header, header: verif-hash, secret_ref: FLW_WEBHOOK_HASH }
  event_type_path: $.event
  events:
    charge.completed:
      canonical: payment.succeeded          # when $.data.status == successful
      reference_path: $.data.tx_ref

```

Connector capabilities

- **Auth schemes:** `api_key_header`, `bearer`, `basic`, `oauth2_client_credentials` (token cached until expiry), `hmac_request_signing` (configurable canonical string and algorithm), `mtls` Later.
- **Templating:** Go `text/template` over canonical input fields. There is no arbitrary code execution: functions are limited to formatting helpers.
- **Response mapping:** JSONPath extraction and value maps into canonical fields. Unmapped required fields fail the conformance check.
- **Safety:** outbound hosts must be HTTPS and on the provider's allowlist, which blocks SSRF to cluster-internal addresses. Secrets are referenced, never inlined. Every request and response is logged with secrets redacted.
- **Versioning:** each saved connector definition is a new immutable version. Bindings point to a specific version, and rollback is one click.

18. Configuration Model

18.1 Configuration data

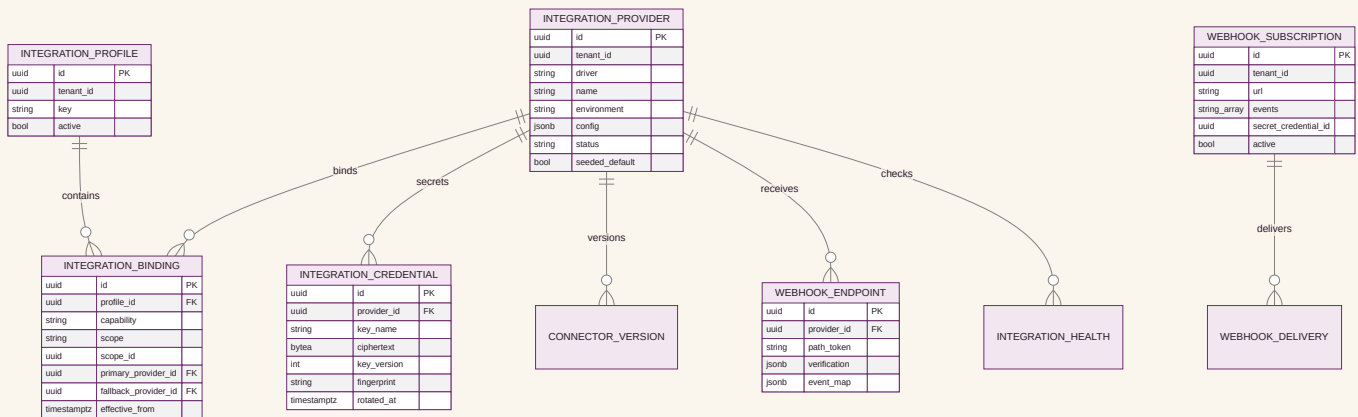


Figure 16. Integration configuration entities (stored in the AELIA database).

18.2 Resolution and precedence

Configuration resolves in layers. Each later layer overrides the one before it for the same key:

- Code defaults.** Each driver's schema defaults.
- Environment bootstrap.** Kubernetes secrets and config: Codevertex base URLs (`SSO_ISSUER_URL` , `NOTIFICATIONS_API_URL` , `SUBSCRIPTIONS_API_URL` , `TREASURY_API_URL` , `VERA_API_URL`), `INTERNAL_SERVICE_KEY` , the `EVENTS_NATS_URL` and the vault master key. These are used by the seeder and as emergency fallback.
- Seeded Codevertex default profile.** Created by the `aelia-api seed integrations` job at install and upgrade. It is idempotent and never overwrites admin edits.
- Platform bindings edited by the Super Admin** in the Integrations Hub.
- Organisation-scoped bindings** Later , for example an agency's own payment account.

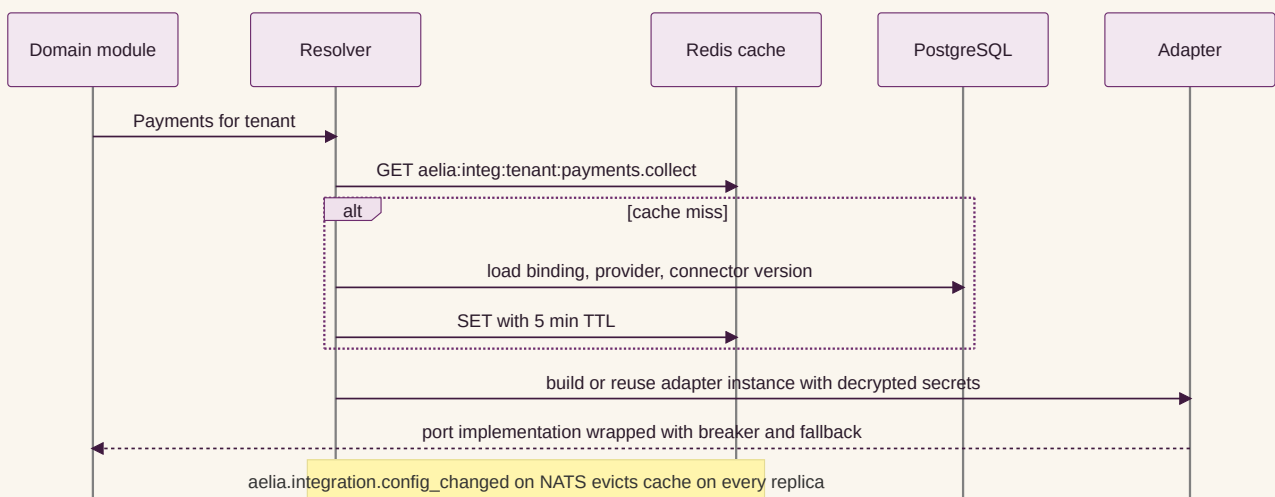


Figure 17. Provider resolution at call time. Configuration changes take effect within seconds, with no redeploy.

18.3 Runtime behaviour

Concern	Behaviour
Health checks	Each active provider is probed every 60 s (driver-defined: OIDC discovery, <code>GET /healthz</code> , a balance call). Results are stored in <code>integration_health</code> and shown in the hub.
Circuit breaker	Per provider, via <code>shared-service-client</code> (gobreaker): trips after 5 consecutive failures, half-open probe after 30 s.
Fallback	If the primary is open-circuited or returns <code>ErrRetryable</code> , the binding's fallback is used, only for capabilities marked fallback-safe (notifications, AI explanations). Money operations never fail over automatically. They queue and alert Finance.
Sandbox vs production	Every provider carries an <code>environment</code> . Staging refuses to bind production providers, and production refuses sandbox ones.
Change safety	A new or changed provider starts as <code>draft</code> . Activation requires a passing "Test connection" and conformance checks, plus a second Super Admin's approval for payment, payout and identity capabilities.
Audit	Every create, update, activate and rotate is written to the audit log with a before/after diff. Secrets appear only as fingerprints.
Credential vault	AES-256-GCM envelope encryption with a rotatable key ring (<code>key_version</code>), following the Codevertex secrets-management standard. Plaintext exists only in process memory. APIs return fingerprints only.

18.4 Integrations Hub (admin UI)

Screen	Contents
Overview	A card per capability showing bound provider, environment, health, last error, and last event received
Provider catalogue	Available drivers with capability badges; "Add provider"
Provider detail	Form generated from the driver's JSON Schema; credential fields (write-only); callback URLs to copy; webhook secret generation; "Test connection"; conformance results; version history
REST connector editor	Operation mapping (form or YAML), a sample request preview, response mapping tester with pasted JSON
Bindings	Capability-to-provider table with primary, fallback, effective date; the approval workflow
Webhooks	Inbound endpoints with verification settings and recent requests; outbound subscriptions with a delivery log and replay
Client configuration	Public settings exposed to <code>aelia-ui</code> (login mode, payment UI mode)

19. Webhooks & Callback URLs

19.1 Inbound webhooks (providers → AELIA)

Each provider gets a dedicated, unguessable endpoint:

POST `https://api.<aelia-domain>/api/v1/integrations/hooks/{providerKey}/{pathToken}`

Step	Behaviour
1. Receive	Raw body retained (needed for signature checks); max 1 MB; 5 s processing budget, then async

Step	Behaviour
2. Verify	Per-provider scheme: <code>hmac_sha512</code> (for example Paystack <code>x-paystack-signature</code>), <code>hmac_sha256</code> with timestamp (<code>t=...</code> , <code>v1=...</code> style), <code>shared_secret_header</code> , <code>jwt</code> (JWKS), IP allowlist (for example M-Pesa callbacks), or <code>none</code> in sandbox only. Constant-time compare. Timestamp tolerance 300 s
3. Dedupe	Key = provider event ID, or a SHA-256 of the body, stored in <code>processed_webhooks</code> for 7 days. Duplicates return 200 without reprocessing
4. Normalise	Adapter <code>ParseWebhook</code> , or the connector <code>event_map</code> , produces canonical events (<code>payment.succeeded</code> , <code>payout.completed</code> , <code>message.delivered</code> , <code>subscription.changed</code> , and so on)
5. Record and ack	Raw payload stored encrypted with its verification result; canonical events written to the outbox; <code>200 OK</code> returned
6. Reconcile	For money events, the worker re-queries <code>CheckStatus</code> before posting to the ledger (trust but verify)

The Codevertex default profile receives most callbacks as **NATS events** (`treasury.*`, `subscription.*`, `auth.*`) because AELIA runs in the same cluster. These are normalised through the same canonical handler, so behaviour is identical whether the provider is Codevertex (events) or external (webhooks).

19.2 Callback and redirect URLs (configured per provider)

URL	Purpose	Default value
OIDC redirect URIs	Return from the IdP after login	<code>https://app.<aelia-domain>/aelia/auth/callback</code> , <code>https://app.<aelia-domain>/auth/callback</code>
Post-logout redirect	Return after IdP logout	<code>https://app.<aelia-domain>/</code>
Payment return URL	Browser return after hosted checkout	<code>https://app.<aelia-domain>/aelia/payments/return?intent={id}</code>
Payment webhook	Server-to-server payment result	Treasury via NATS (default), or <code>.../integrations/hooks/{providerKey}/{token}</code>
Payout result callback	Transfer and B2C results	Treasury via NATS (default), or a provider hook URL
Social OAuth redirect	Creator account connection	<code>https://app.<aelia-domain>/aelia/passport/social/callback/{platform}</code>
Notification delivery receipts	Delivery, read and failure status	Provider hook URL (for non-Codevertex providers)
AI vendor job callback	Generation complete	Provider hook URL

All URLs are displayed ready to copy in the provider detail screen and are derived from the configured public base URL. Changing the domain re-derives them.

19.3 Outbound webhooks (AELIA → third parties)

AELIA can notify external systems (a brand's CRM, AELIA's own BI, a partner agency) about platform events.

```

POST {subscriber_url}
Content-Type: application/json
X-Aelia-Event: campaign.completed
X-Aelia-Delivery: 6b1f... (unique per attempt group)
X-Aelia-Timestamp: 1791234567
X-Aelia-Signature: v1=hex(HMAC_SHA256(secret, timestamp + "." + raw_body))

{ "id": "evt_...", "type": "campaign.completed", "tenant_id": "...", "created_at": "...",
  "data": { "campaign_id": "...", "brand_id": "...", "status": "completed" } }

```

Property	Behaviour
Subscription	URL (HTTPS only), event list (with wildcards such as <code>campaign.*</code>), secret (generated, shown once), active flag, API version
Delivery	At-least-once, via the worker. 10 s timeout. Any 2xx is success
Retries	Exponential backoff (30 s, 2 min, 10 min, 1 h, then every 6 h) for up to 24 h. The subscription is auto-disabled after 3 days of continuous failure, and admins are notified
Replay	Manual replay of any delivery from the log; bulk replay by time range
Security	HMAC-SHA256 signature with timestamp (receivers should reject anything older than 5 min); secret rotation with a 24 h dual-signature window; no internal IDs or PII beyond what the event contract defines

20. Default Codevertex Profile & Provider Swap Scenarios

20.1 Seeded `codevertex-default` profile

Capability	Provider (driver)	Seeded configuration	Secrets (from Kubernetes)
identity	Codevertex SSO (<code>codevertex.sso</code>)	<code>issuer</code> : <code>https://sso.codevertexafrica.com</code> , <code>client_id</code> : <code>aelia-ui</code> , PKCE S256, scopes <code>openid</code> <code>profile</code> <code>email</code> <code>offline_access</code> , tenant slug <code>aelia</code>	<code>INTERNAL_SERVICE</code> <code>E_KEY</code> (API-key validation)
notifications.*	Codevertex Notifications (<code>codevertex.notifications</code>)	<code>base_url</code> from <code>NOTIFICATIONS_API_URL</code> , template namespace <code>aelia/</code> , channels email, SMS, WhatsApp (when AELIA's WhatsApp plan is active), push	<code>NOTIFICATIONS_A</code> <code>PI_KEY</code>
billing	Codevertex Subscriptions (<code>codevertex.subscriptions</code>)	<code>base_url</code> from <code>SUBSCRIPTIONS_API_URL</code> , consumer prefix <code>aelia:brand:</code> , feature codes <code>aelia_*</code>	<code>INTERNAL_SERVICE</code> <code>E_KEY</code>
payments.collect	Codevertex Treasury (<code>codevertex.treasury</code>)	<code>base_url</code> from <code>TREASURY_API_URL</code> , <code>source_service</code> : <code>aelia</code> , methods <code>mpesa</code> , <code>paystack</code> , <code>manual</code> , pay-page base <code>https://books.codevertexafrica.com/pay</code> , events via NATS <code>treasury.></code>	<code>INTERNAL_SERVICE</code> <code>E_KEY</code>
payments.payout	Codevertex Treasury (<code>codevertex.treasury</code>)	Payee type <code>creator</code> , rails <code>mpesa_b2c</code> , <code>paystack_transfer</code> , OTP-gated approval	as above

Capability	Provider (driver)	Seeded configuration	Secrets (from Kubernetes)
invoicing.tax	Codevertex Treasury eTIMS (<code>codevertex.treasury_etims</code>)	Commission and fee invoices only	as above
ai.runtime	MarketFlow Vera (<code>codevertex.vera</code>)	Agent profile <code>aelia-match</code> , tool set <code>aelia.*</code>	<code>INTERNAL_SERVICE_KEY</code>
storage.objects	Codevertex MinIO (<code>codevertex.minio</code>)	Bucket <code>aelia-media-{env}</code> , SSE enabled	MinIO access keys
social.*	Built-in connectors	Disabled until AELIA registers its own developer apps (v1)	App IDs and secrets owned by AELIA
ai.video	—	Unbound until v1 vendor selection	Vendor API key owned by AELIA

20.2 Swap scenario A: move SSO to another OIDC provider

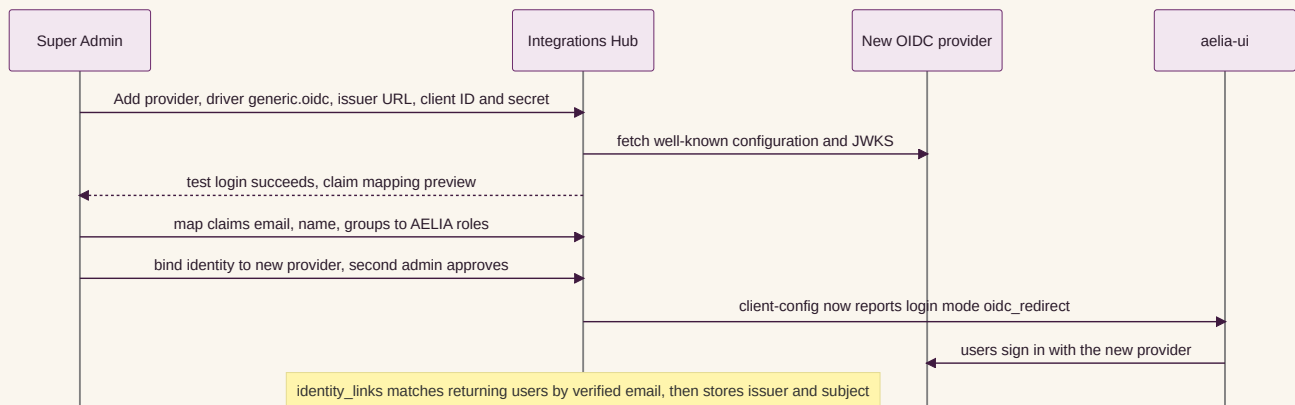


Figure 18. Replacing the identity provider. AELIA user IDs, roles and data are unchanged, because `identity_links` maps (issuer, subject) to the AELIA user.

20.3 Swap scenario B: add a direct payment provider

- Add the provider.** The Super Admin adds a provider with driver `vendor.flutterwave` (a built-in adapter) or `generic.rest` (the connector in §17.3), in the `sandbox` environment.
- Wire up callbacks.** The hub shows the callback and webhook URLs, and AELIA pastes them into the provider's dashboard. The hub generates or accepts the webhook verification secret.
- Test in sandbox.** "Test connection" runs a sandbox `CreateIntent`, `CheckStatus` and a simulated webhook. The conformance suite must pass.
- Switch to production.** The provider is cloned to `production` with live keys and bound to `payments.collect`, effective from a date. Finance and a second Super Admin approve.
- Handle in-flight intents.** Intents already in flight finish on the provider that created them. The provider ID is stored on every intent, so reconciliation stays per-provider.

20.4 Swap scenario C: notifications and billing

- Notifications.** Bind `notifications.sms` to `vendor.africastalking` under AELIA's own account, and keep email on Codevertex Notifications. Templates exist once in AELIA's template registry (`aelia/*`, versioned, per

locale). For non-Codevertex drivers AELIA renders them locally. For Codevertex the template key is passed through.

- **Billing.** Bind `billing` to `builtin.plans`, so plans, features and limits live in AELIA tables and are edited in the admin console, or to `vendor.stripe_billing`. The domain keeps calling `HasFeature` and `GetLimit` unchanged.

Part E · API Specification

How systems communicate: the AELIA public REST API, its conventions, the endpoint catalogue, the event catalogue, and the documentation register for every shared service and external integration.

21. API Conventions

Aspect	Convention
Base URL	<code>https://api.<aelia-domain>/api/v1/{tenant}</code> (tenant slug <code>aelia</code>). Staging uses <code>https://api.staging.<aelia-domain></code> . A sandbox environment for external integrators mirrors production with sandbox providers
Style	Resource-oriented REST, JSON (UTF-8), <code>snake_case</code> fields, ISO-8601 UTC timestamps, UUID v7 identifiers
Money	<code>{ "amount": 2000000, "currency": "KES" }</code> : integer minor units plus ISO-4217 code; never floats
Authentication	<code>Authorization: Bearer <JWT></code> from the bound Identity Provider (users); <code>X-API-Key</code> plus <code>X-Tenant-ID</code> (service-to-service); outbound-webhook signatures (§19.3)
Authorization	Permission codes <code>aelia.{module}.{action}</code> (actions <code>view</code> , <code>view_own</code> , <code>add</code> , <code>change</code> , <code>change_own</code> , <code>delete</code> , <code>manage</code>), checked by middleware and by resource ownership
Middleware order	Rate limit → authentication (JWT or API key) → subscription gate for mutations (<code>RequireActiveSubscriptionForMutationsWithGrace</code>) → JIT user provisioning → tenant and organisation context (<code>X-Organisation-ID</code> for brand-scoped calls) → permission
Idempotency	<code>Idempotency-Key</code> header required on money-moving POSTs (funding, payouts, refunds) and accepted on all POSTs; stored 24 h per key and route; replays return the original response
Pagination	Cursor-based: <code>?limit=25&cursor=...</code> ; the response carries <code>next_cursor</code> (shared <code>pagination</code> package)
Filtering and sorting	<code>?status=funded&created_after=...&sort=-created_at</code>
Concurrency	<code>ETag</code> on mutable resources; <code>If-Match</code> required on state-changing PATCH/PUT; <code>412</code> on mismatch
Errors	<code>{ "error": { "code": "campaign_not_fundable", "message": "...", "details": {...}, "request_id": "..." } }</code>
Rate limits	Per user 120 req/min, per IP 300 req/min, per API key per plan (<code>shared-ratelimit</code>); headers <code>X-RateLimit-Limit</code> , <code>X-RateLimit-Remaining</code> , <code>Retry-After</code>
Versioning	URI major version (<code>/v1</code>); additive changes only within a major; deprecation announced with a <code>Sunset</code> header at least 6 months ahead
Real-time	Server-Sent Events at <code>GET /api/v1/{tenant}/stream</code> (messages, notifications, status changes), authenticated with a short-lived stream token
Uploads	<code>POST /uploads</code> returns a presigned PUT URL and object key; the client uploads directly; <code>POST /uploads/{key}/complete</code> triggers scanning
Documentation	OpenAPI 3.1 generated from handlers; Swagger UI at <code>/v1/docs</code> (public tags), with the full spec for authenticated staff

21.1 Standard error codes

HTTP	Code	Meaning
400	validation_failed	Field errors in <code>details.fields</code>
401	unauthenticated	Missing, expired or invalid token or API key
403	forbidden	Permission missing, or resource not owned
403	subscription_inactive , feature_not_available , usage_limit_exceeded , plan_upgrade_required	Codevertex gating codes, surfaced unchanged (for example the brand plan's active-campaign limit)
404	not_found	Resource not visible to the caller
409	invalid_state_transition	For example approving content on an unfunded campaign
409	idempotency_conflict	Same key with a different payload
412	precondition_failed	ETag mismatch
422	verification_required , kra_pin_required , terms_not_accepted	Business-rule blocks
428	otp_required	Finance approvals need an OTP
429	rate_limited	Retry after the indicated time
502 / 503	provider_unavailable	A bound provider failed; safe to retry with the same idempotency key

22. Endpoint Catalogue

All paths are relative to `/api/v1/{tenant}`. The “Release” column follows the functional requirement served.

22.1 Identity, organisations and Passports

Method and path	Purpose	Permission	Release
GET <code>/me</code>	Current user, roles, organisations, creator profile summary	authenticated	MVP
POST <code>/me/roles</code>	Activate the Brand or Creator role for the current user	authenticated	MVP
POST <code>/organisations</code>	Register a brand organisation	authenticated	MVP
GET, PATCH <code>/organisations/{id}</code>	View and update the brand profile (Brand Passport)	<code>aelia.organisations.view / change_own</code>	MVP
POST <code>/organisations/{id}/members</code> · DELETE <code>.../members/{uid}</code>	Invite or remove brand team members	<code>aelia.organisations.manage_own</code>	MVP
POST <code>/verification-cases</code> · GET <code>/verification-cases/{id}</code>	Submit and track verification with documents	owner	MVP
GET, PUT <code>/creators/me</code>	Creator Passport: profile, niches, skills, languages	<code>aelia.passport.change_own</code>	MVP

Method and path	Purpose	Permission	Release
PUT /creators/me/rates · PUT /creators/me/availability · POST /creators/me/portfolio	Rates and packages, calendar, portfolio	aelia.passport.change_own	MVP
PUT /creators/me/payout-method	M-Pesa number or bank account (validated via PayoutProvider)	aelia.passport.change_own	MVP
GET /creators/me/social/{platform}/connect · DELETE /creators/me/social/{id}	Start OAuth connection; revoke	aelia.passport.change_own	v1
GET /creators/{id}	Public Passport for verified brands	aelia.passport.view	MVP

22.2 Discovery, matching and campaigns

Method and path	Purpose	Permission	Release
GET /creators?niche=&platform=&location=&followers_min=&rate_max=&available_from=	Creator search	aelia.discovery.view	MVP
POST /shortlists · POST /shortlists/{id}/items	Shortlists	aelia.discovery.add	MVP
POST /campaigns · GET /campaigns · GET , PATCH /campaigns/{id}	Brief builder: draft and edit	aelia.campaigns.add / change_own	MVP
POST /campaigns/{id}/submit	Submit the brief for matching	aelia.campaigns.change_own	MVP
POST /campaigns/{id}/composer	Lane options with indicative costs	aelia.campaigns.view_own	v1
POST /campaigns/{id}/brief-assist	AI Brief Builder suggestions	aelia.campaigns.change_own	v1
GET /campaigns/{id}/recommendations	Curated MVP or ranked v1 shortlist with rationale	aelia.campaigns.view_own	MVP
POST /campaigns/{id}/selections	Brand selects creators (proposed)	aelia.campaigns.change_own	MVP
POST /admin/campaigns/{id}/selections/{sid}/confirm	AELIA confirms or adjusts the match	aelia.match.manage	MVP
POST /campaigns/{id}/invitations · POST /opportunities/{id}/applications	Invite creators; creators apply to open calls	brand / creator	MVP
GET /opportunities	Creator opportunity feed	aelia.opportunities.view	MVP
POST /campaigns/{id}/cancel	Cancel under agreement terms	aelia.campaigns.change_own	MVP
POST /campaigns/{id}/rebook	New campaign pre-filled from a completed one	aelia.campaigns.add	v1

22.3 Agreements, workspace and content

Method and path	Purpose	Permission	Release
GET /terms/current · POST /terms/{id}/accept	Platform terms and acceptance	authenticated	MVP
POST /campaigns/{id}/engagements · POST /engagements/{id}/accept	Generate and accept per-party engagement records	brand / creator	MVP
GET /engagements/{id}/document	Rendered agreement PDF with hash	party	MVP
GET /campaigns/{id}/threads · POST /threads/{id}/messages	Campaign messaging	participant	MVP
GET /campaigns/{id}/timeline	Activity timeline	participant	MVP
POST /deliverables/{id}/submissions	Submit a draft version	aelia.content.ad d_own	MVP
POST /submissions/{id}/revision-requests · POST /submissions/{id}/approve	Review decision	brand approver	MVP
POST /deliverables/{id}/publication-proof	Live URL and screenshot	creator	MVP
POST /deliverables/{id}/metrics	Manual performance entry	creator	MVP
GET /content-library	Approved assets with licence metadata	brand	v1

22.4 Payments, tax and reporting

Method and path	Purpose	Permission	Release
GET /campaigns/{id}/quote	Fee and tax preview from current settings	brand	MVP
POST /campaigns/{id}/funding	Create a funding intent (Idempotency-Key required)	aelia.payments. add_own	MVP
POST /funding/{id}/initiate	Start M-Pesa STK or card checkout via the PaymentProvider	brand	MVP
POST /funding/{id}/bank-transfer-proof	Upload proof of bank transfer	brand	MVP
POST /admin/funding/{id}/verify	Finance verifies manual payment (OTP)	aelia.finance.m anage	MVP
GET /payouts · GET /payouts/{id}	Creator payouts and status	creator / finance	MVP
POST /admin/payouts/batches · POST /admin/payouts/batches/{id}/approve	Payout batch approval (OTP)	aelia.finance.m anage	v1
POST /admin/refunds	Refund under agreement or dispute decision	aelia.finance.m anage	MVP
GET /admin/ledger/entries · GET /admin/reconciliation?period=	Ledger explorer and reconciliation report	aelia.finance.v iew	MVP

Method and path	Purpose	Permission	Release
<code>GET /tax/certificates</code> · <code>GET /admin/tax/withholding-register?period=</code>	Creator certificates; WHT register export	creator / finance	v1 / MVP
<code>GET /reports/campaigns/{id}</code> · <code>GET /reports/brand</code> · <code>GET /reports/creator</code>	Performance and financial reports (JSON, CSV, PDF)	owner	v1

22.5 Reputation, administration and integrations

Method and path	Purpose	Permission	Release
<code>POST /campaigns/{id}/reviews</code>	Two-sided review	participant	v1
<code>GET /admin/verification-queue</code> · <code>POST /admin/verification-cases/{id}/decision</code>	Verification queue	<code>aelia.verification.manage</code>	MVP
<code>POST /disputes</code> · <code>POST /admin/disputes/{id}/decision</code>	Dispute case and decision with financial outcome	participant / <code>aelia.disputes.manage</code>	v1
<code>GET, PUT /admin/settings/{key}</code>	Tax, fee, plan, template and match-weight settings (effective-dated)	<code>aelia.settings.manage</code>	MVP
<code>GET /admin/audit-log</code>	Audit explorer	<code>aelia.audit.view</code>	v1
<code>GET /admin/integrations</code> · <code>POST /admin/integrations/providers</code> · <code>PATCH /admin/integrations/providers/{id}</code>	Providers	<code>aelia.integrations.manage</code>	MVP
<code>POST /admin/integrations/providers/{id}/credentials</code> · <code>.../test</code> · <code>.../activate</code>	Write-only secrets; test; activate	<code>aelia.integrations.manage</code> (+ second approver)	MVP
<code>PUT /admin/integrations/bindings/{capability}</code>	Bind capability to provider and fallback	<code>aelia.integrations.manage</code>	MVP
<code>POST /admin/webhooks/subscriptions</code> · <code>GET .../deliveries</code> · <code>POST .../deliveries/{id}/replay</code>	Outbound webhooks	<code>aelia.integrations.manage</code>	v1
<code>GET /integrations/client-config</code>	Public client settings (login and payment UI modes)	public	MVP
<code>POST /integrations/hooks/{providerKey}/{pathToKen}</code>	Inbound provider webhooks	signature	MVP

23. Event Catalogue

23.1 Events published by AELIA (stream `aelia`)

Subject	Emitted when	Key payload	Consumers
<code>aelia.user.onboarded</code>	A role profile is completed	<code>user_id</code> , <code>role</code>	notifications, reporting

Subject	Emitted when	Key payload	Consumers
<code>aelia.verification.decided</code>	Verification approved or rejected	<code>case_id</code> , <code>subject_type</code> , <code>decision</code> , <code>reason_code</code>	notifications, passport
<code>aelia.campaign.submitted</code>	Brief submitted	<code>campaign_id</code> , <code>organisation_id</code> , <code>type</code> , <code>budget</code>	admin queue, match
<code>aelia.match.shortlist_published</code>	Curated or ranked shortlist ready	<code>campaign_id</code> , <code>count</code> , <code>method</code>	notifications
<code>aelia.campaign.creator_confirmed</code>	AELIA confirms a selection	<code>campaign_id</code> , <code>creator_id</code>	notifications, agreements
<code>aelia.agreement.accepted</code>	A party accepts its engagement	<code>engagement_id</code> , <code>party</code> , <code>version</code> , <code>hash</code>	campaign state machine
<code>aelia.campaign.funded</code>	Funding verified	<code>campaign_id</code> , <code>funding_id</code> , <code>amount</code>	ledger, payouts (upfront), notifications
<code>aelia.content.submitted</code> / <code>revision_requested</code> / <code>approved</code>	Content review loop	<code>deliverable_id</code> , <code>version</code> , <code>approver_id</code>	notifications, payouts
<code>aelia.content.published</code>	Publication proof recorded	<code>deliverable_id</code> , <code>url</code>	performance
<code>aelia.payout.requested</code> / <code>completed</code> / <code>failed</code>	Payout lifecycle	<code>payout_id</code> , <code>creator_id</code> , <code>gross</code> , <code>wht</code> , <code>net</code>	notifications, tax, reporting
<code>aelia.refund.completed</code>	Refund posted	<code>refund_id</code> , <code>amount</code>	ledger, notifications
<code>aelia.campaign.completed</code>	All deliverables approved and paid	<code>campaign_id</code>	reviews, reporting, rebooking
<code>aelia.review.submitted</code>	A review is submitted	<code>review_id</code> , <code>subject</code>	reputation
<code>aelia.dispute.opened</code> / <code>decided</code>	Dispute lifecycle	<code>dispute_id</code> , <code>outcome</code>	ledger, notifications
<code>aelia.tax.certificate_issued</code>	Monthly WHT certificate issued	<code>creator_id</code> , <code>period</code> , <code>amount</code>	notifications
<code>aelia.integration.config_changed</code>	Provider or binding changed	<code>capability</code> , <code>provider_id</code>	resolver cache eviction
<code>aelia.integration.provider_unhealthy</code>	Health check failing or circuit open	<code>provider_id</code> , <code>error</code>	admin alerts

23.2 Events consumed by AELIA

Subject	Source	AELIA action
<code>auth.user.created</code> , <code>auth.user.login</code> (plain NATS)	auth-api	Warm the user link; update the last-login snapshot
<code>treasury.payment.succeeded</code> / <code>failed</code>	treasury-api	Mark funding verified or failed (after <code>CheckStatus</code> confirmation); post ledger
<code>treasury.payout.completed</code> (and failure events)	treasury-api	Complete or fail the payout; post ledger; notify

Subject	Source	AELIA action
<code>treasury.refund.completed</code>	treasury-api	Complete the refund; post ledger
<code>treasury.etims.invoice_transmitted</code> / <code>transmission_failed</code>	treasury-api	Mark the commission invoice as transmitted, or alert Finance
<code>subscription.created</code> / <code>upgraded</code> / <code>downgraded</code> / <code>cancelled</code> / <code>renewed</code>	subscriptions-api	Refresh cached entitlements for the tenant or brand consumer

23.3 Notification templates (`aelia/*`)

One template per notifying event and channel, for example `aelia/verification_approved` , `aelia/shortlist_ready` , `aelia/invitation_received` , `aelia/agreement_to_accept` , `aelia/funding_received` , `aelia/content_submitted` , `aelia/revision_requested` , `aelia/content_approved` , `aelia/payout_sent` , `aelia/deadline_reminder` , `aelia/review_request` , `aelia/wht_certificate_ready` . WhatsApp variants use Meta-approved `_btn` templates with link buttons.

24. Shared Service & External API Documentation Register

Requirements spec §4.2 requires every shared service and external integration to be documented to one standard. Each shared service and external integration is documented against the §4.2 fields. Both tables below are kept in the repository (`docs/integrations/`) and in the Integrations Hub.

A. Purpose, interface and environments

Service	Purpose	Endpoints	Authentication	Sandbox / test	Production	Ownership / access
Codevertex SSO	Login, identity, MFA	OIDC authorize, token, refresh, me, discovery (§14)	PKCE; RS256 JWT via JWKS	Staging SSO	<code>sso.codevertexafrica.com</code>	Codevertex operates; AELIA OIDC client configured
Codevertex Notifications	Email, SMS, WhatsApp, push	<code>/tenantId/notifications/messages</code> ; push tokens	JWT or X-API-Key	Staging	<code>notificationsapi.codevertexafrica.com</code>	Codevertex operates; AELIA sender identities
Codevertex Subscriptions	Plans and entitlements	<code>/tenants/{id}/subscription</code> ; entitlements	X-API-Key + X-Tenant-ID	Staging	<code>pricingapi.codevertexafrica.com</code>	Codevertex operates
Codevertex Treasury	Collection, payouts, GL, eTIMS	<code>/payments/intents*</code> ; payout API	X-API-Key + X-Tenant-ID	Sandbox gateways; eTIMS sandbox	<code>booksapi.codevertexafrica.com</code>	Codevertex operates; merchant accounts in AELIA's name

Service	Purpose	Endpoints	Authentication	Sandbox / test	Production	Ownership / access
MarketFlow Vera	AI agent runtime	Agent-run API	X-API-Key	Staging	marketflowai.codevertexafrica.com	Codevertex operates
Paystack / M-Pesa (via Treasury)	Licensed payment rails	Managed by Treasury	Keys held in Treasury gateway config	Paystack test keys; Daraja sandbox	Live keys	AELIA-owned merchant accounts
Social platforms	Account verification, metrics	Meta Graph, TikTok, YouTube Data, LinkedIn	OAuth 2.0 per creator	Developer apps in test mode	Apps approved through platform review	AELIA-owned developer apps
AI video vendor	Clip generation	Vendor REST (async jobs)	API key	Trial keys	Live key	AELIA-owned vendor account

B. Behaviour, limits and operations

Service	Request / response	Error codes	Rate limits	Versioning	Monitoring	Dependencies
Codevertex SSO	JSON; JWT claims	OAuth 2.0 errors	Per client	/api/v1	Hub health every 60 s	–
Codevertex Notifications	JSON; 202 accepted with requestId	400, 401, 429	Per channel and plan	/v1	Hub health; delivery statistics	Channel provider accounts
Codevertex Subscriptions	JSON	Gating codes (\$21.1)	Cached 60 s	/api/v1	Hub health	auth-api
Codevertex Treasury	JSON; intent object	4xx, 5xx, 428 OTP	Per tenant	/api/v1	Hub health; daily reconciliation	Paystack, Daraja, KRA
MarketFlow Vera	JSON	4xx, 5xx	Per-tenant token budget	/v1	Hub health	LLM provider
Paystack / M-Pesa	Treasury abstraction	Mapped by Treasury	Provider limits	Provider versions	Via Treasury	CBK-licensed providers
Social platforms	JSON	Platform-specific, mapped to canonical	Platform quotas	Graph API and platform versions	Token-expiry alerts	Platform app review
AI video vendor	JSON; async job	Vendor-specific, mapped	Vendor quotas	Vendor versions	Job-failure alerts	Vendor

Part F · Database & Data Model Specification

How platform data is structured: the AELIA database, its entities and relationships, ownership against the Codevertex services, and retention.

25. Data Architecture

Aspect	Specification
Database	One PostgreSQL 16 database <code>aelia</code> (schema <code>public</code>), owned by AELIA, reached through PgBouncer in transaction mode
Schema management	Ent schemas in Go, with Atlas versioned migrations committed to the repository and linted in CI (<code>atlas migrate lint</code>). Destructive changes need a two-step expand/contract migration
Keys	UUID v7 primary keys (time-ordered). Every table has <code>tenant_id</code> , <code>created_at</code> , <code>updated_at</code> ; soft delete (<code>deleted_at</code>) where records must be kept for audit
Money	<code>amount_minor bigint</code> plus <code>currency char(3)</code> . The ledger uses integer minor units only
Enumerations	Postgres text with check constraints (portable, migration-friendly)
JSON	<code>jsonb</code> only for provider payload snapshots, configuration and flexible metadata. Queried fields are always real columns
Search	Postgres full-text (<code>tsvector</code>) plus trigram indexes on creator name, bio and niches for MVP. The search port allows a dedicated engine Later
Encryption	Documents and media in object storage with server-side encryption. PII columns (ID number, KRA PIN, payout account) encrypted at application level (AES-256-GCM, rotatable key ring); a blind index (HMAC) where equality search is needed
Audit	Append-only <code>audit_log</code> (actor, action, entity, before/after diff, IP, request ID), hash-chained per day to be tamper-evident
Read models	Reporting projections (<code>metric_snapshots</code> , <code>campaign_stats</code>) are maintained by event consumers, and heavy reports read from them

26. Entity-Relationship Diagrams

26.1 Identity, organisations and Passports

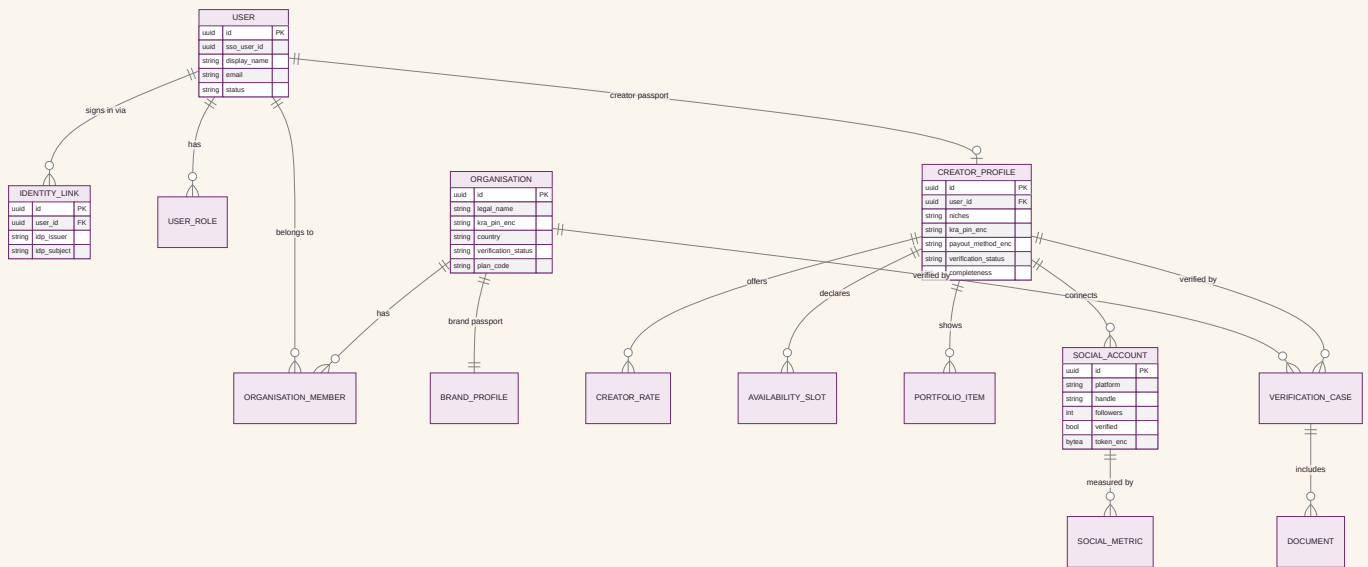


Figure 19. Identity, organisations and Passports.

26.2 Campaigns, agreements and content

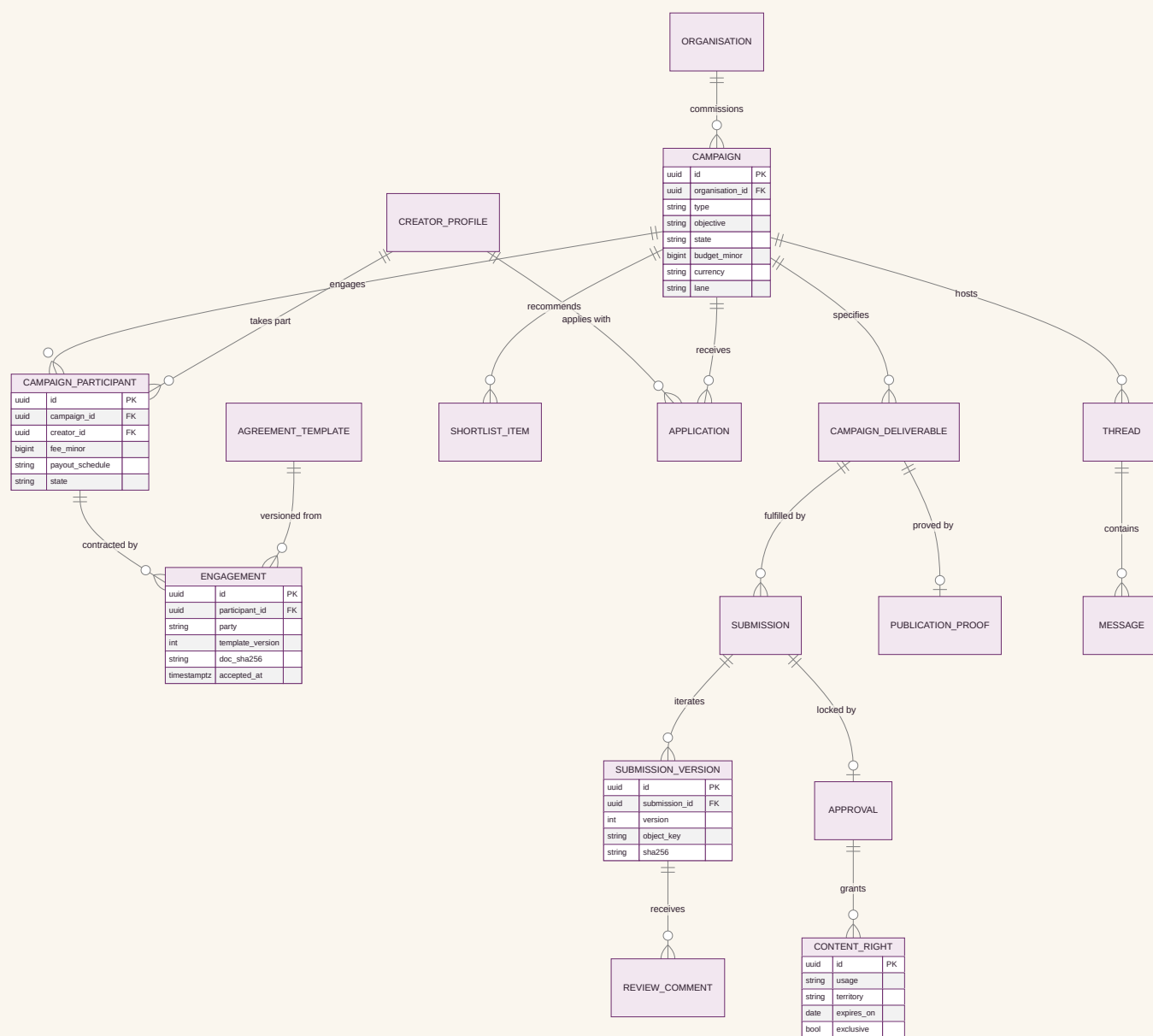


Figure 20. Campaigns, agreements and content. One campaign links two engagement records per creator: AELIA–Brand and AELIA–Creator.

26.3 Payments, ledger and tax

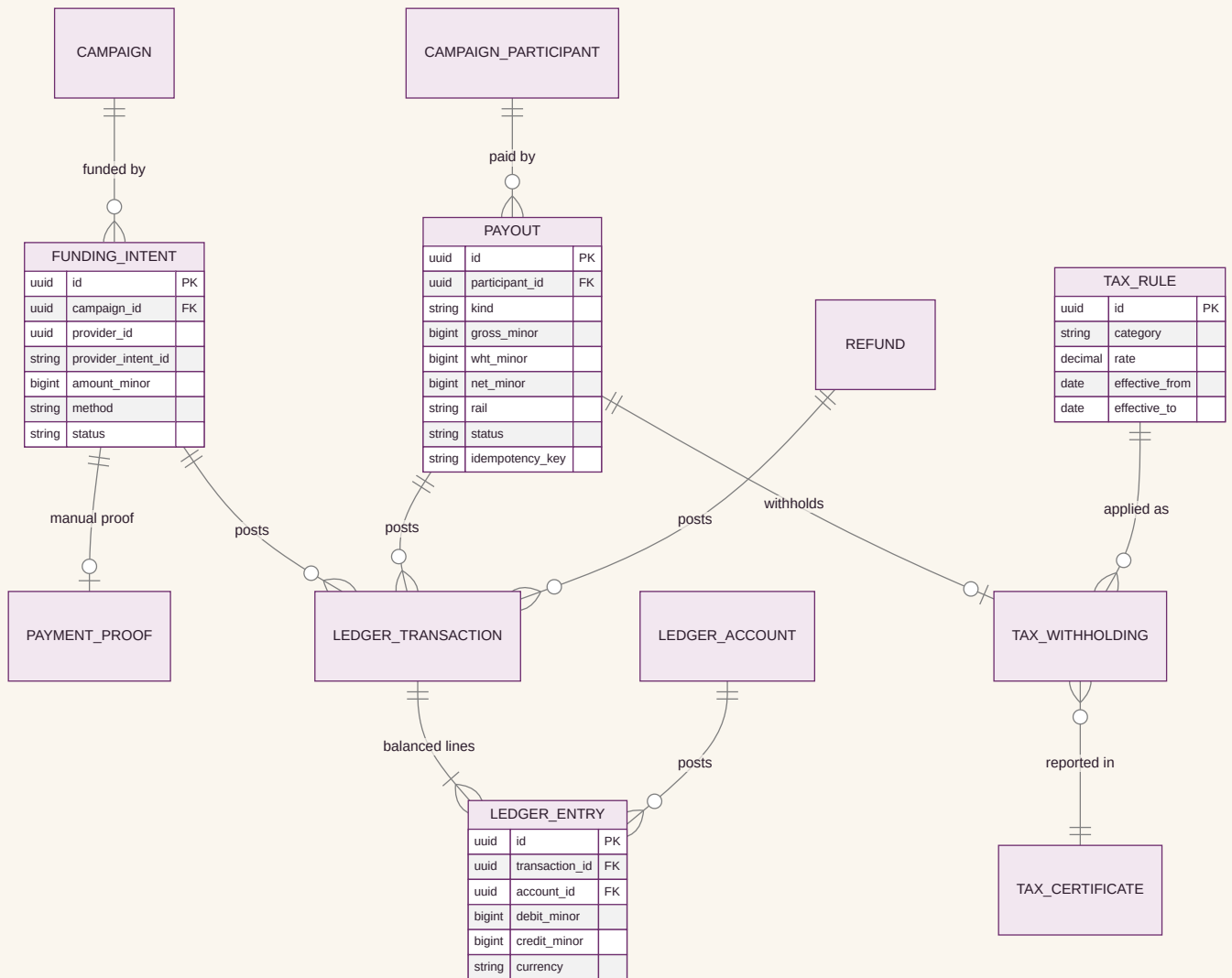


Figure 21. Payments, double-entry ledger and tax.

27. Entity Dictionary, Ownership & Retention

27.1 Core entities

Entity	Description	Key rules
users	AELIA user linked to the IdP identity; display snapshot	Created just-in-time on the first valid token; never stores passwords
identity_links	(issuer, subject) to user mapping	Unique per issuer and subject; enables IdP replacement
organisations , organisation_members	Brand organisations and their team	A brand needs verified status to fund; member roles are owner, approver, finance, viewer
creator_profiles and children	Creator Passport: profile, rates, availability, portfolio, social accounts, metrics	KRA PIN and payout method are encrypted; self-declared vs verified flags

Entity	Description	Key rules
<code>verification_cases</code> , <code>documents</code>	KYC/KYB workflow and evidence	Documents are private objects; the decision is audited
<code>campaigns</code> , <code>campaign_deliverables</code>	Brief and deliverable specification	<code>state</code> follows the machine in §28; the budget currency is fixed at funding
<code>campaign_participants</code>	A creator (or in-house or AI lane) engaged on a campaign	Fee, payout schedule and per-participant state
<code>shortlists</code> , <code>match_runs</code> , <code>match_scores</code>	Curated and computed recommendations	Scores are stored with a weights snapshot for explainability
<code>agreement_templates</code> , <code>terms_acceptances</code> , <code>engagements</code>	Versioned legal templates and per-party acceptance	Immutable once accepted; the PDF hash is stored
<code>submissions</code> , <code>submission_versions</code> , <code>approvals</code> , <code>content_rights</code> , <code>publication_proofs</code>	Content workflow and rights chain	The approved version is immutable; rights reference the engagement terms
<code>threads</code> , <code>messages</code> , <code>attachments</code>	Campaign-scoped messaging	Contact details masked before agreement
<code>funding_intents</code> , <code>payment_proofs</code> , <code>payouts</code> , <code>refunds</code>	Money movements, mirrored with provider references	Idempotency keys unique; status is driven by verified provider events
<code>ledger_accounts</code> , <code>ledger_transactions</code> , <code>ledger_entries</code>	Double-entry ledger	Sum of debits = sum of credits per transaction; append-only; corrections by reversal
<code>tax_rules</code> , <code>tax_withholdings</code> , <code>tax_certificates</code>	Configurable tax and withholding	Rule version stamped on each withholding
<code>reviews</code> , <code>reputation_scores</code>	Two-sided reviews	Double-blind reveal
<code>disputes</code> , <code>dispute_events</code>	Dispute cases	The decision triggers ledger postings
<code>settings</code> , <code>feature_flags</code>	Effective-dated configuration	Every change audited
<code>integration_*</code> , <code>webhook_*</code>	Integrations Hub (§18)	Secrets are ciphertext only
<code>outbox_events</code> , <code>processed_events</code> , <code>processed_webhooks</code>	Reliable events and idempotency	Pruned after publication or TTL
<code>audit_log</code>	Tamper-evident audit	Retained 7 years

27.2 Data ownership versus Codevertex services

Data	System of record	AELIA stores	Must NOT be duplicated in AELIA
Login credentials, MFA factors, sessions	IdentityProvider (Codevertex SSO)	<code>sso_user_id</code> , issuer and subject, display snapshot	Passwords, MFA secrets
Brand, creator, campaign, content, agreement, review, dispute data	AELIA	Everything	—
Commission ledger, payouts, withholding	AELIA (business ledger)	Everything	—

Data	System of record	AELIA stores	Must NOT be duplicated in AELIA
Payment transactions, gateway state, GL journals, eTIMS invoices	Treasury	Intent and payout references plus status snapshot	Card data, gateway credentials, GL journals
Plan subscriptions and entitlements	BillingProvider (Codevertex Subscriptions)	Plan code snapshot, cached entitlements (60 s)	Plan catalogue master data
Notification delivery logs	NotificationProvider	Message request ID and status	Provider logs
CRM contact (optional enrichment)	MarketFlow	Nullable <code>crm_contact_id</code> and name snapshot	CRM records

27.3 Retention schedule

Data class	Retention	Disposal
Financial records (ledger, payouts, withholding, receipts, certificates)	7 years after the financial year (tax and accounting law)	Archive, then delete
Agreements and terms acceptances	7 years after campaign completion	Archive, then delete
Verification documents (ID, incorporation)	Account life + 2 years; rejected applications 6 months	Hard delete from object storage
Campaign content and messages	Account life + 2 years, or the licence term if longer	Delete or anonymise
Social tokens and metrics	Until revoked or account closure; metrics 3 years	Tokens deleted immediately on revoke
Audit log	7 years	Delete
Application logs	30 days	Rotate
Closed accounts (DSR erasure)	Personal data erased or anonymised within 30 days, except where retention is legally required	Anonymise

Part G · Workflows & State Machines

The lifecycles that govern campaigns, participants, content and disputes, and the swimlane showing how money and approvals cross between brand, platform and creator.

28. Campaign Lifecycle

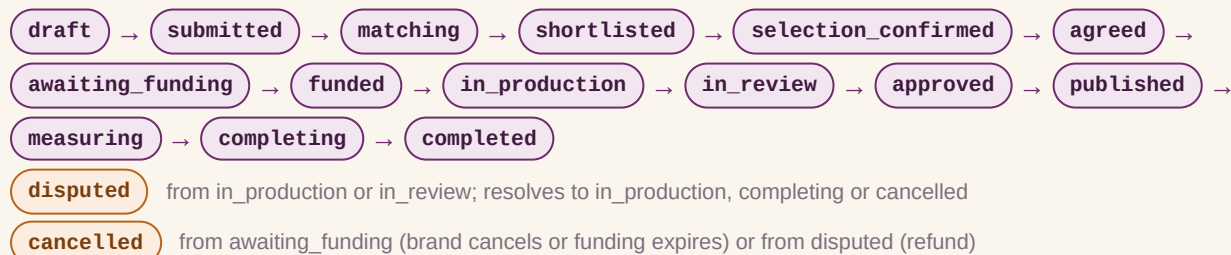
28.1 Transaction swimlane

	1 · Brief	2 · Match	3 · Agree	4 · Fund	5 · Produce and approve	6 · Pay and review
Brand	Post brief: objective, audience, deliverables, budget	Review the shortlist and pick one or more creators	Accept the AELIA campaign order	Fund via M-Pesa, card or bank transfer	Review, request revisions, approve	Rate the creator; rebook
AELIA platform	Brief checked; matching starts	AELIA Match shortlist; human confirmation and introduction	Generate linked engagement records	Verify the provider event; set the commission aside	Lock the approved version; evaluate payout triggers	Release the payout per schedule, minus WHT; certificates
Creator	Passport discoverable	Receive the invitation; accept or apply	Accept the AELIA creator engagement	Starting fee released if scheduled	Create, submit, revise; publish with disclosure	Receive net pay; rate the brand

Figure 22. Three-lane transaction swimlane. The platform lane never holds money itself: funds sit with the licensed provider, and AELIA holds the intent, ledger and payout records.

28.2 Campaign state machine

The developer brief's 17 states are consolidated into a campaign-level machine plus a per-participant machine (§28.3). Together they cover every original state.



From	Event	To
draft	Brand submits the brief	submitted
submitted	Admin picks it up	matching
matching	Shortlist published (curated or ranked)	shortlisted
shortlisted	Brand selects and AELIA confirms	selection_confirmed
selection_confirmed	All engagements accepted	agreed → awaiting_funding

From	Event	To
awaiting_funding	Provider event verified, or Finance approves the bank transfer	funded → in_production
in_production	First submission	in_review
in_review	Revision requested	in_production
in_review	All deliverables approved	approved
approved	Publication proofs recorded (influence)	published → measuring
approved	Content-only campaign	completing
measuring	Metrics window closes	completing
completing	All payouts settled	completed
in_production , in_review	Dispute opened	disputed
awaiting_funding	Brand cancels or funding expires	cancelled

Figure 23. Campaign state machine. Transitions are server-enforced; each emits an `aelia.campaign.*` event and a timeline entry.

Guard	Rule
submitted	Brand verified; current terms accepted; entitlement allows another active campaign
selection_confirmed	An AELIA admin confirmation record exists for each selected participant (MAT-04)
funded	Verified provider event and a <code>CheckStatus</code> confirmation, or a Finance OTP approval for bank transfer
approved	Every deliverable has an approval with a locked version hash
completed	Every participant's payouts are <code>completed</code> and the ledger transaction for the campaign balances
Cancellation	Refund amount computed from the engagement's cancellation terms and the production stage reached

28.3 Participant and payout lifecycle

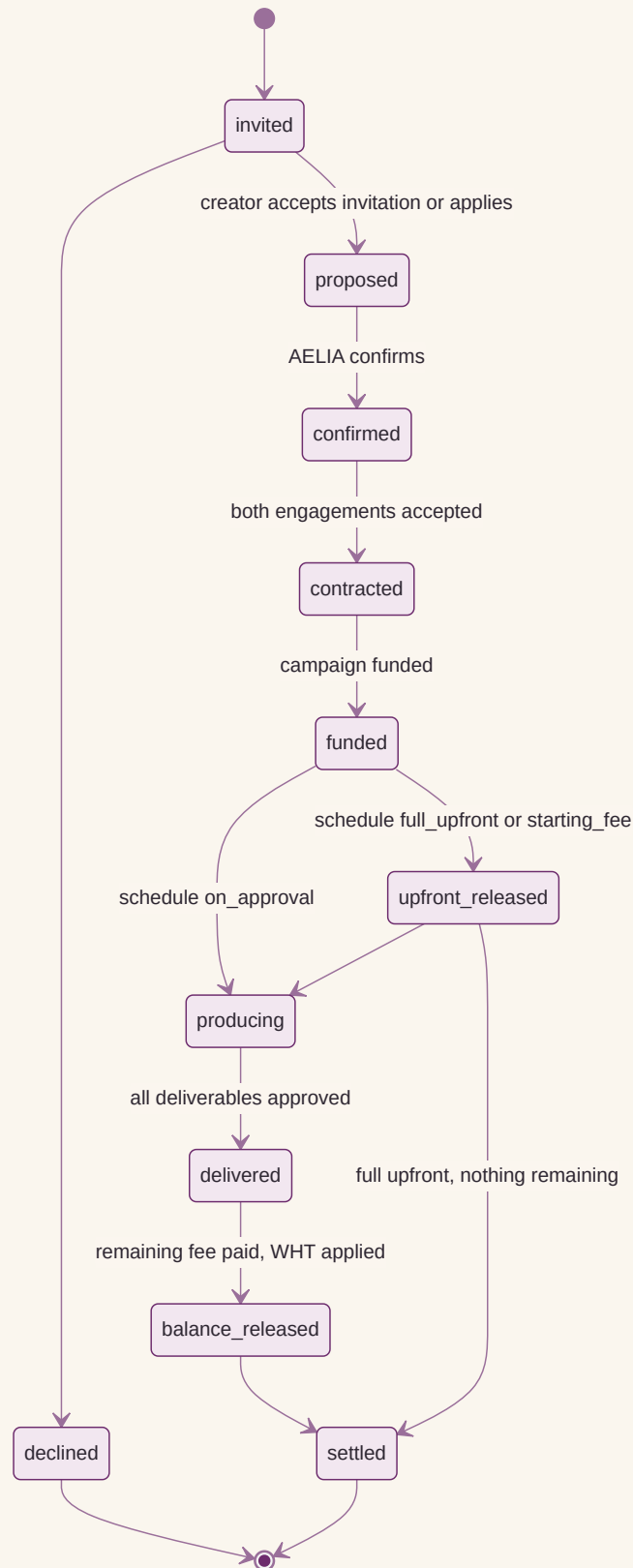


Figure 24. Per-participant lifecycle, including the three payout schedules.

29. Content, Review & Dispute Workflows

29.1 Content submission and approval

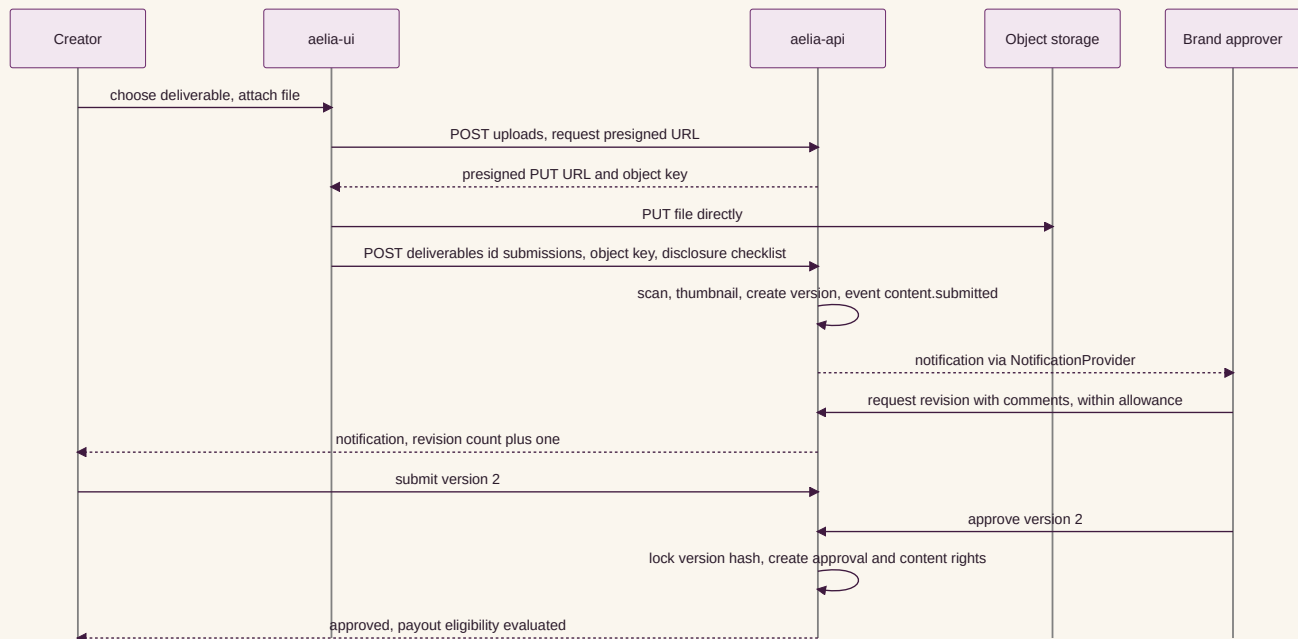


Figure 25. Submission, revision and approval with direct-to-storage uploads.

29.2 Disputes

Stage	Behaviour	SLA
Open	Either party opens a dispute from the workspace with a category (non-delivery, quality, late delivery, non-approval, rights misuse, payment) and evidence	—
Hold	Pending payouts for that participant are frozen; the campaign moves to disputed	Immediate
Mediation	An AELIA admin reviews the thread, timeline, versions and agreement terms; requests statements	5 business days
Decision	Outcome: continue, release in full, partial refund with partial release, or full refund. A written reason is recorded	10 business days
Settlement	Ledger postings and PaymentProvider refund and PayoutProvider release, all idempotent; both parties notified	2 business days

Part H · UI/UX Design Specification

Screens, navigation and the user experience for brands, creators and AELIA operations, delivered as one installable, mobile-first web app.

30. Experience Architecture

30.1 UX principles

- 1. Mobile-first.** Creators work mainly on phones. Every creator journey completes one-handed on a 360 px screen, and brand and admin views are optimised for laptops while still working on phones.
- 2. One next action.** Every dashboard leads with the single most important pending task (fund, approve, submit, verify).
- 3. Money is always explicit.** Every amount shows gross, commission, tax and net, with the rule that produced it.
- 4. Trust signals everywhere.** Verified badges, “AELIA curated” labels, disclosure reminders and agreement links sit next to the action they relate to.
- 5. Status over notification.** A campaign’s current state and its next owner are always visible, with notifications linking back to it.
- 6. Brand-consistent.** The AELIA design system (tokens for colour, type and spacing) is built on shared Codevertex UI primitives, with AELIA’s own brand identity.

30.2 Information architecture

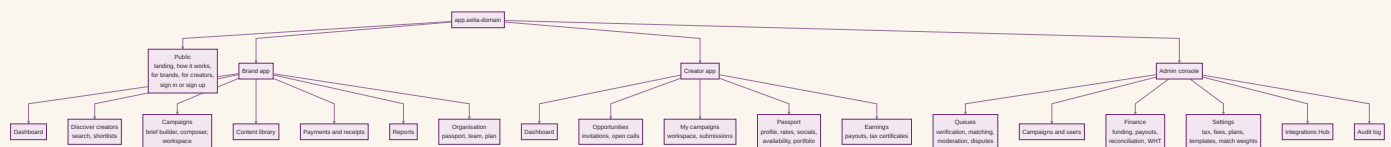


Figure 26. Sitemap. Role-based navigation shows only the areas a user's roles allow; a role switcher appears for multi-role users.

30.3 Key screen illustrations

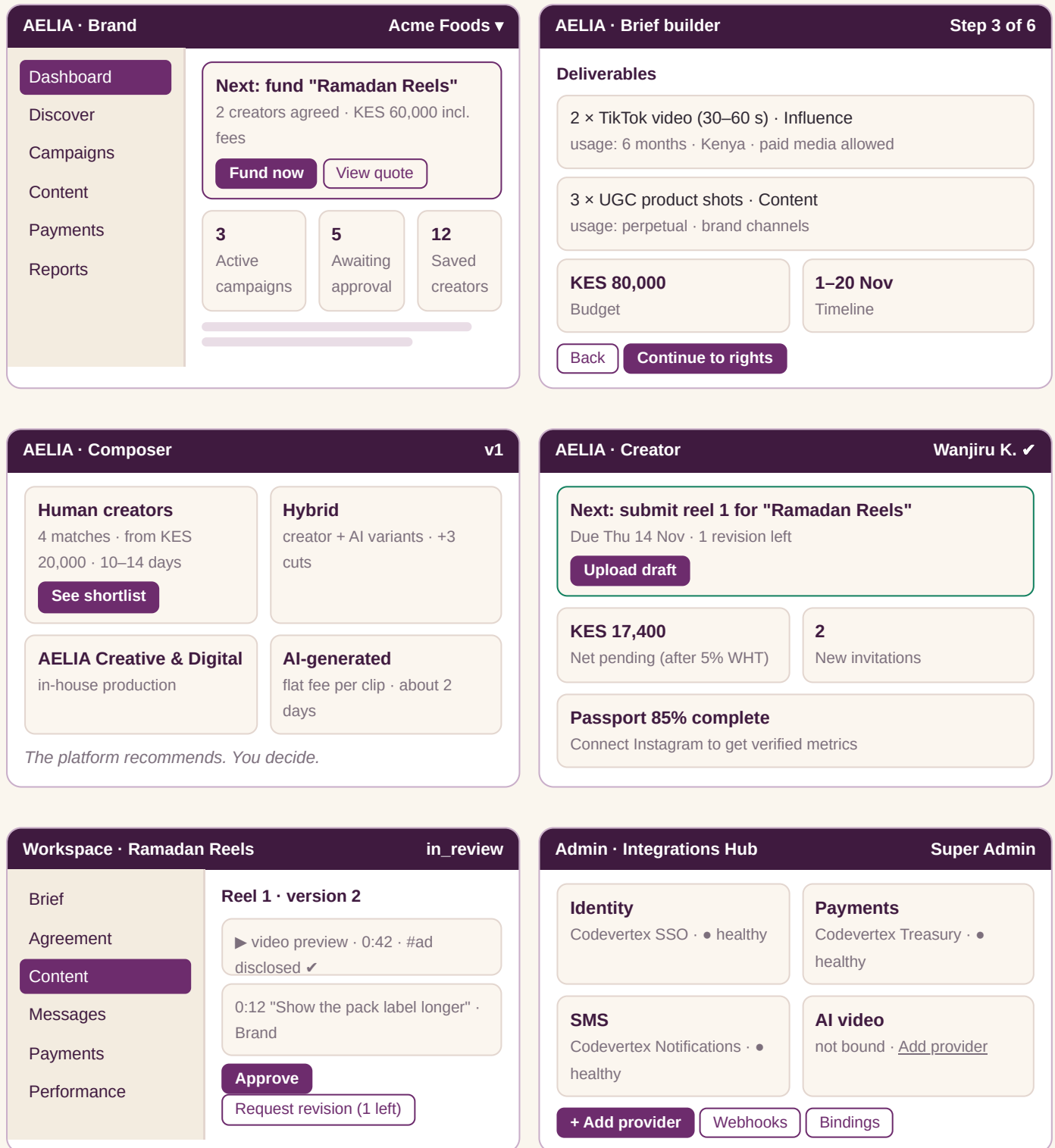


Figure 27. Illustrative wireframes (not final visual design): brand dashboard, brief builder, Campaign Composer, creator dashboard, campaign workspace, Integrations Hub.

30.4 Screen inventory

Area	Screens	Release
Public	Landing; for brands; for creators; how it works; pricing page (content set by AELIA); sign in or sign up (IdP); terms and privacy	MVP
Onboarding	Role choice; brand organisation wizard; creator Passport wizard; verification upload; payout method; KRA PIN	MVP

Area	Screens	Release
Brand	Dashboard; discover (search, filters, creator card, Passport); shortlists; brief builder (6 steps: objective, audience, deliverables, rights, budget and timeline, review); recommendations; agreement review; funding (method choice, M-Pesa prompt, card checkout, bank proof); workspace; content library; payments and receipts; team; plan	MVP ; composer, AI brief, reports, content library v1
Creator	Dashboard; opportunities; invitation and application; engagement acceptance; workspace; submit and revise; publication proof and metrics; earnings and payouts; Passport editor; availability	MVP ; social connect, certificates, reviews v1
Admin	Verification queue; matching queue and shortlist builder; campaign admin; finance (funding verification, payouts, reconciliation, WHT register); settings; Integrations Hub	MVP ; disputes, moderation, audit explorer, webhooks v1

30.5 Design system and PWA behaviour

Topic	Specification
Tokens	Colour, type scale, spacing, radius and elevation as CSS variables. Light theme at MVP, dark theme v1 . Minimum contrast 4.5:1 for text
Components	Built on @bengo-hub/shared-ui-lib primitives: buttons, inputs, tables, modals, toasts, steppers, status pills, money display, file uploader, timeline, chat
Status language	One colour and label per state across campaign, participant, payout and verification, with an icon so colour is never the only signal
Responsive breakpoints	360, 768, 1024 and 1440 px. Bottom tab bar on mobile, side navigation on desktop
PWA	Web manifest with the AELIA icon set; install prompt after first successful action; service worker caching the app shell and static assets; offline read of recent campaigns and messages; queued message send when offline
Push	Web push via the NotificationProvider's push channel; opt-in after onboarding
Performance budgets	Largest Contentful Paint < 2.5 s on mid-range Android over 4G; JS < 250 kB gzipped per route
Accessibility	WCAG 2.1 AA v1 : keyboard navigation, focus states, ARIA labels, captions required on uploaded video where provided
Analytics	Product analytics events (page views, funnel steps) sent to AELIA's own analytics property, owned and accessed by AELIA

Part I · Security & Data Protection Specification

Security and privacy controls, from authentication through to breach response, aligned with the Codevertex platform standards and Kenya's Data Protection Act, 2019.

31. Security Architecture

31.1 Threat model summary

Asset	Main threats	Key controls
Campaign funds and payouts	Payout redirection, duplicate payout, forged payment callbacks, insider fraud	Licensed rails only; verified and re-queried provider events; idempotency keys; OTP and dual control on Finance approvals; payout-method change cool-off (24 h) with notification; ledger reconciliation
Personal data (IDs, KRA PINs, contacts)	Data breach, over-exposure between users	Field encryption; strict ownership checks; contact masking; least-privilege roles; audit
Accounts	Credential stuffing, session hijack	IdP-managed login with MFA for staff; short-lived access tokens (15 min) with refresh rotation; rate limits
Content and rights	Leak of unreleased brand content, rights misuse	Private objects with short-lived presigned URLs; watermarking of drafts v1 ; rights records
Integration credentials	Secret theft, rogue provider configuration	Encrypted vault; fingerprints only; dual approval for identity and money providers; SSRF-safe connector
Platform availability	DDoS, abuse	Cloudflare edge and WAF; ingress rate limits; HPA

31.2 Authentication and authorization

Control	Specification
User authentication	Via the bound IdentityProvider: OIDC Authorization Code + PKCE (S256); tokens validated against JWKS (RS256), issuer and audience; refresh-token rotation
MFA	Required for <code>aelia_admin</code> , <code>aelia_finance</code> and <code>super_admin</code> : the admin middleware checks the MFA assertion (<code>amr</code> / <code>acr</code>); optional for other users v1
Service-to-service	<code>X-API-Key</code> (hashed at rest by auth-api, validated and cached for 5 min) always with an explicit <code>X-Tenant-ID</code> ; per-service keys with rotation
Authorization model	Codevertex three-layer pattern: IdP roles → AELIA roles → permissions <code>aelia.{module}.{action}</code> , plus resource ownership (brand membership, campaign participation) enforced in the service layer
Sensitive actions	OTP step-up (<code>428 otp_required</code>) for payout approval, manual-funding verification, refund, payout-method change, integration activation
Sessions	httpOnly, Secure, SameSite=Lax cookies for the web app's refresh token; CSRF tokens on cookie-authenticated mutations

Role and permission matrix (summary)

Permission area	Brand owner	Brand member	Creator	AELIA Admin	Finance	Super Admin
Organisation and team	manage own	view own	–	view, verify	view	manage
Creator Passport	view (verified brands)	view	manage own	view, verify	view	manage
Campaigns	manage own	per brand role	view own participation	manage	view	manage
Content approval	approve	if approver	submit own	moderate	–	–
Funding	fund own	if finance role	–	view	verify, refund	view
Payouts	view own campaigns	view	view own	view	approve (OTP)	view
Settings, tax, fees	–	–	–	view	tax and fees	manage
Integrations Hub	–	–	–	view health	view	manage (dual approval)
Audit log	–	–	–	view	view finance	view all

31.3 Application, data and infrastructure security

Area	Controls
Input and output	Server-side validation on every field; parameterised queries (Ent); output encoding; strict Content-Security-Policy; upload type and size limits; malware scanning of uploads
API security	OWASP API Top 10 controls; object-level authorization tests for every endpoint; rate limiting (<code>shared-ratelimit</code>); request-size limits; no stack traces in errors
Encryption in transit	TLS 1.2+ everywhere (cert-manager Let's Encrypt behind Cloudflare); HSTS
Encryption at rest	Object storage server-side encryption; application-level AES-256-GCM for PII and secrets with a rotatable key ring; database volume encryption per the hosting standard
Secrets management	Kubernetes secrets created through the Codevertex secret scripts, with Sealed Secrets for critical values; no secrets in images or repositories; secret scanning in CI
Webhooks	Signature verification, timestamp tolerance, deduplication, raw-payload retention (§19)
Dependencies	Dependency and container image scanning in CI; high or critical CVEs block the release
Logging	Structured logs without PII or secrets (redaction middleware); audit log for security-relevant events
Edge	Cloudflare WAF managed rules; bot protection on sign-up and login; ingress <code>limit-rps</code> and <code>limit-connections</code>
Kubernetes	Non-root containers; read-only root filesystem where possible; resource limits; NetworkPolicy restricting <code>aelia</code> namespace egress to required services and provider hosts
Testing	SAST and secret scan on every PR; authenticated DAST on staging before G2 and G4; security and data-protection review in P5; an independent penetration test is recommended before v1 at AELIA's discretion

31.4 Ownership and access to critical assets

Requirements spec §4.1: critical AELIA assets must not exist only inside Codevertex-controlled accounts.

Asset or resource	Requirement	Implementation
Domain and DNS	AELIA controlled	Registered in AELIA's name. Codevertex is granted DNS-edit delegation
Source code	AELIA access and control	Repositories in a GitHub organisation both parties can access from day one, with AELIA as an owner-level member
Database	AELIA access and control	Dedicated <code>aelia</code> database; read access for AELIA's nominated technical contact; export on request
Analytics	AELIA access and control	Analytics property owned by AELIA; Codevertex has user access
Payment accounts	Appropriate AELIA access	Paystack, M-Pesa paybill or till, bank accounts and KRA PIN or eTIMS all in AELIA's name. Treasury stores their configuration
API credentials	Controlled access	Social developer apps and AI vendor accounts registered by AELIA; secrets entered through the Integrations Hub vault
Cloud environment	Controlled access	Codevertex-managed cluster; AELIA gets environment documentation and read access to deployment manifests
Documentation	AELIA access	This SRDD, API docs, runbooks and ADRs in the shared repository
Backups	AELIA access	Encrypted backups with a documented restore procedure; a copy available to AELIA on request
App store accounts	AELIA controlled when applicable	Created in AELIA's name when native apps are built Later

32. Data Protection & Privacy

32.1 Roles and registration

- **AELIA is the data controller** for brand, creator and campaign personal data. **Codevertex is the data processor**, hosting and operating the platform and shared services under a written data-processing agreement.
- Both register with the **Office of the Data Protection Commissioner (ODPC)** where the Data Protection Act, 2019 requires it. This is a Phase 0 legal task.
- Sub-processors are listed in the DPA and in the privacy notice: payment providers, notification channel providers, AI vendors and social platforms. Adding a provider through the Integrations Hub prompts a sub-processor review checklist.

32.2 Privacy-by-design controls

Principle	Platform control
Lawfulness and consent	Versioned privacy notice and terms acceptance at sign-up; separate, withdrawable consents for marketing messages and for social-account data processing; <code>consent_log</code> with version, purpose, time and channel
Purpose limitation	Social data used only for Passport metrics and matching; AI vendors receive only the brief content needed for generation
Data minimisation	Contact details masked until an agreement exists; brands see creator KYC status, never documents

Principle	Platform control
Accuracy	Self-service profile editing; verification re-checks on material changes
Storage limitation	Retention schedule (§27.3) enforced by scheduled jobs
Integrity and confidentiality	Part I security controls
Accountability	Audit log; a Data Protection Impact Assessment (DPIA) for matching and social data before v1; a processing register
Cross-border transfers	Providers outside Kenya (for example AI and social platforms) are covered by transfer safeguards recorded in the processing register

32.3 Data subject rights

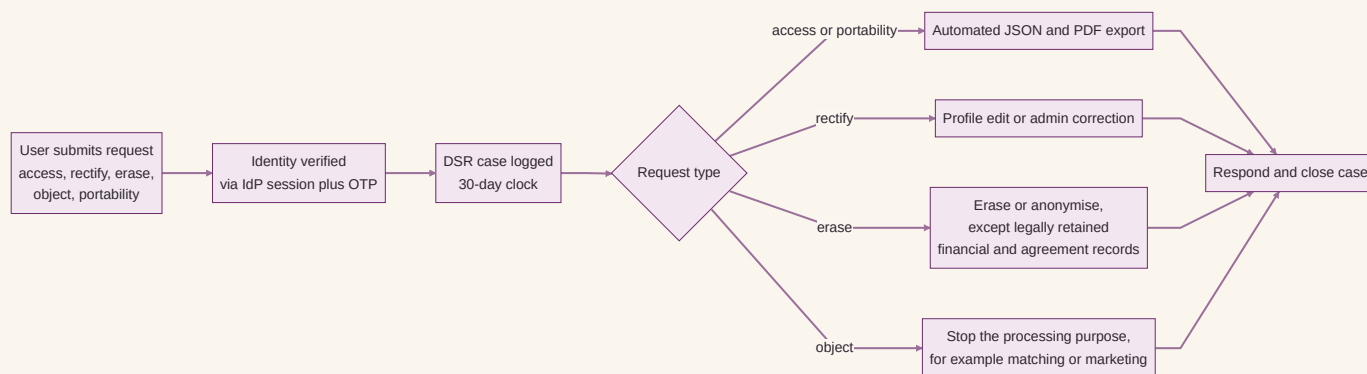


Figure 28. Data-subject-request workflow, completed within the statutory period.

32.4 Breach response

Step	Action	Timing
Detect and contain	Alert triage, isolate the affected component, revoke keys or rotate secrets	Immediately
Assess	Scope, data categories, affected subjects, risk to individuals	Within 24 h
Notify controller	Codevertex (processor) notifies AELIA (controller)	Without undue delay, within 48 h of awareness
Notify regulator	AELIA notifies the ODPC	Within 72 h of becoming aware
Notify subjects	Where there is a real risk of harm, with guidance	Without undue delay
Review	Post-incident review, corrective actions, runbook update	Within 10 business days

32.5 Sponsored-content disclosure and brand safety

- **Disclosure at submission.** Every influence deliverable carries a required disclosure specification (for example `#ad` or the platform's paid-partnership label). The creator confirms it at submission, and it is verified on the publication proof.
- **Brand-safety reporting.** A brand-safety report path is available to users and admins, and feeds the moderation queue.
- **Automated scanning** Later. Content-safety scanning plugs in through the `ModerationProvider` port.

Part J · Payment Integration Specification

The payment and transaction architecture: licensed-rail collection, the commission ledger, creator payouts with withholding, and reconciliation. It is delivered through the Codevertex Treasury service by default and can be swapped through the `PaymentProvider` and `PayoutProvider` ports.

33. Payments, Ledger & Tax

33.1 Principles

- Licensed rails only.** Holding client funds in Kenya requires a Payment Service Provider licence from the Central Bank of Kenya, and AELIA does not operate escrow or a stored-value wallet. Funds are collected and disbursed by licensed providers (Paystack, M-Pesa) through merchant accounts **in AELIA's name**. Treasury holds the gateway configuration, and AELIA records intents, ledger entries and payouts.
- Fund, then produce.** A campaign is funded after terms are agreed and before production starts. Commission is recognised at funding; the creator's share is released according to the agreed payout schedule.
- Split at the processor where supported.** On card and Paystack collection, a configured transaction split routes the platform share and the creator-payable share at processor level in real time. Where a split is not available (M-Pesa, bank transfer), the ledger segregates the creator-payable liability until payout. This behaviour is a provider capability flag (`SupportsSplit`).
- Everything is configurable.** Commission rate, who bears it (deducted from the creator's fee or added to the brand's price), processing-fee pass-through, AI-lane fees, tax rules and payout schedules are effective-dated settings, and nothing is hard-coded.
- Idempotent and reconcilable.** Every money operation carries an idempotency key. Every provider event is verified and re-queried, and every period reconciles to the shilling.

33.2 Funding flows

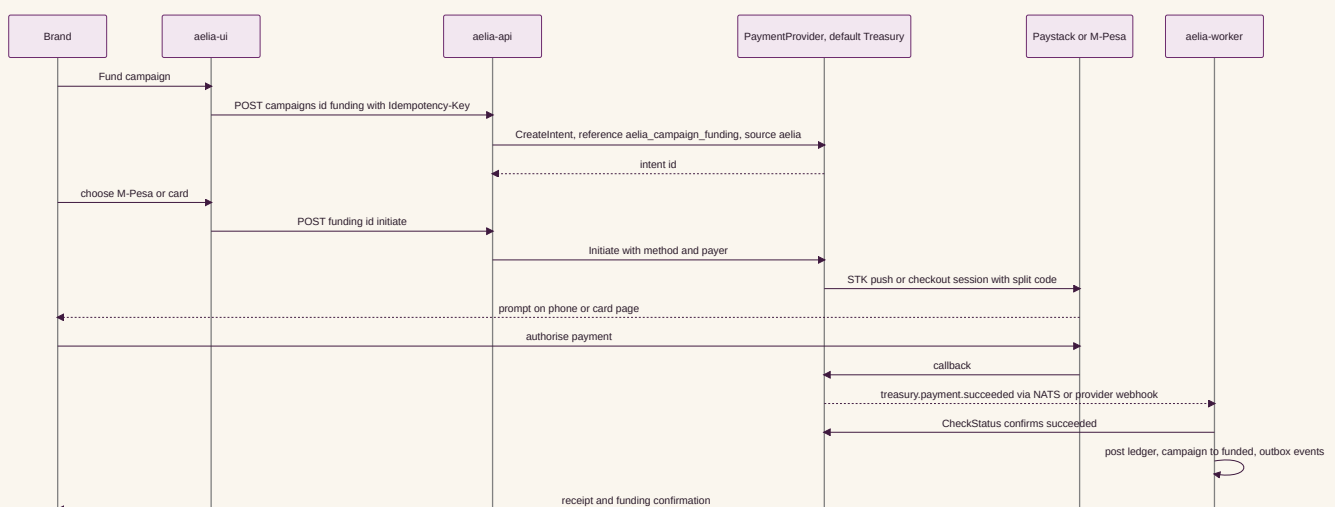


Figure 29. Card and M-Pesa funding through the Payment port. With the default profile, Treasury is the provider and events arrive on NATS.

Method	Flow	Confirmation	Release
M-Pesa (STK push)	Treasury <code>initiate</code> with <code>payment_method: mpesa</code> and the phone number; the customer approves on their handset	<code>treasury.payment.succeeded</code> plus <code>check-status</code>	MVP
Card or Paystack channels	Treasury <code>initiate</code> with <code>payment_method: paystack</code> returns <code>authorization_url</code> ; TreasuryPaymentModal or a redirect	Paystack signed callback via Treasury, plus <code>check-status</code>	MVP
Bank transfer	The brand gets AELIA bank details and a reference, then uploads proof; Finance verifies against the bank statement; Treasury <code>confirm-manual</code>	Finance OTP approval; ledger posts on approval	MVP
Other currencies and rails	New PaymentProvider bindings per tenant or country	Per provider	Later

33.3 Payouts and Treasury extensions

Payouts are created by the worker when a participant reaches a payout trigger in its schedule:

Schedule	Trigger	Amount released
Full payment on approval (default for first-time pairings)	All the participant's deliverables approved	100% of the creator share, net of withholding
Starting fee, then balance	Engagement accepted and campaign funded; then on approval	A configured percentage (for example 30–50%) as a starting fee, then the balance. WHT is applied proportionally at each release
Full payment upfront (eligibility-gated)	Engagement accepted and campaign funded	100%. Available only to creators meeting configured Passport thresholds (completion and on-time rate)

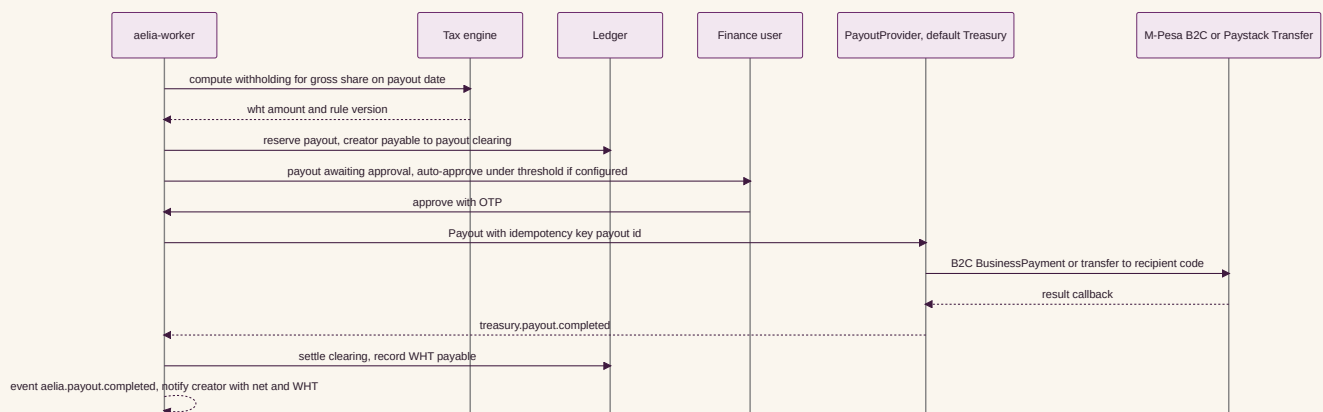


Figure 30. Creator payout with withholding, approval and idempotent disbursement.

Treasury extensions required (delivered by Codevertex in P3):

Extension	Description
creator payee type	Added to the payout engine alongside the existing payee types (riders, suppliers, staff, equity holders), with recipient creation for M-Pesa numbers, Pochi tills and bank accounts
S2S payout API	<code>POST /api/v1/{tenant}/payouts</code> with <code>payee_type</code> , <code>payee_ref</code> , <code>amount</code> , <code>currency</code> , <code>rail</code> , <code>reference_type: aelia_payout</code> , <code>reference_id</code> , and a required <code>Idempotency-Key</code> ; status query and <code>treasury.payout.completed</code> or failure events

Extension	Description
Three-way split	The existing platform-and-tenant service-charge split, extended to brand payment → platform commission → creator-payable for AELIA's reference types
Funding reference types	<code>aelia_campaign_funding</code> and <code>aelia_refund</code> , with <code>source_service: aelia</code> , so GL auto-posting maps to AELIA's chart of accounts
Commission invoicing	eTIMS invoice for AELIA's commission and fee revenue, triggered by an AELIA event

33.4 Double-entry ledger

AELIA's business ledger is the source of truth for platform revenue, creator liabilities and withholding. Treasury's GL receives summarised postings through its normal event-driven auto-posting.

Account	Type	Purpose
<code>provider_clearing:{provider}</code>	Asset	Funds collected at the provider, not yet settled to AELIA's bank
<code>bank:{account}</code>	Asset	Settled funds
<code>creator_payable:{creator}</code>	Liability	The creator's share awaiting release
<code>payout_clearing</code>	Liability	Payouts submitted, awaiting a provider result
<code>wht_payable</code>	Liability	Withholding due to KRA
<code>vat_payable</code>	Liability	VAT on fees where applicable
<code>brand_refund_payable:{brand}</code>	Liability	Refunds due
<code>commission_revenue</code>	Revenue	Platform commission
<code>fee_revenue:{type}</code>	Revenue	AI-lane, subscription and other fees
<code>processing_fees</code>	Expense	Provider charges (or recovered, per setting)

Worked test vector (configured commission 8%, borne by the creator; WHT 5% of gross):

Event	Debit	Credit	Amount (KES)
Brand funds a KES 20,000 creator fee	<code>provider_clearing</code>	<code>creator_payable</code>	18,400
	<code>provider_clearing</code>	<code>commission_revenue</code>	1,600
Content approved; payout submitted	<code>creator_payable</code>	<code>payout_clearing</code>	17,400
	<code>creator_payable</code>	<code>wht_payable</code>	1,000
Provider confirms payout	<code>payout_clearing</code>	<code>provider_clearing</code>	17,400
Monthly WHT remittance	<code>wht_payable</code>	<code>bank</code>	1,000

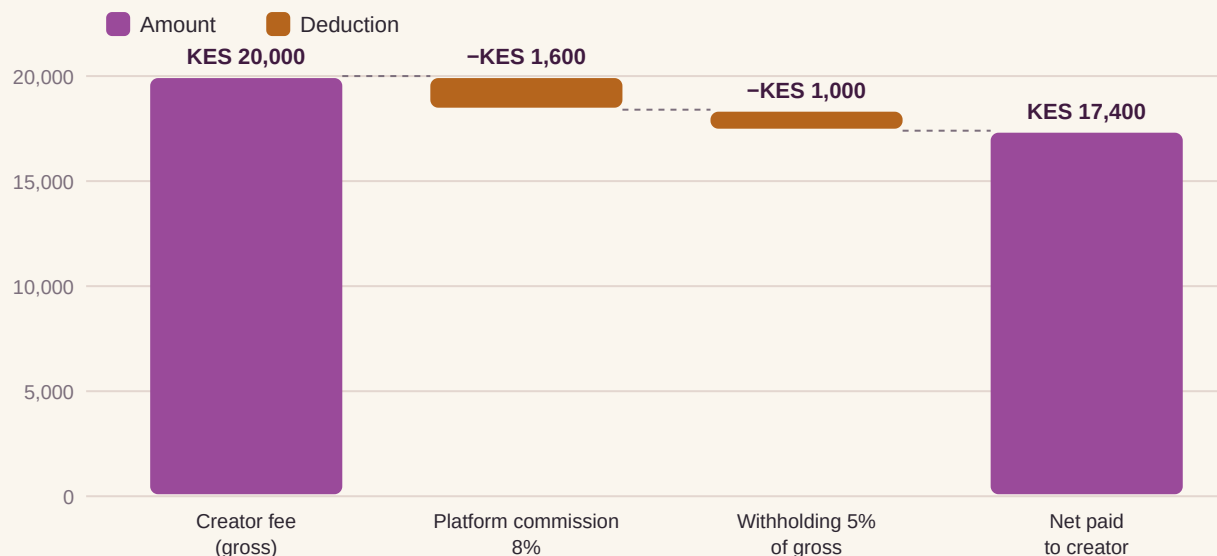


Figure 31. Payout calculation for the test vector above: gross KES 20,000 → net KES 17,400. Rates are settings, and this vector is part of the automated test suite.

33.5 Tax engine

Element	Specification
Rules	<code>tax_rules(category, rate, basis, threshold, effective_from, effective_to, country)</code> . Categories map to work types (digital content monetisation, professional services, and so on) as configured by Finance
Default rule	Digital-content withholding at 5% of the gross creator fee , applied at payout, with the rule version stamped on each withholding
Preconditions	Creator KRA PIN validated before any payout (TAX-02)
Outputs	Withholding register export for remittance (MVP); monthly per-creator withholding certificates (PDF) v1 ; eTIMS invoices for AELIA's commission and fees via Treasury v1
Change control	Rate changes are effective-dated. Funded campaigns keep the quote they were funded on, and withholding always follows the rule in force on the payout date

33.6 Reconciliation and controls

Control	Frequency	Pass criterion
Ledger balance check	Every posting	Debits = credits per transaction; the posting is rejected otherwise
Provider reconciliation	Daily (automated)	Every provider transaction matches exactly one funding intent or payout, and vice versa; exceptions go to the Finance queue
Bank reconciliation	Daily for manual transfers; monthly overall	Settled amounts match bank statements
Period close	Monthly	Ledger trial balance, WHT register and provider statements agree to the shilling (the G2 exit criterion)
Stuck-state sweeper	Every 15 min	Intents and payouts pending beyond their SLA are re-queried with <code>CheckStatus</code> , then escalated
Segregation of duties	Always	The creator of a payout batch cannot approve it; integration changes to money providers need dual approval

Part K · Test & UAT Plan

How the platform is tested and accepted: test levels, integration conformance, pilot UAT, and the acceptance criteria for each gate.

34. Test Strategy

34.1 Test levels

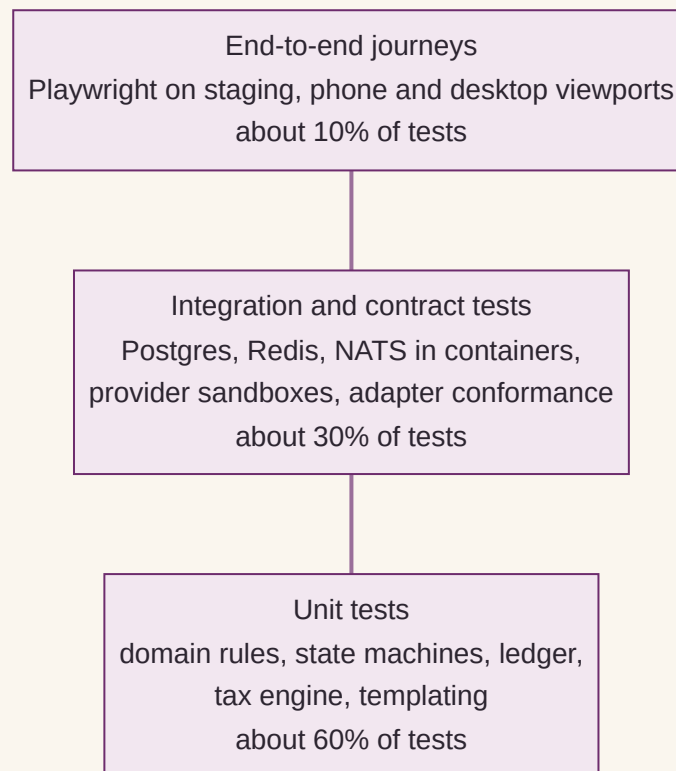


Figure 32. Test pyramid. Money, tax and state-machine code has at least 70% coverage.

Level	Scope	Tooling	When
Unit	Domain services, state-machine guards, ledger postings, tax rules, fee quotes, connector templating	Go <code>testing</code> , testify; Vitest for UI logic	Every commit (CI)
Integration	Repositories and migrations, outbox relay, consumers, webhook verification, idempotency	Testcontainers (PostgreSQL, Redis, NATS)	Every PR
Contract and conformance	Every adapter against its port contract (§34.4)	Shared conformance suite, provider sandboxes and recorded fixtures	Every PR touching adapters; nightly against live sandboxes
API	OpenAPI conformance, authorization matrix (every endpoint × every role), error codes	Generated tests plus a role-matrix harness	Every PR
End-to-end	The 16-step brand and creator journeys; funding and payouts in sandbox	Playwright	Nightly on staging; before each gate

Level	Scope	Tooling	When
Performance	NFR-02 targets: 1k concurrent users, search at 10k creators, webhook bursts	k6	P5 (before G4); after major changes
Security	SAST, secret scan, dependency and image scan, authenticated DAST, access review	CI scanners, OWASP ZAP	Every PR (static); before G2 and G4 (dynamic)
Resilience	Provider outage and circuit breaker, NATS redelivery, duplicate webhooks, pod loss	Fault injection in staging	P5
Recovery	Backup restore to a clean environment within 4 h	Restore drill	Before G5, then quarterly
Accessibility	WCAG 2.1 AA checks on core screens	axe and manual review	v1

34.2 Critical test scenarios (money and state)

#	Scenario	Expected result
T-01	Duplicate <code>treasury.payment.succeeded</code> delivery	Funding posted once; the second is acknowledged and ignored
T-02	Forged or unsigned payment webhook	401; no state change; security log entry
T-03	Provider says succeeded but <code>check-status</code> says pending	Funding stays pending; re-queried by the sweeper
T-04	Payout retried after a timeout	Same idempotency key; exactly one disbursement
T-05	Starting-fee schedule with a partial refund on dispute	Ledger balances; WHT proportional; refund equals policy
T-06	Tax rate changed between funding and payout	Quote unchanged; withholding follows the rule on the payout date; rule version stamped
T-07	Payout method changed within 24 h of a payout	Payout held; creator and Finance notified
T-08	Content approval on an unfunded campaign	409 <code>invalid_state_transition</code>
T-09	Brand member without approver role approves content	403 <code>forbidden</code>
T-10	Primary notification provider down	Fallback provider used for notifications; money operations never fail over
T-11	Identity provider swapped to generic OIDC in staging	Existing users log in and keep their data through <code>identity_links</code>
T-12	Month-end reconciliation with 100 synthetic campaigns	Trial balance, WHT register and provider statements agree to the shilling

34.3 Environments and test data

- **Staging.** Mirrors production topology with sandbox providers: the Paystack test mode, the Daraja sandbox, the Treasury eTIMS sandbox, social apps in development mode and the AI vendor on trial keys.

- **Test data.** Synthetic seed data (brands, creators, campaigns) is generated by a seeding command. Production data is never copied to lower environments.
- **Pilot preparation.** Pilot brands and creators are onboarded in production directly after G2. Staging keeps demo tenants for training.

34.4 Integration adapter conformance suite

Every adapter, whether Codevertex, vendor or a `generic.rest` connector, must pass the suite for its port before activation (INT-03). The same suite runs in CI for built-in drivers and from the Integrations Hub's "Test connection" for configured providers.

Port	Conformance checks
IdentityProvider	Discovery and JWKS reachable; PKCE flow; token signature, issuer and audience validation; refresh; user-info claim mapping; logout redirect
NotificationProvider	Send per supported channel to a test recipient; idempotent resend; status query or delivery receipt parsing; template variables rendered
BillingProvider	Entitlements fetch; feature check; limit check; plan change reflected within the cache TTL
PaymentProvider	Create intent; initiate (sandbox); status mapping of every provider status to canonical; webhook signature verify (valid, invalid, replayed); refund; idempotency
PayoutProvider	Recipient creation; payout with an idempotency key (a repeat returns the same payout); success and failure callbacks; status mapping
AIVideoVendor / Social / Storage	Auth; create and poll a job or fetch a profile; error mapping; cost quote; revoke (social); presign put and get (storage)

35. User Acceptance Testing & Acceptance Criteria

35.1 UAT approach

- **Who.** UAT runs with the pilot's real brands and creators plus AELIA operations staff, led by the AELIA product owner (accountable) with Codevertex support.
- **Scripts.** Scripts follow the journeys in §5, with each script referencing the FR IDs it verifies. Defects are triaged by severity: critical and high must be fixed before sign-off; medium and low are scheduled.
- **Sign-off.** Sign-off is recorded per gate by the designated AELIA product owner.

UAT script	Journey	Verifies
UAT-01	Brand registers, is verified, invites a team member	REG-01...07, BPP-01
UAT-02	Creator builds a Passport, sets rates and payout method, is verified	CPP-01...06, REG-04, TAX-02
UAT-03	Brand creates a brief; admin curates a shortlist; brand selects; admin confirms	CMP-01...04, DIS-05, MAT-01, MAT-04
UAT-04	Terms accepted by both parties; campaign funded by M-Pesa	AGR-01...04, PAY-01, PAY-03
UAT-05	Campaign funded by bank transfer with proof; Finance verifies	PAY-02, ADM-03
UAT-06	Creator submits, brand requests a revision, then approves; publication proof recorded	CNT-01...05
UAT-07	Payout with WHT under each payout schedule; creator sees net and tax	PAY-04, TAX-01, TAX-03

UAT script	Journey	Verifies
UAT-08	Messaging and notifications at every step (in-app, email, SMS)	COM-01...03, COM-07
UAT-09	Month-end: reconciliation and WHT register	PAY-03, RPT-03, TAX-03
UAT-10 v1	Ranked matching, reviews, rebooking, reports, certificates, AI clip order	MAT-02...05, REP-*, RBK-02, RPT-02, TAX-04, AIC-02...04
UAT-11 v1	Super Admin adds a sandbox provider, binds it, receives a webhook, replays an outbound webhook	INT-02...06
UAT-12 v1	Dispute opened, mediated and settled with a partial refund	ADM-04, PAY-05

35.2 Gate acceptance criteria

Gate	Week	Acceptance criteria
G1 · Environments live	2	CI/CD deploys every merge to staging; SSO login works for test brand and creator accounts; Codevertex default integration profile seeded and healthy; monitoring and alerts active
G2 · MVP pilot live	13	UAT-01...09 passed; a real campaign funded, delivered, approved and paid out in production; ledger reconciles to the shilling; no open critical or high security findings; runbook for pilot support published
G3 · Feature complete	22	All v1 requirements implemented; every brief gets a ranked shortlist; every AI clip is costed automatically; conformance suite green for every bound provider
G4 · UAT signed off, v1 live	24	UAT-10...12 passed; load test meets NFR-02; security and data-protection review complete; v1 released to production
G5 · Handover	26	Documentation and training delivered; restore proven within 4 h; zero open high or critical findings; two weeks of close post-launch support completed

35.3 Pilot validation (business acceptance, months 3–6)

Pilot targets are in §7.3. The platform must report every indicator directly from its own records: brands and creators verified, campaigns completed, paid transactions, on-time delivery rate, rebooking intent, and the economic validation data.

Part L · Deployment & Support Runbook

How the platform is deployed, operated, monitored, backed up and supported, and the technical risks that the plan manages.

36. Deployment Architecture

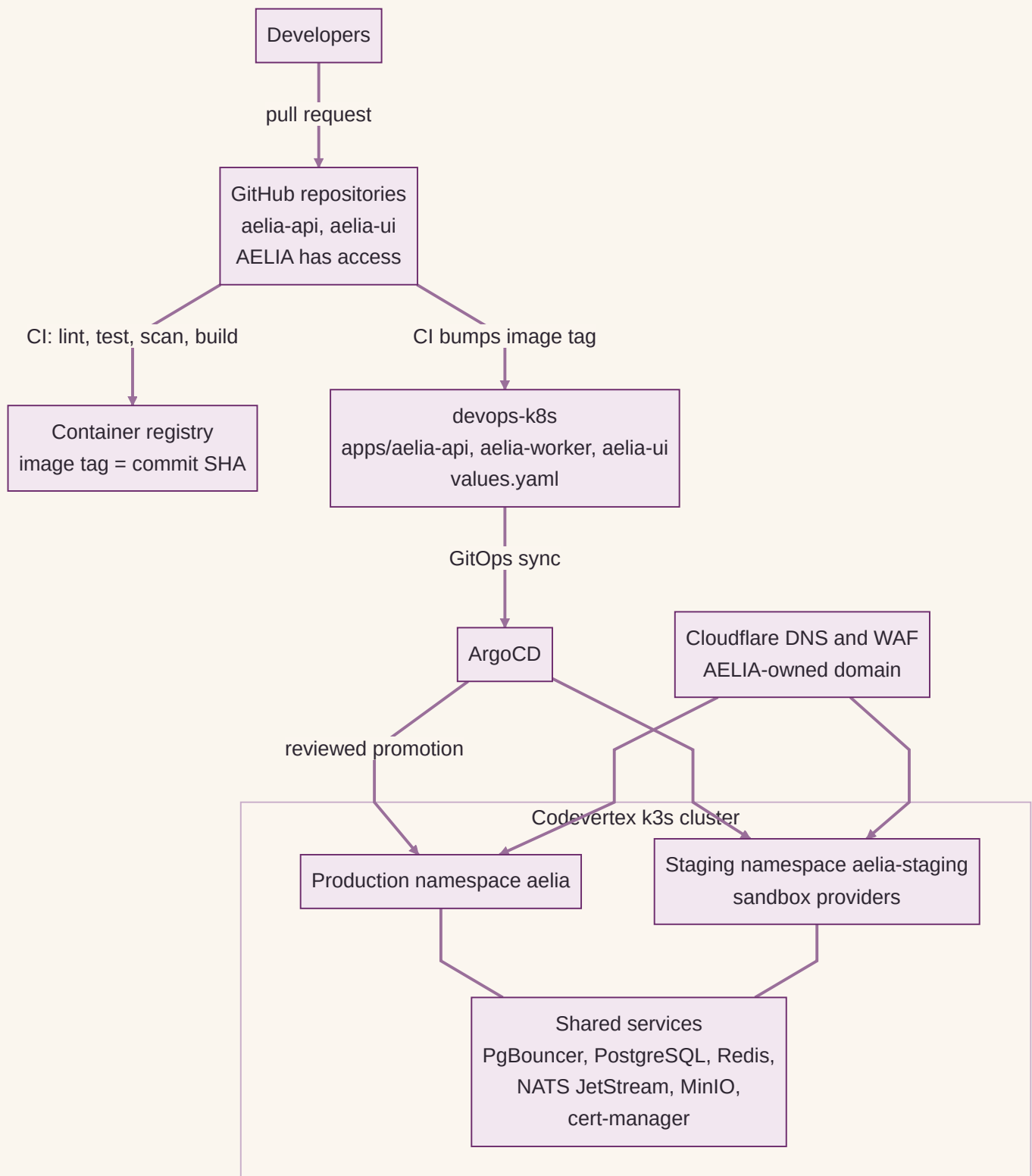


Figure 33. Build and deployment pipeline. Production is served on AELIA's own domain through Cloudflare and cluster ingress with automatic TLS.

36.1 Environments

Environment	Hosts	Providers	Data	Deployment
Development	Local (Docker Compose) and preview builds	Mocks and sandboxes	Synthetic	On demand
Staging	<code>app.staging.<aelia-domain></code> , <code>api.staging.<aelia-domain></code>	Sandboxes (Paystack test, Daraja sandbox, eTIMS sandbox)	Synthetic and demo	Auto on merge to <code>main</code>
Production	<code>app.<aelia-domain></code> , <code>api.<aelia-domain></code>	Live, in AELIA's name	Live	Reviewed promotion (tag); change window agreed with AELIA

36.2 Kubernetes and configuration

Item	Specification
Chart	The Codevertex generic chart <code>charts/app</code> , registered in <code>devops-k8s</code> as <code>apps/aelia-api</code> , <code>apps/aelia-worker</code> and <code>apps/aelia-ui</code> (<code>app.yaml</code> plus <code>values.yaml</code>)
Replicas and availability	<code>replicaCount: 2</code> , PodDisruptionBudget <code>minAvailable: 1</code> , HPA on CPU (target 70%); VPA in recommendation mode
Probes	<code>/healthz</code> (liveness), <code>/readyz</code> (readiness: DB, Redis and NATS reachable)
Resources (initial)	api: 100m to 500m CPU, 128 to 512 Mi; worker: 50m to 300m CPU, 128 to 384 Mi; ui: 50m to 300m CPU, 128 to 384 Mi. Tuned from pilot metrics
Migrations	Container entrypoint runs <code>migrate</code> (Atlas, direct <code>POSTGRES_MIGRATE_URL</code>), then <code>serve</code> ; expand/contract for breaking changes
Configuration	Env vars: <code>DATABASE_URL</code> , <code>POSTGRES_MIGRATE_URL</code> , <code>REDIS_URL</code> , <code>EVENTS_NATS_URL</code> , <code>PUBLIC_BASE_URL</code> , <code>SSO_ISSUER_URL</code> , <code>NOTIFICATIONS_API_URL</code> , <code>SUBSCRIPTIONS_API_URL</code> , <code>TREASURY_API_URL</code> , <code>VERA_API_URL</code> , <code>OBJECT_STORAGE_*</code> , <code>VAULT_MASTER_KEY</code> (secret), <code>INTERNAL_SERVICE_KEY</code> (secret)
Ingress	cert-manager <code>letsencrypt-prod</code> ; CORS limited to the AELIA app origins; rate-limit annotations; body size 50 MB for the API (uploads go direct to storage)
Jobs	<code>seed-integrations</code> (post-install and upgrade hook); CronJobs inside the worker for reminders, metrics refresh, reconciliation, certificates, retention

36.3 Release process

1. Merge to `main` after review and green CI. The image is built and pushed, CI updates the staging `image.tag`, and ArgoCD syncs.
2. QA on staging, including automated end-to-end tests. The sprint demo happens here.
3. Promote by tagging a release. CI updates the production `image.tag`, ArgoCD syncs with a rolling update and zero downtime, and release notes go to AELIA.
4. **Rollback:** revert the image tag in `devops-k8s`, and ArgoCD rolls back. Expand/contract migrations keep the previous version compatible.

37. Operations & Support

37.1 Monitoring and alerting

Signal	Source	Alert
Availability	External uptime checks on <code>app</code> and <code>api</code> every minute	Down for 2 minutes → on-call (Slack and email via <code>fleet-health-watcher</code>)
Errors and latency	Structured logs (zap via <code>httpware</code>) with <code>request_id</code> , <code>tenant_id</code> , <code>user_id</code> ; <code>/metrics</code> endpoints	5xx rate above 2% over 5 min; p95 above 800 ms over 10 min
Money health	Worker metrics: stuck intents and payouts, reconciliation exceptions, webhook verification failures	Any stuck item past SLA; any failed verification burst
Integrations	Integrations Hub health (every 60 s), circuit-breaker state	Provider unhealthy for 5 minutes → Super Admin and on-call
Event bus	Outbox backlog and consumer lag	Outbox rows older than 2 minutes; lag above 1,000 messages
Capacity	Pod CPU and memory; database connections; storage growth	Above 80% sustained for 15 min

37.2 Backup and recovery

Item	Policy
Database	Daily full backups plus WAL archiving where available; encrypted; retained 30 days (daily) and 12 months (monthly); copy available to AELIA
Object storage	Versioned bucket with replication to a secondary location; 30-day version retention
Configuration	GitOps manifests in <code>devops-k8s</code> ; integration configuration inside the database backup; secrets escrowed per the secrets standard
Targets	RTO 4 h and RPO 24 h (NFR-04); restore drill before G5, then quarterly

37.3 Incident management and support

Severity	Definition	Response	Target resolution or workaround
Critical	Platform down, payments or payouts failing, data breach suspected	Within 2 h, any day	4 h
High	Core journey blocked for many users (for example submissions failing)	Within 4 business hours	1 business day
Medium	Feature degraded with a workaround	Within 1 business day	Next sprint
Low	Cosmetic issue or question	Within 2 business days	Backlog

Support after launch covers:

- round-the-clock uptime monitoring and alerts
- security patches, updates and certificate renewals
- daily backups and quarterly restore tests

- bug fixes, and running the AI-generated content pipeline
- small improvements, and a monthly health report to AELIA

The first two weeks after v1 launch are close post-launch support with daily check-ins. On partnership transition, Codevertex provides 90 days of transition support with full runbooks.

37.4 Runbook index (delivered in P5)

Runbook	Contents
RB-01 Deploy and rollback	Promotion, rollback, migration failure handling
RB-02 Payment incident	Stuck funding or payouts, provider outage, manual reconciliation, customer communication
RB-03 Provider swap	Adding and binding a provider, rollback to the Codevertex default profile
RB-04 Secret rotation	Vault key rotation, provider credential rotation, webhook secret rotation
RB-05 Restore	Database and object restore, verification checklist
RB-06 Security incident and breach	Containment, assessment, notification timeline (§32.4)
RB-07 Data subject requests	Export, erasure, verification
RB-08 Month-end close	Reconciliation, WHT register, certificates, eTIMS checks

38. Technical Risks, Assumptions & Dependencies

Risk	Severity	Mitigation	Owner
Payment and tax account approvals (Paystack, M-Pesa paybill, KRA eTIMS) take longer than planned	High	Applications start in week 1; payments are built and tested in sandbox meanwhile; the Treasury integration already exists	AELIA, with Codevertex
Legal sign-off on the split-payment and funds-flow design slips past month 1	High	Counsel engaged in week 1; the design supports both processor split and ledger-segregated modes as configuration	AELIA
Treasury extensions (creator payee, payout API, three-way split) delay P3	Medium	Scheduled early in P3 by the Treasury owner; the <code>PayoutProvider</code> port allows an interim direct adapter	Codevertex
Social platform API permissions or app review limit metrics access	Medium	Manual metrics in MVP; apps submitted for review in P4; adapters degrade gracefully	Codevertex, with AELIA app ownership
AI video vendor pricing or API changes	Medium	Cost per clip tracked automatically; brand price is a setting; the <code>AIVideoVendor</code> port allows switching vendors	Codevertex
Scope growth during the build	Medium	Change control with a 3-day impact estimate; the Future Enhancements Backlog	Both
Too few pilot brands or creators for a meaningful UAT	Medium	Recruitment through AELIA Marketing clients and Academy creators from month 2	AELIA
Shared-cluster resource contention	Low	Resource requests and limits; HPA; move to dedicated capacity if pilot metrics require it	Codevertex

Risk	Severity	Mitigation	Owner
Team member unavailable	Low	Code reviews, runbooks, backup from the wider Codevertex engineering team	Codevertex

Assumptions:

- A named AELIA product owner is available for weekly decisions and fortnightly demos.
- AELIA provides brand assets and copy for public pages, legal texts (terms, privacy notice, agreement templates) and business accounts in its name.
- Scope follows §7; changes go through change control.

Exclusions (this SRDD's delivery plan):

- Native mobile apps and all other **Later** items.
- The Aelia Brand House website.
- Independent third-party penetration testing (recommended, at AELIA's discretion).
- Legal and regulatory filings, content production and paid advertising.

39. Appendix A · Requirements Traceability Matrix

Spec requirement (source)	SRDD requirement IDs	Design reference	Verified by
User Registration & Verification	REG-01...10	§12.4, §31.2, Part F 26.1	UAT-01, UAT-02
Creator Passport	CPP-01...10	§13, §17.1 SocialPlatformConnector	UAT-02, UAT-10
Brand Passport	BPP-01...04	§13, Part F 26.1	UAT-01
Creator Discovery	DIS-01...06	§22.2	UAT-03
Creator Matching	MAT-01...06	§13, §17.1 AIMatchProvider	UAT-03, UAT-10
Campaign Creation	CMP-01...08	§22.2, §28	UAT-03
Campaign Workspace	WSP-01...05	§30.3	UAT-06, UAT-08
Content Marketplace and Submission	CNT-01...08	§29.1	UAT-06
Contracts and Rights	AGR-01...06	Part F 26.2	UAT-04
Communication	COM-01...07	§14 Notifications, §23.3	UAT-08
Payments	PAY-01...10	Part J	UAT-04, 05, 07, 09; T-01...12
Tax (feasibility correction)	TAX-01...07	§33.5	UAT-07, UAT-09
Performance Tracking	PRF-01...04	§22.3	UAT-06, UAT-10
Reputation	REP-01...03	§22.5	UAT-10
Rebooking	RBK-01...03	§22.2	UAT-10
Reporting	RPT-01...05	§13 reporting	UAT-09, UAT-10
Administration	ADM-01...08	§30.2, §29.2	UAT-05, UAT-12
AI content lane (feasibility)	AIC-01...06	§17.1 AIVideoVendor	UAT-10

Spec requirement (source)	SRDD requirement IDs	Design reference	Verified by
Configurable integrations (this SRDD)	INT-01...07	Part D	UAT-11; §34.4
Technical, data and trust requirements (spec Part 4)	NFR-01...16	Parts C, F, I, L	§34, §35
Critical ownership and access (spec §4.1)	NFR-15	§31.4	G5 checklist
API documentation standard (spec §4.2)	–	§24	G5 checklist
Pilot validation (spec §5.1)	–	§7.3	§35.3
Definition of Done (spec §5.2)	–	§8.3	Every sprint

40. Appendix B · References & Sign-Off

40.1 References

Reference	Location
AELIA System Requirements & MVP Specification v1.0	Aelia Holdings, September 2026
AELIA Creator Commerce Technical Partnership & Feasibility Report	Codevertex Africa, 21 September 2026
AELIA Creator Commerce Project Proposal (technical and delivery sections)	Codevertex Africa, 26 September 2026
Codevertex microservice, event and data-ownership architecture	shared-docs/docs/architecture/
Codevertex SSO integration, login flow and authorization pattern	shared-docs/docs/architecture/
Notifications REST API; Treasury payment workflow; Paystack, M-Pesa and eTIMS references	shared-docs/docs/integrations/
Platform engineering standards: S2S conventions, idempotency and outbox, secrets management	shared-docs/internal/platform-standards/

40.2 Document approval

Signing below approves this SRDD v1.0 as the baseline for development starting 5 October 2026. Later changes follow the change-control process in §8.4.

FOR AELIA HOLDINGS LIMITED · PRODUCT OWNER

NAME

SIGNATURE

DATE

FOR CODEVERTEX AFRICA LIMITED · TECHNICAL LEAD

NAME

SIGNATURE

DATE